

16. Buses

16.1 Overview

Table 16.1 lists the bus specifications, Figure 16.1 shows the bus configuration, and Table 16.2 lists the addresses assigned for each bus.

Table 16.1 Bus Specifications

Bus Type		Description
CPU bus	Instruction bus	- Connected to the CPU (for instructions) - Connected to on-chip memory (RAM, expansion RAM, ECCRAM, code flash memory) - Operates in synchronization with the system clock (ICLK)
	Operand bus	- Connected to the CPU (for operands) - Connected to on-chip memory (RAM, expansion RAM, ECCRAM, code flash memory) - Operates in synchronization with the system clock (ICLK)
Memory bus	Memory bus 1	Connected to RAM
	Memory bus 2	Connected to code flash memory
	Memory bus 3	Connected to expansion RAM and ECCRAM
Internal main bus	Internal main bus 1	- Connected to the CPU - Operates in synchronization with the system clock (ICLK)
	Internal main bus 2	- Connected to the DMAC, DTC, and extended bus master - Connected to on-chip memory (RAM, expansion RAM, ECCRAM, code flash memory) - Operates in synchronization with the system clock (ICLK)
Internal peripheral bus	Internal peripheral bus 1	- Connected to peripheral modules (DTC, DMAC, EXDMAC, interrupt controller, and bus error monitoring section) - Operates in synchronization with the system clock (ICLK) (EDMAC operates in synchronization with the BCLK)
	Internal peripheral bus 2	- Connected to peripheral modules (modules other than those connected to internal peripheral buses 1, 3 to 5) - Operates in synchronization with the peripheral-module clock (PCLKB)
	Internal peripheral bus 3	- Connected to peripheral modules (USB, PDC, and standby RAM) - Operates in synchronization with the peripheral-module clock (PCLKB)
	Internal peripheral bus 4	- Connected to peripheral modules (EDMAC, ETHERC, PMGI, EPTPC, GPTW, MTU, SCLi, RSPI) - Operates in synchronization with the peripheral-module clock (PCLKA)
	Internal peripheral bus 5	- Connected to peripheral modules (GLCDC, DRW2D) - Operates in synchronization with the peripheral-module clock (PCLKA)
	Internal peripheral bus 6	- Connected to code flash (in P/E) and data flash memory - Operates in synchronization with the FlashIF clock (FCLK)
External bus	CS area	- Connected to the external devices - Operates in synchronization with the external-bus clock (BCLK)
	SDRAM area	- Connected to the SDRAM - Operates in synchronization with the SDRAM clock (SDCLK)

P/E: Programming/Erase

BCLK (external-bus clock): 120 MHz (max.) The CSC (CS area controller) and the EXDMAC operate in synchronization with the BCLK.

SDCLK (SDRAM clock): 80 MHz (max.) The SDRAMC (SDRAM area controller) operates in synchronization with the SDCLK.

BCLK pin output: The frequency is the same as the BCLK as default. 1/2 BCLK can be supplied by setting the BCLK pin output select bit (BCKCR.BCLKDIV) in the external bus clock control register. For details, refer to section 9, Clock Generation Circuit.

Note: The BCLK and the SDCLK should be operated with the same frequency when the SDRAM is in use.

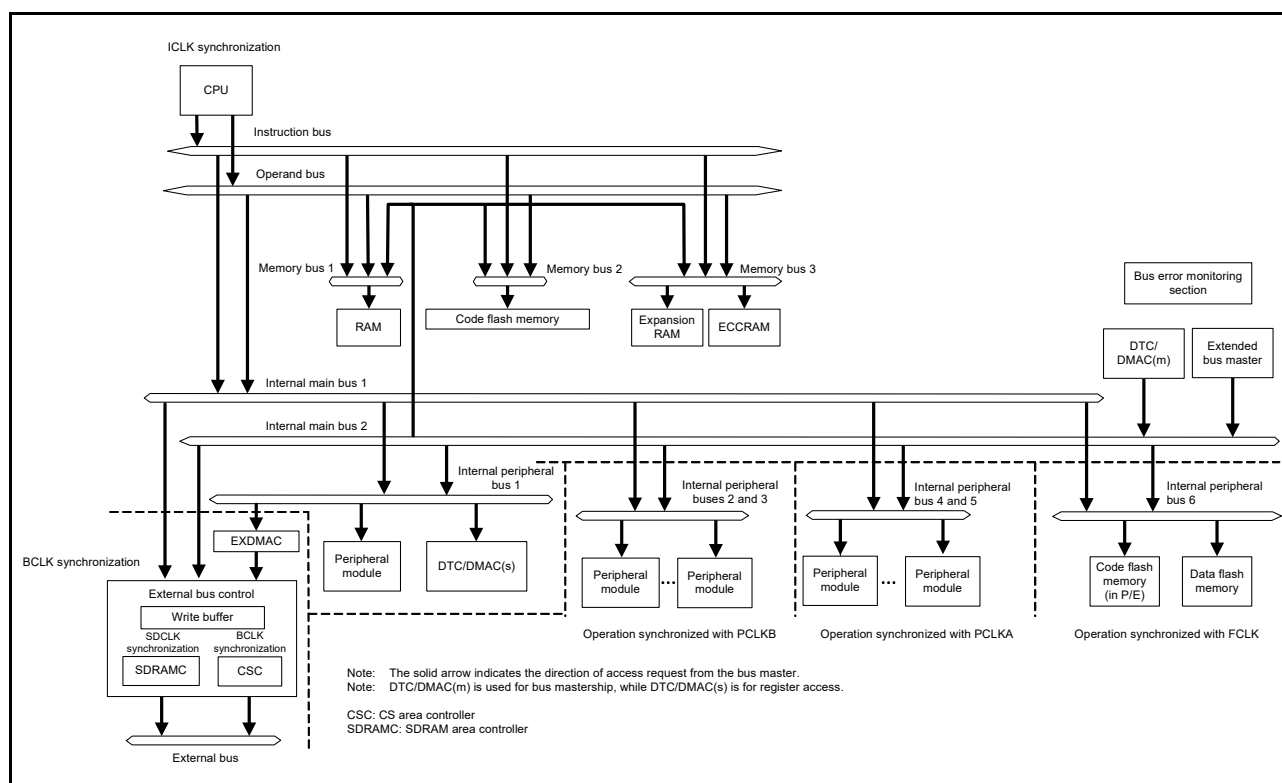


Figure 16.1 Bus Configuration

Table 16.2 Addresses Assigned for Each Bus

Address	Bus		Area	
	On-Chip ROM Enabled	On-Chip ROM Disabled	On-Chip ROM Enabled	On-Chip ROM Disabled
0000 0000h to 0007 FFFFh	Memory bus 1		RAM	
0008 0000h to 0008 7FFFh	Internal peripheral bus 1		Peripheral I/O registers	
0008 8000h to 0009 FFFFh	Internal peripheral bus 2			
000A 0000h to 000B FFFFh	Internal peripheral bus 3			
000C 0000h to 000D FFFFh	Internal peripheral bus 4			
000E 0000h to 000F FFFFh	Internal peripheral bus 5			
0010 0000h to 007F FFFFh	Internal peripheral bus 6	Reserved area	Code flash memory (for programming/erasure), data flash memory	Reserved area
0080 0000h to 0087 FFFFh	Memory bus 3		Expansion RAM	
0088 0000h to 00FF 7FFFh			Reserved area	
00FF 8000h to 00FF FFFFh			ECCRAM	
0100 0000h to 07FF FFFFh	External bus		External address space (CS1 to CS7)	
0800 0000h to 0FFF FFFFh			SDRAM area	
1000 0000h to 7FFF FFFFh	Reserved area		Reserved area	
8000 0000h to FEFF FFFFh	Memory bus 2	Reserved area	Code flash memory	Reserved area
FF00 0000h to FFFF FFFFh		External bus		External address space (CS0)

16.2 Description of Buses

16.2.1 CPU Buses

The CPU buses consist of the instruction and operand buses, which are connected to internal main bus 1. As the names suggest, the instruction bus is used to fetch instructions for the CPU, while the operand bus is used for operand access.

The instruction bus and the operand bus are 64 bits.

Connection of the instruction and operand buses to RAM, expansion RAM, ECCRAM and code flash memory provides the CPU with direct access to these areas, i.e. access is not via internal main bus 1. However, only reading is possible in direct access to code flash memory by the CPU; programming and erasure are handled via an internal peripheral bus.

Bus requests for instruction fetching and operand access are arbitrated through internal main bus 1. The order of priority is operand access then instruction fetching.

If instruction fetching and operand access are requested for different buses (memory bus 1, memory bus 2, memory bus 3, and internal main bus 1), the bus-access operations can proceed simultaneously. For example, parallel access to code flash memory and RAM or to code flash memory and external space is possible.

16.2.2 Memory Buses

The memory buses consist of memory bus 1, memory bus 2, and memory bus 3. The RAM is connected to memory bus 1, code flash memory is connected to memory bus 2, and expansion RAM and ECCRAM are connected to memory bus 3. The memory buses are 64 bits. Requests for bus mastership from the CPU buses (instruction fetching and operand) and internal main bus 2 are arbitrated through memory buses 1 to 3.

The priority order of the buses can be set using the memory bus 1 and 3 (RAM, expansion RAM, ECCRAM) priority control bits (BPRA[1:0]), and memory bus 2 (code flash memory) priority control bits (BPRO[1:0]) in the bus priority control register (BUSPRI) for the corresponding memory buses. Memory bus 3 has the same configuration as memory bus 1. When the priority order is fixed, internal main bus 2 has priority over the CPU bus (operand over instruction fetching). When the priority order is toggled, a bus has a lower priority when the request of that bus is accepted.

16.2.3 Internal Main Buses

The internal main buses consist of a bus for use by the CPU (internal main bus 1) and a bus for use by the other bus-master modules, i.e. the DTC, DMAC, and extended bus master (internal main bus 2).

Bus requests for instruction fetching and operand access are arbitrated through internal main bus 1. The order of priority is operand access then instruction fetching.

Requests for bus mastership from the DTC, DMAC, and extended bus master are arbitrated by internal main bus 2. The order of priority is extended bus master, DMAC, and then DTC as listed in **Table 16.3**.

Between the DTC and DMAC, only the one that accepted the transfer request issues the bus mastership request. The priority order of transfer requests between the DTC and DMAC is DMAC0, DMAC1, DMAC2, DMAC3, DMAC4, DMAC5, DMAC6, DMAC7, and then DTC, regardless of the BUSPRI setting.

If the CPU and another bus master are requesting access to different buses (on-chip memory, internal peripheral buses 1 to 6, and external bus), the respective bus-access operations can proceed simultaneously.

However, when the CPU executes the XCHG instruction, requests for bus access from masters other than the CPU are not accepted until data transfer for the XCHG instruction is completed regardless of the bus priority control register (BUSPRI) setting. Furthermore, requests for bus access from masters other than the DTC are not accepted during reading and writing-back of transfer control information for the DTC.

Table 16.3 Order of Priority for Bus Masters

Priority	Internal main bus	Bus Master
High ↑	—	EXDMAC
	2	Extended bus master
		DMAC
		DTC
Low	1	CPU

Note: The above applies when the priority order of the buses is fixed.

The priority order of the internal main bus 1 and other buses (internal main bus 2 and EXDMAC) can be toggled by using the bus priority control register (BUSPRI) (round-robin method).

However, the priority order of EXDMAC to be connected only to the external bus has priority over internal main bus 2 regardless of the setting of the bus priority control register (BUSPRI) (EXDMAC > internal main bus 2).

The extended bus master of three functions of the EDMAC, GLCDC, and DRW2D modules. The EDMAC, GLCDC, and DRW2D can be operated at the same time. The priority order among the EDMAC, GLCDC, and DRW2D in cases of contention of requests for bus mastership can be set in the EBMAPCR register.

16.2.4 Internal Peripheral Buses

Connection of peripheral modules to the internal peripheral buses is as described in Table 16.4.

Table 16.4 Connection of Peripheral Modules to the Internal Peripheral Buses

Type of Bus	Peripheral Modules
Internal peripheral bus 1	DTC, DMAC, EXDMAC, interrupt controller, and bus error monitoring section
Internal peripheral bus 2	Peripheral modules other than those connected to internal peripheral buses 1, 3 to 5
Internal peripheral bus 3	USB, PDC, and standby RAM
Internal peripheral bus 4	EDMAC, ETHERC, PMGI, EPTPC, GPTW, MTU, SCLi, and RSPI
Internal peripheral bus 5	GLCDC and DRW2D
Internal peripheral bus 6	Code flash memory (in P/E) or data flash memory

Requests for bus mastership from the CPU (internal main bus 1) and other bus masters (internal main bus 2) are arbitrated through internal peripheral buses 1 to 6.

The priority order of two internal main buses can be set using the bus priority control register (BUSPRI). The priority order can be set with the internal peripheral bus 1 priority control bits (BUSPRI.BPIB[1:0]), internal peripheral bus 2 and 3 priority control bits (BUSPRI.BPGB[1:0]), internal peripheral bus 4 and 5 priority control bits (BUSPRI.BPHB[1:0]), and internal peripheral bus 6 priority control bits (BUSPRI.BPFB[1:0]) for the corresponding internal peripheral buses. When the priority order is fixed, internal main bus 2 has priority over internal main bus 1. When the priority order is toggled, a bus has a lower priority when the request of that bus is accepted (round-robin method).

The order of accepting requests may change depending on the BUSPRI setting (refer to Figure 16.2).

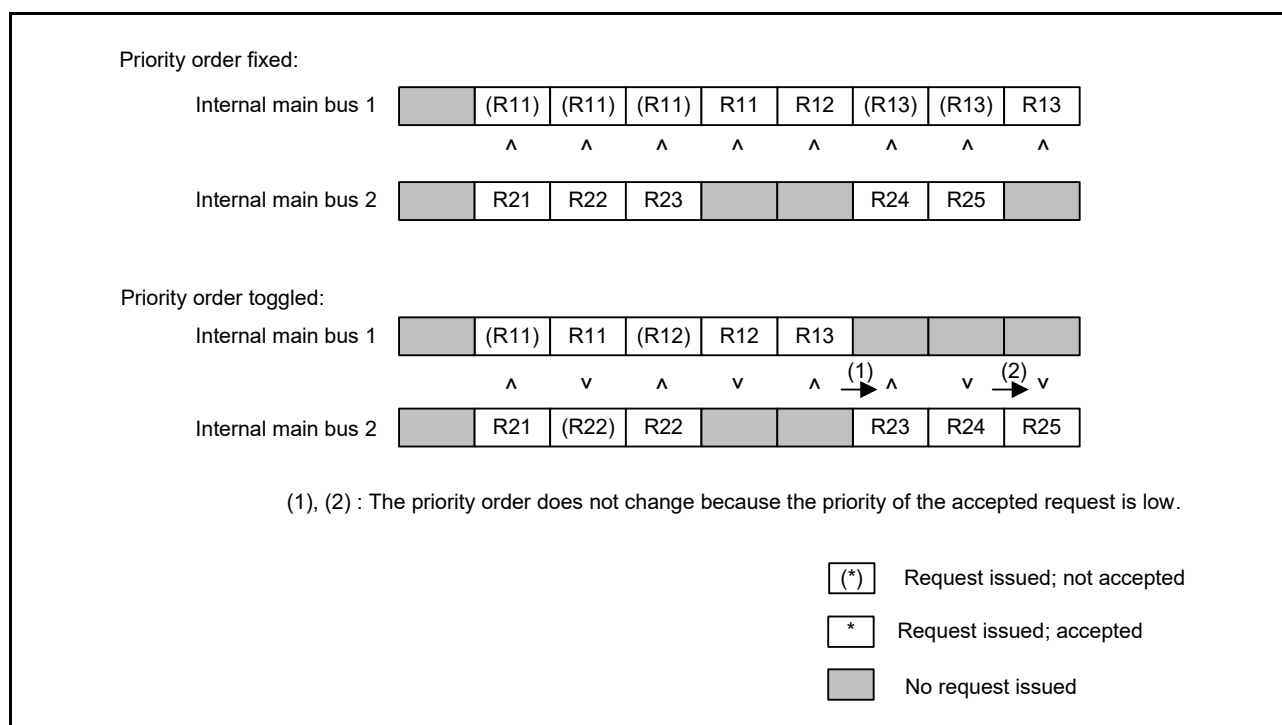


Figure 16.2 Priority Order between Internal Peripheral Bus Accesses

16.2.5 Write Buffer Function (Internal Peripheral Bus)

The internal peripheral bus has the write buffer function, which allows the next round of bus access to start, before the current write access is completed, in write access. However, if the following round of bus access is from the same bus master but to the different internal peripheral bus, it is suspended until the bus operations already in progress are completed. When access to the internal memory is scheduled after the write access to the internal peripheral bus from the CPU, the following round of bus access can be started before the current bus operation is completed and thus the order of accesses may be changed (refer to Figure 16.3).

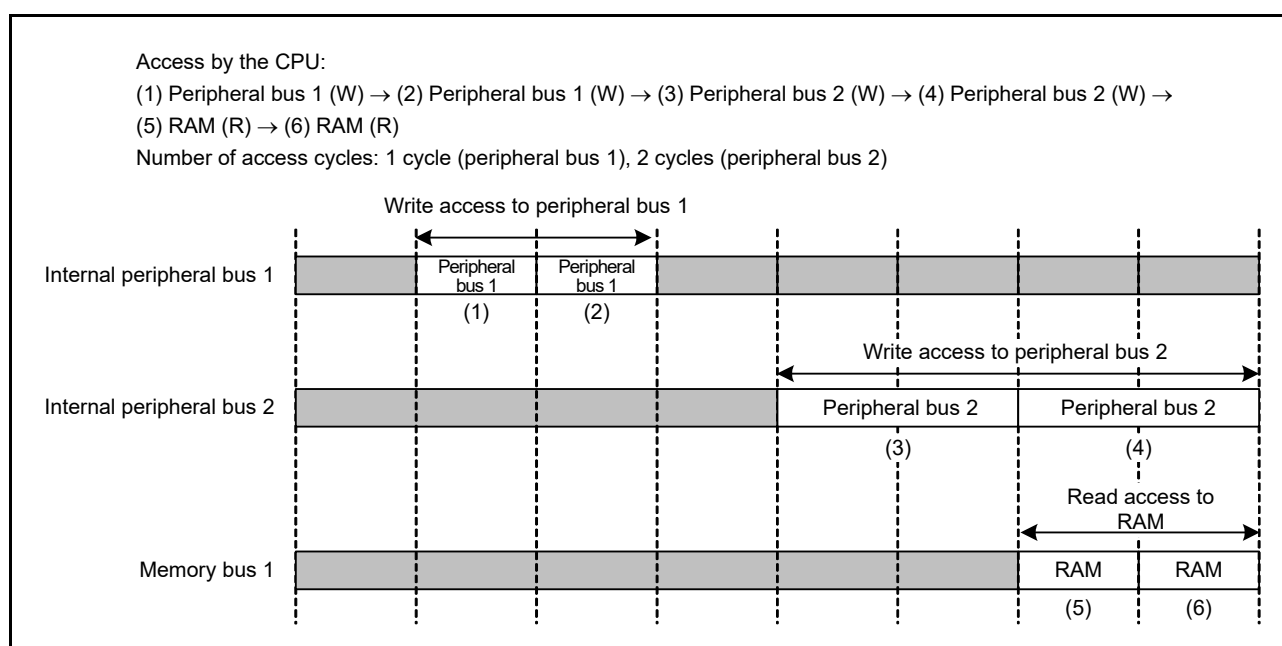


Figure 16.3 Write Buffer Function

16.2.6 External Bus

Table 16.5 lists the specifications of the external bus.

The external bus controller arbitrates requests for bus mastership on the external address space and external bus controller registers (CSC and SDRAMC) from internal main bus 1, internal main bus 2, and EXDMAC. However, the external address space is only accessible from the EXDMAC.

The priority order of these three buses can be set using the external bus priority control bits (BPEB[1:0]) in the bus priority control register (BUSPRI). When the priority order is EXDMAC, fixed, the order is internal main bus 2, and then internal main bus 1. When the priority order is toggled, a bus has a lower priority when the request of that bus is accepted, between internal main bus 1 and the other buses (internal main bus 2 and EXDMAC). However, the order of priority is EXDMAC and then internal main bus 2, regardless of the external bus priority control bits settings.

The order of accepting requests may change depending on the BUSPRI setting (refer to Figure 16.4).

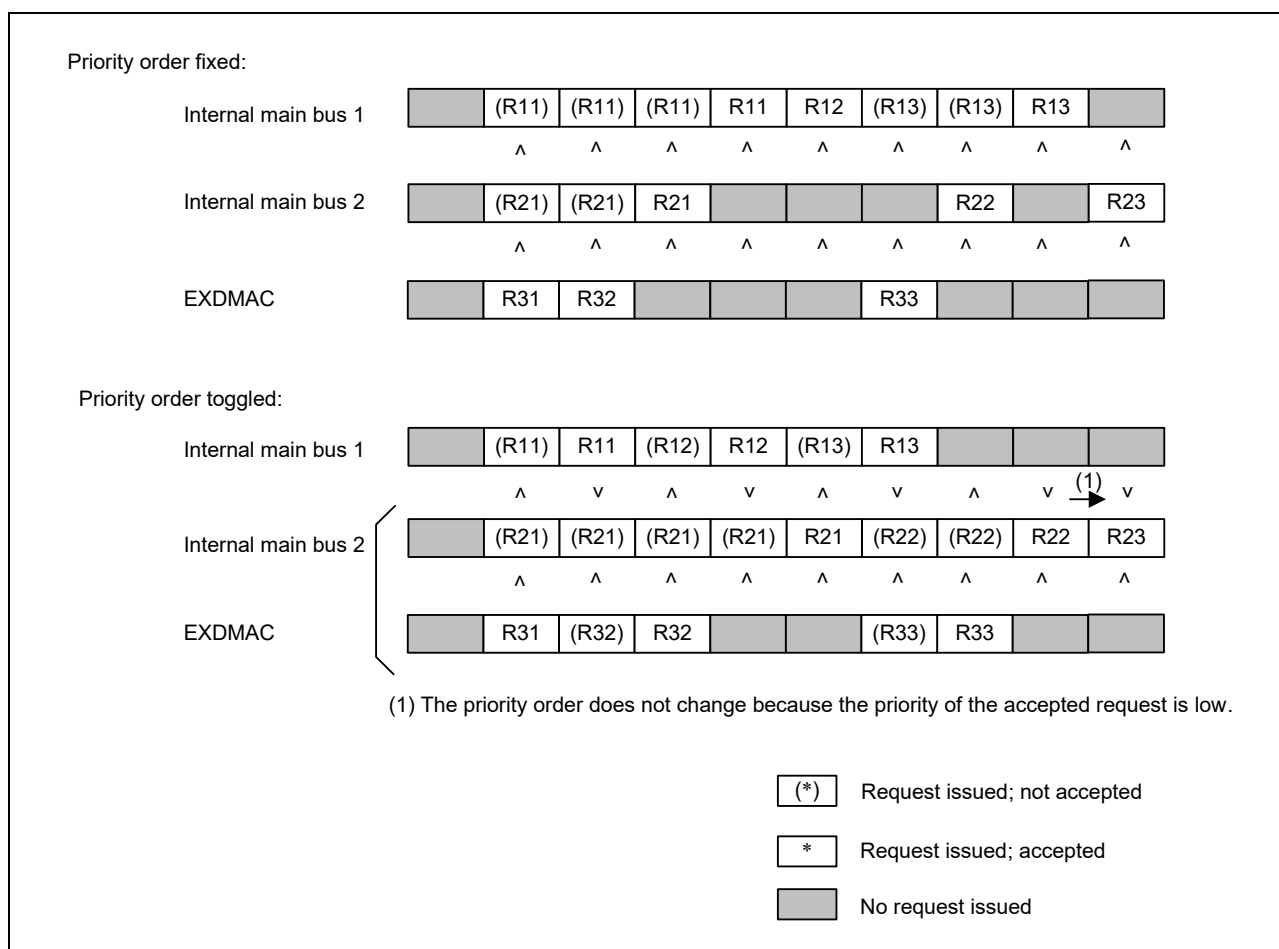


Figure 16.4 Priority Order of Internal Peripheral Bus Accesses

Table 16.5 Specifications of the External Bus

Item	Description
External address space	- An external address space is divided into eight CS areas (CS0 to CS7) and the SDRAM area (SDCS) for management. - Chip select signals can be output for each area. - Bus width can be set for each area. - Separate bus: An 8, 16, or 32-bit bus space is selectable. - Address/data multiplexed bus: An 8 or 16-bit bus space is selectable. - An endian mode can be specified for each area.
CS area controller	- Recovery cycles can be inserted. - Read recovery: Up to 15 cycles - Write recovery: Up to 15 cycles - Cycle wait function: Wait for up to 31 cycles (page access: up to 7 cycles) - Wait control can be used to set up the following. - Timing of assertion and negation for chip-select signals (CS0# to CS7#) - The timing of assertion of the read signal (RD#) and write signals (WR0#/WR#, and WR1# to WR3#) - The timing with which data output starts and ends - Write access mode: Single write strobe mode/byte strobe mode - Separate bus or address/data multiplexed bus can be set for each area.
SDRAM area controller	- Multiplexing output of row address/column address (8, 9, 10, or 11 bits) - Self-refresh and auto-Refresh selectable - CAS latency can be specified from one to three cycles
Write buffer function	When write data from the bus master has been written to the write buffer, write access by the bus master is completed.
Frequency	- The CS area controller (CSC) operates in synchronization with the external-bus clock (BCLK).^{*1} - The SDRAM area controller (SDRAMC) operates in synchronization with the SDRAM clock (SDCLK).

Note 1. The BCLK and the SDCLK should be operated with the same frequency when the SDRAM is in use.

Table 16.6 lists the input/output pins of the external bus.

Table 16.6 Pin Configuration of the External Bus (1/2)

Pin Name	I/O	Description
A23 to A0 ^{*1}	Output	Address output pins
D31 to D0	I/O	Data input/output pins D31 to D0 pins are enabled when the 32-bit bus space is specified. D15 to D0 pins are enabled when the 16-bit bus space is specified. D7 to D0 pins are enabled when the 8-bit bus space is specified.
BC0# ^{*1}	Output	A strobe signal; (the BC0# signal being at the low level) during access to an external address space in single write strobe mode indicates that D7 to D0 are valid. When an 8-bit bus space is specified, this output pin is always held low regardless of write access mode.
BC1#	Output	A strobe signal; (the BC1# signal being at the low level) during access to an external address space in single write strobe mode indicates that D15 to D8 are valid. This pin is not used when the 8-bit bus space is specified.
BC2#	Output	A strobe signal; (the BC2# signal being at the low level) during access to an external address space in single write strobe mode indicates that D23 to D16 are valid. This pin is not used when the 8- or 16-bit bus space is specified.
BC3#	Output	A strobe signal; (the BC3# signal being at the low level) during access to an external address space in single write strobe mode indicates that D31 to D24 are valid. This pin is not used when the 8- or 16-bit bus space is specified.
CS0#	Output	A chip select signal for area 0 (CS0)
CS1#	Output	A chip select signal for area 1 (CS1)
CS2#	Output	A chip select signal for area 2 (CS2)
CS3#	Output	A chip select signal for area 3 (CS3)
CS4#	Output	A chip select signal for area 4 (CS4)
CS5#	Output	A chip select signal for area 5 (CS5)
CS6#	Output	A chip select signal for area 6 (CS6)

Table 16.6 Pin Configuration of the External Bus (2/2)

Pin Name	I/O	Description
CS7#	Output	A chip select signal for area 7 (CS7)
RD#	Output	A strobe signal indicating that reading from an external address space (CS0 to CS7) is in progress
WR0#/WR#	Output	WR0# signal is a strobe signal indicates that (the WR0# signal being at the low level) writing to an external address space is in progress in byte strobe mode, and D7 to D0 are valid. WR# signal is a strobe signal that indicates writing to an external address space is in progress in single write strobe mode. When an 8-bit bus space is specified, this output pin is held low during a write access regardless of write access mode.
WR1#	Output	A strobe signal; (the WR1# signal being at the low level) during writing to an external address space in byte strobe mode indicates that D15 to D8 are valid. This signal is invalid in single write strobe mode. This pin is not used when the 8-bit bus space is specified.
WR2#	Output	A strobe signal; (the WR2# signal being at the low level) during writing to an external address space in byte strobe mode indicates that D23 to D16 are valid. This signal is invalid in single write strobe mode. This pin is not used when the 8- or 16-bit bus space is specified.
WR3#	Output	A strobe signal; (the WR3# signal being at the low level) during writing to an external address space in byte strobe mode indicates that D31 to D24 are valid. This signal is invalid in single write strobe mode. This pin is not used when the 8- or 16-bit bus space is specified.
ALE	Output	Address latch signal when address/data multiplexed bus is selected.
WAIT#	Input	A wait request signal when accessing the external address space (CS0 to CS7) (Low: Wait request)
SDCLK	Output	SDRAM clock
CKE	Output	SDRAM clock enable signal
SDCS#	Output	SDRAM chip select signal
RAS#	Output	SDRAM row address strobe signal
CAS#	Output	SDRAM column address strobe signal
WE#	Output	SDRAM write enable signal
DQM0	Output	SDRAM I/O data mask enable signal for D7 to D0
DQM1	Output	SDRAM I/O data mask enable signal for D15 to D8
DQM2	Output	SDRAM I/O data mask enable signal for D23 to D16
DQM3	Output	SDRAM I/O data mask enable signal for D31 to D24

Note 1. The A0 and BC0# pin functions share the same pin, and either becomes effective according to the area, with the function being A0 in byte strobe mode and BC0# in single write strobe mode. Note that setting the 8-bit external bus width is prohibited in single write strobe mode. For information on other multiplexed pin functions, refer to section 22, I/O Ports.

16.2.7 Parallel Operation

Parallel operation is possible when different bus-master modules are requesting access to different slave modules. For example, if the CPU is fetching an instruction from code flash memory and an operand from RAM, the DMAC is able to handle transfer between a peripheral bus and the external bus at the same time.

An example of parallel operations is shown in Figure 16.5. In this example, the CPU is able to employ the instruction and operand buses for simultaneous access to code flash memory and RAM, respectively. Furthermore, the DMAC simultaneously employs internal main bus 2 for access to a peripheral bus or the external bus during access to RAM and code flash memory by the CPU.

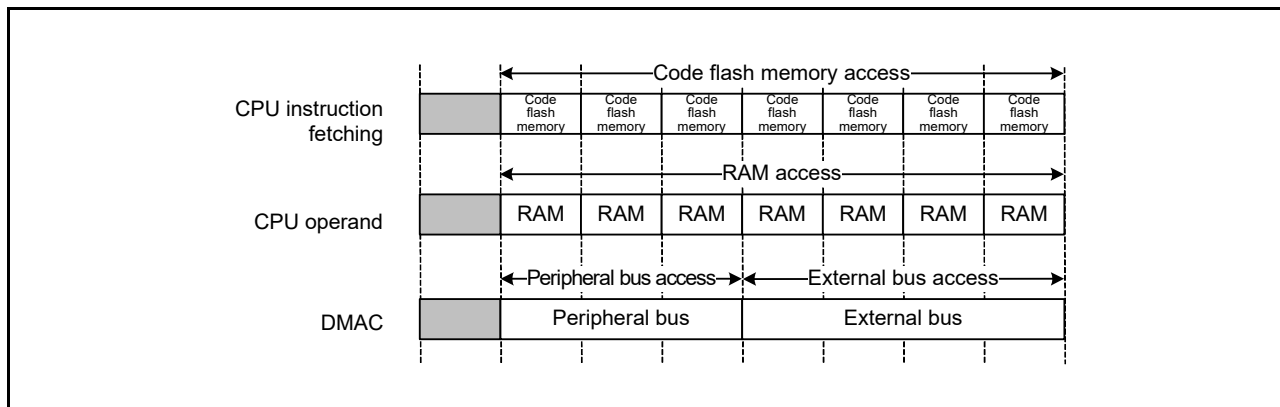


Figure 16.5 Example of Parallel Operations

16.2.8 Bus Settings

- (1) Set the mode of the external bus in the CSn mode register (CSnMOD), CSn wait-control register 1 (CSnWCR1), CSn wait-control register 2 (CSnWCR2), CSn control register (CSnCR), CSn recovery-cycle setting register (CSnREC), CS recovery cycle insertion enable register (CSRECEN), bus error monitoring enable register (BEREN), and bus priority control register (BUSPRI).
- (2) Make settings for pins in the CS output enable register (PFCSE), CS output pin select register 0 (PFCSS0), CS output pin select register 1 (PFCSS1), address output enable register 0 (PFAOE0), address output enable register 1 (PFAOE1), external-bus control register 0 (PFBCR0), and external-bus control register 1 (PFBCR1).
- (3) Set up pins to be used as input port pins.
- (4) Set the external-bus enable bit (EXBE) in the system control register 0 (SYSCR0) to 1 (enabling the external bus).

16.2.9 Restrictions

(1) Prohibition of Access that Spans Areas of Address Space

Single access that spans two areas of the address space is prohibited, and operation of such an access is not guaranteed. Setting must be made so that two areas are not accessed at the same time by a single word or longword access.

Single access that spans two areas of the address space is also prohibited in the EXDMAC block transfer in single address mode or cluster transfer, and operation of such an access is not guaranteed. Setting must be made so that two areas are not accessed at the same time by a single access in the EXDMAC block transfer in single address mode or cluster transfer.

(2) Restrictions in Relation to RMPA and String-Manipulation Instructions

- (a) Although the external space has a per-area ending-switching facility (only for data), the allocation of data to be handled by RMPA or string-manipulation instructions to an area where the endian differs from that of the chip is prohibited, and operation is not guaranteed if this restriction is not observed. If data to be handled by RMPA or string-manipulation instructions are allocated to the external space, they must be allocated to areas where the endian setting is the same as that for the chip.
- (b) The allocation of data to be handled by RMPA or string-manipulation instructions to I/O registers is prohibited, and operation is not guaranteed if this restriction is not observed.

(3) Restriction on Endian

When the endian setting for each area is different from that for the chip, no instruction code can be allocated in the area.

The instruction code should be allocated to the external space whose endian setting is the same as that for the chip.

For the CPU to handle 64-bit operand access, the area for access must have the same endian setting as that of the chip.

64-bit operand access is not possible in the case of areas with a different endian setting from that of the chip.

16.3 Register Descriptions

16.3.1 CSn Control Register (CSnCR) (n = 0 to 7)

Address(es): CS0CR 0008 3802h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	MPXEN	—	—	—	EMOD E	—	—	BSIZE[1:0]	—	—	—	—	EXENB
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1

Value after reset:

Address(es): CS1CR 0008 3812h, CS2CR 0008 3822h, CS3CR 0008 3832h,
CS4CR 0008 3842h, CS5CR 0008 3852h, CS6CR 0008 3862h, CS7CR 0008 3872h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	MPXEN	—	—	—	EMOD E	—	—	BSIZE[1:0]	—	—	—	—	EXENB
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	EXENB	Operation Enable	0: Operation is disabled 1: Operation is enabled	R/W
b3 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b5, b4	BSIZE[1:0]	External Bus Width Select	b5 b4 0 0: A 16-bit bus space is selected 0 1: A 32-bit bus space is selected 1 0: An 8-bit bus space is selected 1 1: Setting prohibited	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	EMODE	Endian Mode	0: Endian of area n is the same as the endian of operating mode. 1: Endian of area n is not the endian of operating mode. (n = 0 to 7)	R/W
b11 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	MPXEN	Address/Data Multiplexed I/O Interface Select	0: Separate bus interface is selected for area n. 1: Address/data multiplexed I/O interface is selected for area n. (n = 0 to 7)	R/W
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Do not write to the CSnCR register while access to the CSn area is in progress.

EXENB Bit (Operation Enable)

This bit enables or disables operation of the respective CS areas.

After this MCU is reset, operation is enabled (EXENB = 1) only for area 0; operation in other areas is disabled (EXENB = 0).

An attempt at access to an area for which operation has been disabled does not lead to access via the external bus.

However, if the illegal address access detection enable bit in the bus error monitoring enable register has been set to enable detection (BEREN.IGAEN = 1), such an attempt will lead to an illegal-access error.

BSIZE[1:0] Bits (External Bus Width Select)

These bits specify the data bus width of each area.

The data bus width of area 0 (CS0) after a reset depends on the setting of the bus width in operating mode.

When the address/data multiplexed I/O interface is selected with the MPXEN bit, setting other than 8- and 16-bit bus spaces is prohibited. If set, the operation cannot be guaranteed.

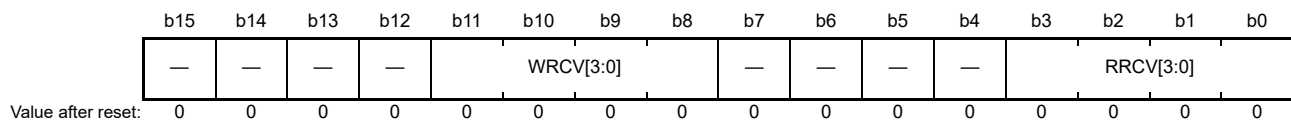
EMODE Bit (Endian Mode)

This bit specifies the endian of each area.

When the endian setting for each area is different from that for the chip, no instruction code can be allocated in the area. The instruction code should be allocated to the external space whose endian setting is the same as that for the chip.

16.3.2 CSn Recovery Cycle Register (CSnREC) (n = 0 to 7)

Address(es): CS0REC 0008 380Ah, CS1REC 0008 381Ah, CS2REC 0008 382Ah, CS3REC 0008 383Ah, CS4REC 0008 384Ah, CS5REC 0008 385Ah, CS6REC 0008 386Ah, CS7REC 0008 387Ah



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	RRCV[3:0]	Read Recovery	b3 b0 0 0 0 0: No recovery cycle is inserted. 0 0 0 1: 1 recovery cycle is inserted. 0 0 1 0: 2 recovery cycles are inserted. 0 0 1 1: 3 recovery cycles are inserted. 0 1 0 0: 4 recovery cycles are inserted. 0 1 0 1: 5 recovery cycles are inserted. 0 1 1 0: 6 recovery cycles are inserted. 0 1 1 1: 7 recovery cycles are inserted. 1 0 0 0: 8 recovery cycles are inserted. 1 0 0 1: 9 recovery cycles are inserted. 1 0 1 0: 10 recovery cycles are inserted. 1 0 1 1: 11 recovery cycles are inserted. 1 1 0 0: 12 recovery cycles are inserted. 1 1 0 1: 13 recovery cycles are inserted. 1 1 1 0: 14 recovery cycles are inserted. 1 1 1 1: 15 recovery cycles are inserted.	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b11 to b8	WRCV[3:0]	Write Recovery	b11 b8 0 0 0 0: No recovery cycle is inserted. 0 0 0 1: 1 recovery cycle is inserted. 0 0 1 0: 2 recovery cycles are inserted. 0 0 1 1: 3 recovery cycles are inserted. 0 1 0 0: 4 recovery cycles are inserted. 0 1 0 1: 5 recovery cycles are inserted. 0 1 1 0: 6 recovery cycles are inserted. 0 1 1 1: 7 recovery cycles are inserted. 1 0 0 0: 8 recovery cycles are inserted. 1 0 0 1: 9 recovery cycles are inserted. 1 0 1 0: 10 recovery cycles are inserted. 1 0 1 1: 11 recovery cycles are inserted. 1 1 0 0: 12 recovery cycles are inserted. 1 1 0 1: 13 recovery cycles are inserted. 1 1 1 0: 14 recovery cycles are inserted. 1 1 1 1: 15 recovery cycles are inserted.	R/W
b15 to b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Do not write to the CSnREC register while access to the CSn area is in progress.

When the preceding bus access is a separate bus access, CSnREC is valid when the recovery cycle insertion is enabled with the separate bus recovery cycle insertion enable bit (RCVENj (j = 0 to 7)) in CSRECEN. When the preceding bus access is an address/data multiplexed bus access, CSnREC is valid when the recovery cycle insertion is enabled with the multiplexed bus recovery cycle insertion enable bit (RCVENMj) in CSRECEN.

RRCV[3:0] Bits (Read Recovery)

These bits specify the number of recovery cycles to be inserted after a read access to the external bus.

When the recovery cycle insertion is enabled and a value except 0000b is written to these bits, one to 15 recovery cycles are inserted in the following cases.

- After a read access to the external bus, a read access is made to the external bus in the same area.
- After a read access to the external bus, a read access is made to the external bus in a different area.
- After a read access to the external bus, a write access is made to the external bus in the same area.
- After a read access to the external bus, a write access is made to the external bus in a different area.

WRCV[3:0] Bits (Write Recovery)

These bits specify the number of recovery cycles to be inserted after a write access to the external bus.

When the recovery cycle insertion is enabled and a value except 0000b is written to these bits, one to 15 recovery cycles are inserted in the following cases.

- After a write access to the external bus, a read access is made to the external bus in the same area.
- After a write access to the external bus, a read access is made to the external bus in a different area.
- After a write access to the external bus, a write access is made to the external bus in the same area.
- After a write access to the external bus, a write access is made to the external bus in a different area.

16.3.3 CS Recovery Cycle Insertion Enable Register (CSRECEN)

Address(es): 0008 3880h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	RCVEN M7	RCVEN M6	RCVEN M5	RCVEN M4	RCVEN M3	RCVEN M2	RCVEN M1	RCVEN M0	RCVEN 7	RCVEN 6	RCVEN 5	RCVEN 4	RCVEN 3	RCVEN 2	RCVEN 1	RCVEN 0
Value after reset:	0	0	1	1	1	1	1	0	0	0	1	1	1	1	1	0

Bit	Symbol	Bit Name	Description	R/W
b0	RCVEN0	Separate Bus Recovery Cycle Insertion Enable 0	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b1	RCVEN1	Separate Bus Recovery Cycle Insertion Enable 1	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b2	RCVEN2	Separate Bus Recovery Cycle Insertion Enable 2	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b3	RCVEN3	Separate Bus Recovery Cycle Insertion Enable 3	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b4	RCVEN4	Separate Bus Recovery Cycle Insertion Enable 4	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b5	RCVEN5	Separate Bus Recovery Cycle Insertion Enable 5	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b6	RCVEN6	Separate Bus Recovery Cycle Insertion Enable 6	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b7	RCVEN7	Separate Bus Recovery Cycle Insertion Enable 7	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b8	RCVENM0	Multiplexed Bus Recovery Cycle Insertion Enable 0	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b9	RCVENM1	Multiplexed Bus Recovery Cycle Insertion Enable 1	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b10	RCVENM2	Multiplexed Bus Recovery Cycle Insertion Enable 2	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b11	RCVENM3	Multiplexed Bus Recovery Cycle Insertion Enable 3	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b12	RCVENM4	Multiplexed Bus Recovery Cycle Insertion Enable 4	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b13	RCVENM5	Multiplexed Bus Recovery Cycle Insertion Enable 5	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b14	RCVENM6	Multiplexed Bus Recovery Cycle Insertion Enable 6	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W
b15	RCVENM7	Multiplexed Bus Recovery Cycle Insertion Enable 7	0: Recovery cycle insertion is disabled. 1: Recovery cycle insertion is enabled.	R/W

Do not write to the CSRECEN register while access to the CSn area is in progress.

RCVENn Bit (Separate Bus Recovery Cycle Insertion Enable n) (n = 0 to 7)

This bit enables or disables the insertion of read recovery cycles when, after a read access to the external bus, a read access is made to the external bus in the same area.

RCVENMn Bit (Multiplexed Bus Recovery Cycle Insertion Enable n) (n = 0 to 7)

This bit enables or disables the insertion of read recovery cycles when, after a read access to the external bus, a read access is made to the external bus in the same area.

Table 16.7 Insertion of Recovery Cycles

Access Type	External Address Space	Insertion of Recovery Cycles	Corresponding Bits (Separate/Multiplexed)
Read access after read access	Same area	Recovery cycles specified by the RRCV[3:0] bits are inserted.	RCVEN0/RCVENM0
	Different area	Recovery cycles specified by the RRCV[3:0] bits are inserted.	RCVEN1/RCVENM1
Read access after write access	Same area	Recovery cycles specified by the RRCV[3:0] bits are inserted.	RCVEN2/RCVENM2
	Different area	Recovery cycles specified by the RRCV[3:0] bits are inserted.	RCVEN3/RCVENM3
Write access after read access	Same area	Recovery cycles specified by the WRCV[3:0] bits are inserted.	RCVEN4/RCVENM4
	Different area	Recovery cycles specified by the WRCV[3:0] bits are inserted.	RCVEN5/RCVENM5
Write access after write access	Same area	Recovery cycles specified by the WRCV[3:0] bits are inserted.	RCVEN6/RCVENM6
	Different area	Recovery cycles specified by the WRCV[3:0] bits are inserted.	RCVEN7/RCVENM7

16.3.4 CSn Mode Register (CSnMOD) (n = 0 to 7)

Address(es): CS0MOD 0008 3002h, CS1MOD 0008 3012h, CS2MOD 0008 3022h, CS3MOD 0008 3032h,
CS4MOD 0008 3042h, CS5MOD 0008 3052h, CS6MOD 0008 3062h, CS7MOD 0008 3072h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	PRMOD	—	—	—	—	—	PWENB	PRENB	—	—	—	—	EWENB	—	—	WRMOD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	WRMOD	Write Access Mode Select	0: Byte strobe mode 1: Single write strobe mode	R/W
b2, b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b3	EWENB	External Wait Enable	0: External wait is disabled 1: External wait is enabled	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	PRENB	Page Read Access Enable	0: Page read access is disabled 1: Page read access is enabled	R/W
b9	PWENB	Page Write Access Enable	0: Page write access is disabled 1: Page write access is enabled	R/W
b14 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15	PRMOD	Page Read Access Mode Select	0: Normal access compatible mode 1: External data read continuous assertion mode	R/W

Do not write to the CSnMOD register while access to the CSn area is in progress.

WRMOD Bit (Write Access Mode Select)

This bit selects a write access operating mode.

Writing 0 to this bit selects byte strobe mode where data write operation is controlled by the WRn# (n = 0 to 3) signal corresponding to the respective byte positions.

Writing 1 to this bit selects single write strobe mode where data write operation is controlled by the BCn# (n = 0 to 3) signal and the WR# signal corresponding to respective byte positions. Note that setting the external bus width of 8 bits is prohibited in single write strobe mode.

Table 16.8 Control Signals for Write Access Mode

Mode	Pin Name							
Write Access Mode	WR3#	WR2#	WR1#	WR0#/ WR#	BC3#	BC2#	BC1#	BC0#
Byte strobe mode	✓	✓	✓	✓ (WR0#)	×	×	×	×
Single write strobe mode	×	×	×	✓ (WR#)	✓	✓	✓	✓

✓: Enabled, ×: Disabled

EWENB Bit (External Wait Enable)

This bit enables or disables external wait.

Writing 1 to this bit selects external wait and allows control of the number of waits in each cycle with the WAIT# signal. In this state, wait cycles are inserted while the WAIT# signal is at the low level.

Writing 0 to this bit disables the WAIT# signal.

PRENB Bit (Page Read Access Enable)

This bit enables or disables page read accesses.

Note: When the address/data multiplexed I/O interface is selected with the MPXEN bit in CSnCR, this bit should not be set to enable page read accesses. Page read accesses are not supported in the address/data multiplexed I/O interface.

PWENB Bit (Page Write Access Enable)

This bit enables or disables page write accesses.

Note: When the address/data multiplexed I/O interface is selected with the MPXEN bit in CSnCR, this bit should not be set to enable page write accesses. Page write accesses are not supported in the address/data multiplexed I/O interface.

PRMOD Bit (Page Read Access Mode Select)

This bit selects a page read access operating mode.

Writing 0 to this bit selects normal access compatible mode where the RD# signal is negated and RD assert wait is inserted each time a piece of data is read. However, when there is no RD assert wait, the RD# signal is negated only in the final transfer of the external bus access.

Writing 1 to this bit selects external data read continuous assertion mode where RD assert wait is inserted and the RD# signal is continuously asserted during this time period.

16.3.5 CSn Wait Control Register 1 (CSnWCR1) (n = 0 to 7)

Address(es): CS0WCR1 0008 3004h, CS1WCR1 0008 3014h, CS2WCR1 0008 3024h, CS3WCR1 0008 3034h,
CS4WCR1 0008 3044h, CS5WCR1 0008 3054h, CS6WCR1 0008 3064h, CS7WCR1 0008 3074h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
—	—	—	CSRWAIT[4:0]				—	—	—	CSWWAIT[4:0]					
0	0	0	0	0	1	1	1	0	0	0	0	0	1	1	1
Value after reset:															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	CSPRWAIT[2:0]			—	—	—	—	—	CSPWWAIT[2:0]		
0	0	0	0	0	1	1	1	0	0	0	0	0	1	1	1
Value after reset:															

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	CSPWWAIT[2:0]	Page Write Cycle Wait Select*1	b2 b0 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 1 1: Wait with a length of 3 clock cycles are inserted. 1 0 0: Wait with a length of 4 clock cycles are inserted. 1 0 1: Wait with a length of 5 clock cycles are inserted. 1 1 0: Wait with a length of 6 clock cycles are inserted. 1 1 1: Wait with a length of 7 clock cycles are inserted.	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b10 to b8	CSPRWAIT[2:0]	Page Read Cycle Wait Select*2	b10 b8 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 1 1: Wait with a length of 3 clock cycles are inserted. 1 0 0: Wait with a length of 4 clock cycles are inserted. 1 0 1: Wait with a length of 5 clock cycles are inserted. 1 1 0: Wait with a length of 6 clock cycles are inserted. 1 1 1: Wait with a length of 7 clock cycles are inserted.	R/W
b15 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Bit Name	Description	R/W
b20 to b16	CSWWAIT[4:0]	Normal Write Cycle Wait Select	b20 b16 0 0 0 0 0: No wait is inserted. 0 0 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 0 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 0 0 1 1: Wait with a length of 3 clock cycles are inserted. 0 0 1 0 0: Wait with a length of 4 clock cycles are inserted. 0 0 1 0 1: Wait with a length of 5 clock cycles are inserted. 0 0 1 1 0: Wait with a length of 6 clock cycles are inserted. 0 0 1 1 1: Wait with a length of 7 clock cycles are inserted. 0 1 0 0 0: Wait with a length of 8 clock cycles are inserted. 0 1 0 0 1: Wait with a length of 9 clock cycles are inserted. 0 1 0 1 0: Wait with a length of 10 clock cycles are inserted. 0 1 0 1 1: Wait with a length of 11 clock cycles are inserted. 0 1 1 0 0: Wait with a length of 12 clock cycles are inserted. 0 1 1 0 1: Wait with a length of 13 clock cycles are inserted. 0 1 1 1 0: Wait with a length of 14 clock cycles are inserted. 0 1 1 1 1: Wait with a length of 15 clock cycles are inserted. 1 0 0 0 0: Wait with a length of 16 clock cycles are inserted. 1 0 0 0 1: Wait with a length of 17 clock cycles are inserted. 1 0 0 1 0: Wait with a length of 18 clock cycles are inserted. 1 0 0 1 1: Wait with a length of 19 clock cycles are inserted. 1 0 1 0 0: Wait with a length of 20 clock cycles are inserted. 1 0 1 0 1: Wait with a length of 21 clock cycles are inserted. 1 0 1 1 0: Wait with a length of 22 clock cycles are inserted. 1 0 1 1 1: Wait with a length of 23 clock cycles are inserted. 1 1 0 0 0: Wait with a length of 24 clock cycles are inserted. 1 1 0 0 1: Wait with a length of 25 clock cycles are inserted. 1 1 0 1 0: Wait with a length of 26 clock cycles are inserted. 1 1 0 1 1: Wait with a length of 27 clock cycles are inserted. 1 1 1 0 0: Wait with a length of 28 clock cycles are inserted. 1 1 1 0 1: Wait with a length of 29 clock cycles are inserted. 1 1 1 1 0: Wait with a length of 30 clock cycles are inserted. 1 1 1 1 1: Wait with a length of 31 clock cycles are inserted.	R/W
b23 to b21	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b28 to b24	CSRWAIT[4:0]	Normal Read Cycle Wait Select	b28 b24 0 0 0 0 0: No wait is inserted. 0 0 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 0 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 0 0 1 1: Wait with a length of 3 clock cycles are inserted. 0 0 1 0 0: Wait with a length of 4 clock cycles are inserted. 0 0 1 0 1: Wait with a length of 5 clock cycles are inserted. 0 0 1 1 0: Wait with a length of 6 clock cycles are inserted. 0 0 1 1 1: Wait with a length of 7 clock cycles are inserted. 0 1 0 0 0: Wait with a length of 8 clock cycles are inserted. 0 1 0 0 1: Wait with a length of 9 clock cycles are inserted. 0 1 0 1 0: Wait with a length of 10 clock cycles are inserted. 0 1 0 1 1: Wait with a length of 11 clock cycles are inserted. 0 1 1 0 0: Wait with a length of 12 clock cycles are inserted. 0 1 1 0 1: Wait with a length of 13 clock cycles are inserted. 0 1 1 1 0: Wait with a length of 14 clock cycles are inserted. 0 1 1 1 1: Wait with a length of 15 clock cycles are inserted. 1 0 0 0 0: Wait with a length of 16 clock cycles are inserted. 1 0 0 0 1: Wait with a length of 17 clock cycles are inserted. 1 0 0 1 0: Wait with a length of 18 clock cycles are inserted. 1 0 0 1 1: Wait with a length of 19 clock cycles are inserted. 1 0 1 0 0: Wait with a length of 20 clock cycles are inserted. 1 0 1 0 1: Wait with a length of 21 clock cycles are inserted. 1 0 1 1 0: Wait with a length of 22 clock cycles are inserted. 1 0 1 1 1: Wait with a length of 23 clock cycles are inserted. 1 1 0 0 0: Wait with a length of 24 clock cycles are inserted. 1 1 0 0 1: Wait with a length of 25 clock cycles are inserted. 1 1 0 1 0: Wait with a length of 26 clock cycles are inserted. 1 1 0 1 1: Wait with a length of 27 clock cycles are inserted. 1 1 1 0 0: Wait with a length of 28 clock cycles are inserted. 1 1 1 0 1: Wait with a length of 29 clock cycles are inserted. 1 1 1 1 0: Wait with a length of 30 clock cycles are inserted. 1 1 1 1 1: Wait with a length of 31 clock cycles are inserted.	R/W
b31 to b29	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. The CSPWWAIT[2:0] value is valid only when the PWENB bit in CSnMOD is set to 1.

Note 2. The CSPRWAIT[2:0] value is valid only when the PRENB bit in CSnMOD is set to 1.

Do not write to the CSnWCR1 register while access to the CSn area is in progress.

Set each of these bits within a range of the restrictions described in section 16.5.7 (1) Limitations on Using Separate Bus Interface or section 16.5.7 (2) Limitations on Using Address/Data Multiplexed Bus Interface, according to the bus interface used. In addition, during the EXDMAC transfer in single address mode, set each of these bits within a range of the restrictions described in section 16.5.7 (5) Limitations on EXDMAC Single Address Transfer Mode.

CSPWWAIT[2:0] Bits (Page Write Cycle Wait Select)

These bits specify the number of wait cycles to be inserted into the second and subsequent accesses during a page write cycle.

This setting is enabled when the PWENB bit in CSnMOD is set to 1.

Note: Be sure to satisfy $1 \leq \text{CSnWCR2.WDON}[2:0] \text{ value} \leq \text{CSnWCR2.WRON}[2:0] \text{ value} \leq \text{CSnWCR1.CSPWWAIT}[2:0] \text{ value}$ and $\text{CSnWCR2.CSON}[2:0] \text{ value} \leq \text{CSnWCR2.WRON}[2:0] \text{ value} \leq \text{CSnWCR1.CSPWWAIT}[2:0] \text{ value}$.

CSPRWAIT[2:0] Bits (Page Read Cycle Wait Select)

These bits specify the number of wait cycles to be inserted into the second and subsequent accesses during a page read cycle.

This setting is enabled when the PRENB bit in CSnMOD is set to 1.

Note: Be sure to satisfy $\text{CSnWCR2.CSON}[2:0] \text{ value} \leq \text{CSnWCR2.RDON}[2:0] \text{ value} \leq \text{CSnWCR1.CSPRWAIT}[2:0] \text{ value}$.

CSWWAIT[4:0] Bits (Normal Write Cycle Wait Select)

These bits specify the number of wait cycles to be inserted into the first access during a normal write cycle or page write cycle.

Note: Be sure to satisfy $1 \leq \text{CSnWCR2.WDON}[2:0] \text{ value} \leq \text{CSnWCR2.WRON}[2:0] \text{ value} \leq \text{CSnWCR1.CSWWAIT}[4:0] \text{ value}$ and $\text{CSnWCR2.CSON}[2:0] \text{ value} \leq \text{CSnWCR2.WRON}[2:0] \text{ value} \leq \text{CSnWCR1.CSWWAIT}[4:0] \text{ value}$.

CSRWAIT[4:0] Bits (Normal Read Cycle Wait Select)

These bits specify the number of wait cycles to be inserted into the first access during a normal read cycle or page read cycle.

Note: Be sure to satisfy $\text{CSnWCR2.CSON}[2:0] \text{ value} \leq \text{CSnWCR2.RDON}[2:0] \text{ value} \leq \text{CSnWCR1.CSRWAIT}[4:0] \text{ value}$.

16.3.6 CSn Wait Control Register 2 (CSnWCR2) (n = 0 to 7)

Address(es): CS0WCR2 0008 3008h, CS1WCR2 0008 3018h, CS2WCR2 0008 3028h, CS3WCR2 0008 3038h,
CS4WCR2 0008 3048h, CS5WCR2 0008 3058h, CS6WCR2 0008 3068h, CS7WCR2 0008 3078h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	CSON[2:0]			—	WDON[2:0]			—	WRON[2:0]			—	RDON[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	AWAIT[1:0]		—	WDOFF[2:0]		—	CSWOFF[2:0]			—	CSROFF[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	CSROFF[2:0]	Read-Access CS Extension Cycle Select	b2 b0 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 1 1: Wait with a length of 3 clock cycles are inserted. 1 0 0: Wait with a length of 4 clock cycles are inserted. 1 0 1: Wait with a length of 5 clock cycles are inserted. 1 1 0: Wait with a length of 6 clock cycles are inserted. 1 1 1: Wait with a length of 7 clock cycles are inserted.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6 to b4	CSWOFF[2:0]	Write-Access CS Extension Cycle Select	b6 b4 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 1 1: Wait with a length of 3 clock cycles are inserted. 1 0 0: Wait with a length of 4 clock cycles are inserted. 1 0 1: Wait with a length of 5 clock cycles are inserted. 1 1 0: Wait with a length of 6 clock cycles are inserted. 1 1 1: Wait with a length of 7 clock cycles are inserted.	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b10 to b8	WDOFF[2:0]	Write Data Output Extension Cycle Select	b10 b8 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 1 1: Wait with a length of 3 clock cycles are inserted. 1 0 0: Wait with a length of 4 clock cycles are inserted. 1 0 1: Wait with a length of 5 clock cycles are inserted. 1 1 0: Wait with a length of 6 clock cycles are inserted. 1 1 1: Wait with a length of 7 clock cycles are inserted.	R/W
b11	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b13, b12	AWAIT[1:0]	Address Cycle Wait Select	b13 b12 0 0: No wait is inserted. 0 1: Wait with a length of 1 clock cycle is inserted. 1 0: Wait with a length of 2 clock cycles are inserted. 1 1: Wait with a length of 3 clock cycles are inserted.	R/W
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b18 to b16	RDON[2:0]	RD Assert Wait Select	b18 b16 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 1 1: Wait with a length of 3 clock cycles are inserted. 1 0 0: Wait with a length of 4 clock cycles are inserted. 1 0 1: Wait with a length of 5 clock cycles are inserted. 1 1 0: Wait with a length of 6 clock cycles are inserted. 1 1 1: Wait with a length of 7 clock cycles are inserted.	R/W
b19	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Bit Name	Description	R/W
b22 to b20	WRON[2:0]	WR Assert Wait Select	b22 b20 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 1 1: Wait with a length of 3 clock cycles are inserted. 1 0 0: Wait with a length of 4 clock cycles are inserted. 1 0 1: Wait with a length of 5 clock cycles are inserted. 1 1 0: Wait with a length of 6 clock cycles are inserted. 1 1 1: Wait with a length of 7 clock cycles are inserted.	R/W
b23	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b26 to b24	WDON[2:0]	Write Data Output Wait Select	b26 b24 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 1 1: Wait with a length of 3 clock cycles are inserted. 1 0 0: Wait with a length of 4 clock cycles are inserted. 1 0 1: Wait with a length of 5 clock cycles are inserted. 1 1 0: Wait with a length of 6 clock cycles are inserted. 1 1 1: Wait with a length of 7 clock cycles are inserted.	R/W
b27	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b30 to b28	CSON[2:0]	CS Assert Wait Select	b30 b28 0 0 0: No wait is inserted. 0 0 1: Wait with a length of 1 clock cycle is inserted. 0 1 0: Wait with a length of 2 clock cycles are inserted. 0 1 1: Wait with a length of 3 clock cycles are inserted. 1 0 0: Wait with a length of 4 clock cycles are inserted. 1 0 1: Wait with a length of 5 clock cycles are inserted. 1 1 0: Wait with a length of 6 clock cycles are inserted. 1 1 1: Wait with a length of 7 clock cycles are inserted.	R/W
b31	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Do not write to the CSnWCR2 register while access to the CSn area is in progress.

Set each of these bits within a range of the restrictions described in section 16.5.7 (1) Limitations on Using Separate Bus Interface or section 16.5.7 (2) Limitations on Using Address/Data Multiplexed Bus Interface, according to the bus interface used. In addition, during the EXDMAC transfer in single address mode, set each of these bits within a range of the restrictions described in section 16.5.7 (5) Limitations on EXDMAC Single Address Transfer Mode.

CSROFF[2:0] Bits (Read-Access CS Extension Cycle Select)

These bits specify the number of wait cycles to be inserted in a time period from the end of a wait cycle (RD# signal negated) until the CSn# signal (n = 0 to 7) is negated in read access mode.

CSWOFF[2:0] Bits (Write-Access CS Extension Cycle Select)

These bits specify the number of wait cycles to be inserted in a time period from the end of a wait cycle (WRn# signal (n = 0 to 3) negated) until the CSn# signal (n = 0 to 7) is negated in write access mode.

Note: Be sure to satisfy CSnWCR2.WDOFF[2:0] value ≤ CSnWCR2.CSWOFF[2:0] value.

WDOFF[2:0] Bits (Write Data Output Extension Cycle Select)

These bits specify the number of wait cycles to be inserted in a time period from the end of a wait cycle (WRn# signal (n = 0 to 3) negated) until the write data output is completed in write access mode.

When the EXDMAC is in single-address transfer mode, although the output of write-data from the chip does not proceed, the value for cycles of delay until output of the write data in divided-up page access over the bus becomes effective.

Note: Be sure to satisfy CSnWCR2.WDOFF[2:0] value ≤ CSnWCR2.CSWOFF[2:0] value.

AWAIT[1:0] Bits (Address Cycle Wait Select)

These bits specify the number of wait cycles to be inserted into an address output cycle with the address/data multiplexed I/O interface.

Note: CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.AWAIT[1:0] value
 For read access, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSRWAIT[4:0] value.
 For write access, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value and CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.

RDON[2:0] Bits (RD Assert Wait Select)

These bits specify the number of wait cycles to be inserted before the RD# signal is asserted.

Note: For normal read access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSRWAIT[4:0] value.
 For page read access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSPRWAIT[2:0] value.
 Note: For read access by the EXDMAC in single-address transfer mode, as well as satisfying the above conditions, set the CSnWCR2.RDON[2:0] bits to one or a greater value.
 Note: When the address/data multiplexed I/O interface is selected, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSRWAIT[4:0] value.

WRON[2:0] Bits (WR Assert Wait Select)

These bits specify the number of wait cycles to be inserted before the WRn# signal (n = 0 to 3) is asserted.

Note: For normal write access, satisfy 1 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value and CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.
 For page write access, satisfy 1 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value and CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value.
 Note: For write access by the EXDMAC in single-address transfer mode, as well as satisfying the above conditions, set the CSnWCR2.WRON[2:0] bits to one or a greater value.
 Note: When the address/data multiplexed I/O interface is selected, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.

WDON[2:0] Bits (Write Data Output Wait Select)

These bits specify the number of wait cycles to be inserted before the write data is output.

Note: For normal write access, satisfy 1 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.
 For page write access, satisfy 1 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value.
 Note: When the address/data multiplexed I/O interface is selected, satisfy CSnWCR2.AWAIT[1:0] value + 2 $\leq$ CSnWCR2.WDON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.

CSON[2:0] Bits (CS Assert Wait Select)

These bits specify the number of wait cycles to be inserted before the CSn# signal (n = 0 to 7) is asserted.

Note: For normal read access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSRWAIT[4:0] value.
 For page read access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.RDON[2:0] value $\leq$ CSnWCR1.CSPRWAIT[2:0] value.

For normal write access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSWWAIT[4:0] value.

For page write access, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.WRON[2:0] value $\leq$ CSnWCR1.CSPWAIT[2:0] value.

Note: When the address/data multiplexed I/O interface is selected, satisfy CSnWCR2.CSON[2:0] value $\leq$ CSnWCR2.AWAIT[1:0] value.

16.3.7 SDC Control Register (SDCCR)

Address(es): 0008 3C00h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	BSIZE[1:0]	—	—	—	—	EXENB
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	EXENB	Operation Enable	0: Operation is disabled 1: Operation is enabled	R/W
b3 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b5, b4	BSIZE[1:0]	SDRAM Bus Width Select	b5 b4 0 0: A 16-bit bus space is selected 0 1: A 32-bit bus space is selected 1 0: An 8-bit bus space is selected 1 1: Setting prohibited	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

EXENB Bit (Operation Enable)

This bit enables or disables the operation of the SDRAM address space. After reset, this bit is set to 0 (operation disabled). An attempt at access to an area for which operation has been disabled does not lead to SDRAM access. If the illegal address access detection enable bit (IGAEN) in the bus error monitoring enable register (BEREN) has been set to 1 (detection enabled), such an attempt will lead to a bus error.

16.3.8 SDC Mode Register (SDCMOD)

Address(es): 0008 3C01h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	EMODE
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	EMODE	Endian Mode	0: Endian of SDRAM address space is the same as the endian of operating mode. 1: Endian of SDRAM address space is not the endian of operating mode.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Writing to this register is possible only once after release from the reset state. Operation is not guaranteed after write access more than once is attempted.

EMODE Bit (Endian Mode)

This bit specifies the endian of the SDRAM address space.

When the endian setting for each area is different from that for the chip, no instruction code can be allocated in the area. The instruction code should be allocated to the external space whose endian setting is the same as that for the chip.

16.3.9 SDRAM Access Mode Register (SDAMOD)

Address(es): 0008 3C02h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	BE
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	BE	Continuous Access Enable	0: Continuous access is disabled 1: Continuous access is enabled	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Set SDAMOD while the conditions listed in Table 16.15, Conditions for Register Modification, are satisfied. The operation is not guaranteed if this register is set while these conditions are not satisfied.

BE Bit (Continuous Access Enable)

This bit enables or disables continuous access to the SDRAM access space.

Note: When the SDRAM area is accessed from bus masters other than EXDMAC, continuous access is always disabled regardless of the setting.

16.3.10 SDRAM Self-Refresh Control Register (SDSELF)

Address(es): 0008 3C10h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	SFEN
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	SFEN	SDRAM Self-Refresh Enable	0: Self-refresh is disabled 1: Self-refresh is enabled	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Set SDSELF while the conditions listed in Table 16.15, Conditions for Register Modification, are satisfied. The operation is not guaranteed if this register is set while these conditions are not satisfied.

SFEN Bit (SDRAM Self-Refresh Enable)

This bit controls self-refresh operation.

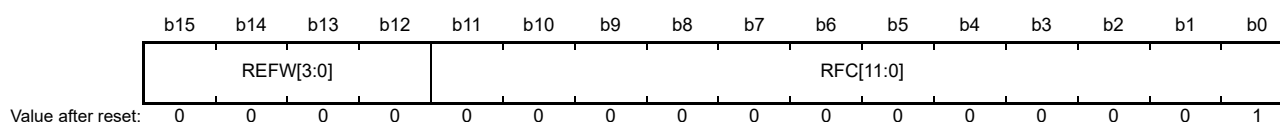
Setting this bit to 1 performs auto-refresh cycle operation, after which self-refresh operation begins.

Clearing this bit to 0 ends self-refresh operation, and auto-refresh operation resumes afterwards.

If this bit was set to 1, the value written to this bit is reflected when self-refresh operation starts. If this bit was set to 0, the value written to this bit has already been reflected when auto-refresh operation starts following the end of self-refresh operation.

16.3.11 SDRAM Refresh Control Register (SDRF CR)

Address(es): 0008 3C14h



Bit	Symbol	Bit Name	Description	R/W
b11 to b0	RFC[11:0]	Auto-Refresh Request Interval Setting	b11 b0 0 0 0 0 0 0 0 0 0 0 0 0: Setting prohibited 0 0 0 0 0 0 0 0 0 0 0 1: 2 cycles 0 0 0 0 0 0 0 0 0 1 0: 3 cycles : 1 1 1 1 1 1 1 1 1 1 1 1: 4096 cycles	R/W
b15 to b12	REFW[3:0]	Auto-Refresh Cycle/ Self-Refresh Clearing Cycle Count Setting	b15 b12 0 0 0 0: 1 cycle 0 0 0 1: 2 cycles 0 0 1 0: 3 cycles 0 0 1 1: 4 cycles 0 1 0 0: 5 cycles 0 1 0 1: 6 cycles 0 1 1 0: 7 cycles 0 1 1 1: 8 cycles 1 0 0 0: 9 cycles 1 0 0 1: 10 cycles 1 0 1 0: 11 cycles 1 0 1 1: 12 cycles 1 1 0 0: 13 cycles 1 1 0 1: 14 cycles 1 1 1 0: 15 cycles 1 1 1 1: 16 cycles	R/W

RFC[11:0] Bits (Auto-Refresh Request Interval Setting)

These bits specify the auto-refresh request interval.

These bits can be written to at any time, regardless of the state of the auto-refresh operation enable (RFEN) bit in SDRFEN.

If auto-refresh is enabled, the value written to these bits is reflected after the end of auto-refresh cycles. The refresh counter operates in SDCLK.

REFW[3:0] Bits (Auto-Refresh Cycle/ Self-Refresh Clearing Cycle Count Setting)

These bits specify the number of auto-refresh cycles and the number of self-refresh clearing cycles.

These bits can be written to at any time, regardless of the state of the auto-refresh operation enable (RFEN) bit in SDRFEN.

If an auto-refresh cycle is in progress, the value written to these bits while auto-refresh is enabled takes effect after the cycle completes.

Note: Auto-refresh requests are not accepted while SDRAM is being accessed; they must wait until the access completes, so the auto-refresh interval may become enlarged in some cases. Set the RFC[11:0] bits to an auto-refresh request interval value that satisfies the auto-refresh interval specification of the SDRAM being used. Furthermore, make sure to set the auto-refresh request interval to a duration longer than the auto-refresh cycle. Note that the auto-refresh interval cannot be automatically adjusted when the frequency is changed during operation; in this case, perform self-refresh operation and set the auto-refresh interval appropriate for the frequency again.

- Auto-Refresh Request Interval and RFC Set Value

SDRAMC (SDRAM area controller) includes a 12-bit refresh counter that generates auto-refresh requests at fixed intervals. The following equation is used to calculate the set value for the RFC[11:0] bits from the auto-refresh request interval.

$$\text{RFC} = (\text{Auto-refresh request interval} / \text{SDCLK cycle}) - 1$$

Note: Auto-refresh requests are not accepted while SDRAM is being accessed; they must wait until the access completes. However, the counter value is updated regardless of whether or not the request was accepted. Note that if two or more auto-refresh requests are generated while SDRAM is being accessed, the second and subsequent requests are ignored.

16.3.12 SDRAM Auto-Refresh Control Register (SDRFEN)

Address(es): 0008 3C16h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	RFEN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	RFEN	Auto-Refresh Operation Enable	0: Auto-refresh operation is disabled 1: Auto-refresh operation is enabled	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

RFEN Bit (Auto-Refresh Operation Enable)

Setting this bit to 0 while auto-refreshing is enabled causes RFEN to be set to 0 and auto-refresh operation to halt after the end of the auto-refresh cycle. However, if RFEN is again set to 1 before the end of the auto-refresh cycle, auto-refreshing continues and the RFEN bit is not set to 0. Setting the RFEN bit to 1 while auto-refresh is disabled starts auto-refresh operation, and refresh requests are then generated at fixed intervals determined by a counter. The interval at which refresh requests are generated is determined by the value of the auto-refresh request interval setting (RFC[11:0]) bits in the SDRAM refresh control register (SDRFCR).

Refresh requests are not accepted while SDRAM is being accessed; they must wait until the access completes.

If an SDRAM access and a refresh request are generated at the same time, the refresh request takes precedence.

16.3.13 SDRAM Initialization Sequence Control Register (SDICR)

Address(es): 0008 3C20h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	INIRQ
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	INIRQ	Initialization Sequence Start	0: Invalid 1: Initialization sequence starts	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Writing to this register is possible only once after release from the reset state. Operation is not guaranteed after write access more than once is attempted.

INIRQ Bit (Initialization Sequence Start)

Setting this bit to 1 causes the SDRAM initialization sequence to start and automatically sets the initialization status bit (INIST) in the SDRAM status register (SDSR) to 1. The INIST bit is cleared automatically after the initialization sequence ends.

The value written to the INIRQ bit is not retained.

If access to an external address space or an external bus controller register occurs after the initialization sequence is started, the access is suspended until the initialization sequence ends.

Note: Set the INIRQ bit to start the SDRAM initialization sequence while the conditions listed in Table 16.15, Conditions for Register Modification, are satisfied. The operation is not guaranteed if this bit is set while these conditions are not satisfied.

16.3.14 SDRAM Initialization Register (SDIR)

Address(es): 0008 3C24h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	PRC[2:0]		ARFC[3:0]				ARFI[3:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	ARFI[3:0]	Initialization Auto-Refresh Interval	b3 b0 0 0 0 0: 3 cycles 0 0 0 1: 4 cycles 0 0 1 0: 5 cycles 0 0 1 1: 6 cycles 0 1 0 0: 7 cycles 0 1 0 1: 8 cycles 0 1 1 0: 9 cycles 0 1 1 1: 10 cycles 1 0 0 0: 11 cycles 1 0 0 1: 12 cycles 1 0 1 0: 13 cycles 1 0 1 1: 14 cycles 1 1 0 0: 15 cycles 1 1 0 1: 16 cycles 1 1 1 0: 17 cycles 1 1 1 1: 18 cycles	R/W
b7 to b4	ARFC[3:0]	Initialization Auto-Refresh Count	b7 b4 0 0 0 0: Setting prohibited 0 0 0 1: 1 time 0 0 1 0: 2 times 0 0 1 1: 3 times 0 1 0 0: 4 times 0 1 0 1: 5 times 0 1 1 0: 6 times 0 1 1 1: 7 times 1 0 0 0: 8 times 1 0 0 1: 9 times 1 0 1 0: 10 times 1 0 1 1: 11 times 1 1 0 0: 12 times 1 1 0 1: 13 times 1 1 1 0: 14 times 1 1 1 1: 15 times	R/W
b10 to b8	PRC[2:0]	Initialization Precharge Cycle Count	b10 b8 0 0 0: 3 cycles 0 0 1: 4 cycles 0 1 0: 5 cycles 0 1 1: 6 cycles 1 0 0: 7 cycles 1 0 1: 8 cycles 1 1 0: 9 cycles 1 1 1: 10 cycles	R/W
b15 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Writing to this register is possible only once after release from the reset state. Operation is not guaranteed after write access more than once is attempted.

ARFI[3:0] Bits (Initialization Auto-Refresh Interval)

These bits specify the interval at which auto-refresh commands are issued in the SDRAM initialization sequence.

ARFC[3:0] Bits (Initialization Auto-Refresh Count)

These bits specify the number of times auto-refresh is to be performed in the SDRAM initialization sequence.

PRC[2:0] Bits (Initialization Precharge Cycle Count)

These bits specify the number of precharge cycles in the SDRAM initialization sequence.

Note: Make settings that satisfy the specifications of the connected SDRAM before starting the initialization sequence.

16.3.15 SDRAM Address Register (SDADR)

Address(es): 0008 3C40h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	MXC[1:0]	
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	MXC[1:0]	Address Multiplex Select	b1 b0 0 0: 8-bit shift 0 1: 9-bit shift 1 0: 10-bit shift 1 1: 11-bit shift	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Set SDADR while the conditions listed in Table 16.15, Conditions for Register Modification, are satisfied. The operation is not guaranteed if this bit is set while these conditions are not satisfied.

MXC[1:0] Bits (Address Multiplex Select)

These bits select the size of the shift toward the lower half of the row address in row address/column address multiplexing. These bits also select the row address bits to be used for comparison in the SDRAMC continuous access operation.

For details, refer to Table 16.20, Address Multiplexing.

16.3.16 SDRAM Timing Register (SDTR)

Address(es): 0008 3C44h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	RAS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	RCD[1:0]		RP[2:0]		WR	—	—	—	—	—	—	CL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	CL[2:0]	SDRAMC Column Latency	b2 b0 0 0 0: Setting prohibited 0 0 1: 1 cycle 0 1 0: 2 cycles 0 1 1: 3 cycles 1 0 0: Setting prohibited 1 0 1: Setting prohibited 1 1 0: Setting prohibited 1 1 1: Setting prohibited	R/W
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	WR	Write Recovery Interval	0: 1 cycle 1: 2 cycles	R/W
b11 to b9	RP[2:0]	Row Precharge Interval	b11 b9 0 0 0: 1 cycle 0 0 1: 2 cycles 0 1 0: 3 cycles 0 1 1: 4 cycles 1 0 0: 5 cycles 1 0 1: 6 cycles 1 1 0: 7 cycles 1 1 1: 8 cycles	R/W
b13, b12	RCD[1:0]	Row Column Latency	b13b12 0 0: 1 cycle 0 1: 2 cycles 1 0: 3 cycles 1 1: 4 cycles	R/W
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b18 to b16	RAS[2:0]	Row Active Interval	b18 b16 0 0 0: 1 cycle 0 0 1: 2 cycles 0 1 0: 3 cycles 0 1 1: 4 cycles 1 0 0: 5 cycles 1 0 1: 6 cycles 1 1 0: 7 cycles 1 1 1: Setting prohibited	R/W
b31 to b19	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

SDTR specifies the timing for read and write accesses to SDRAM. For details, refer to section 16.6.12.3, Timing Register Settings and Access Timing.

Set SDTR while the conditions listed in Table 16.15, Conditions for Register Modification, are satisfied. The operation is not guaranteed if this bit is set while these conditions are not satisfied.

Writing to this register is possible only once after release from the reset state. Operation is not guaranteed after write access more than once is attempted.

CL[2:0] Bits (SDRAMC Column Latency)

These bits specify the column latency of the SDRAM controller. This setting only affects the latency setting on the SDRAM controller side. To specify the column latency for externally connected SDRAM, use the SDRAM mode register (SDMOD), which is described below.

WR Bit (Write Recovery Interval)

This bit specifies the interval that must elapse between the SDRAM write command (WRIT) and deactivation (PALL).

RP[2:0] Bits (Row Precharge Interval)

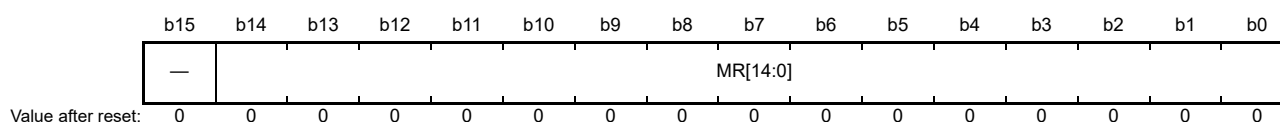
These bits specify the minimum number of cycles that must elapse between the SDRAM deactivation command (PALL) and the next valid command.

RAS[2:0] Bits (Row Active Interval)

These bits specify the minimum interval that must elapse between the SDRAM row activation command (ACTV) and deactivation (PALL). The value specified by these bits should be less than or equal to the sum of the row-column latency (RCD[1:0]) and column latency (CL[2:0]) settings.

16.3.17 SDRAM Mode Register (SDMOD)

Address(es): 0008 3C48h



Bit	Symbol	Bit Name	Description	R/W
b14 to b0	MR[14:0]	Mode Register Setting	Writing to these bits: Mode register set command is issued	R/W
b15	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

SDMOD specifies the value to be written to the SDRAM mode register.

Writing to SDMOD causes a mode register set command to be issued automatically to SDRAM.

Set SDMOD while the conditions listed in Table 16.15, Conditions for Register Modification, are satisfied. The operation is not guaranteed if this bit is set while these conditions are not satisfied.

MR[14:0] Bits (Mode Register Setting)

Writing to these bits causes a mode register set command to be issued to SDRAM. The setting of the MR[14:0] bits is output to the lower bits of the address. For details, refer to section 16.6.11, Setting Mode Register.

If access to an external address space or an external bus controller register occurs after writing to mode register, the access is suspended until the mode register set command is issued.

Note: The following points should be kept in mind regarding SDMOD settings.

- Make sure to set a burst length of 1 for SDRAM. Operation cannot be guaranteed with settings other than burst length 1.
- The SDRAM column latency must match the setting of the SDRAMC column latency setting bits (CL[2:0]) in the SDRAM timing register (SDTR). Operation cannot be guaranteed if the latency settings do not agree.
- Whenever writing is to proceed, confirm that all of the status bits (SDSR.SRFST, INIST, and MRSST) in the SDRAM status register are 0 beforehand.

16.3.18 SDRAM Status Register (SDSR)

Address(es): 0008 3C50h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	SRFST	INIST	—	—	MRSST
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	MRSST	Mode Register Setting Status	0: Mode register setting not in progress 1: Mode register setting in progress	R
b2, b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b3	INIST	Initialization Status	0: Initialization sequence not in progress 1: Initialization sequence in progress	R
b4	SRFST	Self-Refresh Transition/Recovery Status	0: Transition/recovery not in progress 1: Transition/recovery in progress	R
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

MRSST Bit (Mode Register Setting Status)

When set to 1, this bit indicates that SDRAM mode register setting is in progress. If SDSA is accessed during the mode register setting operation, the CPU processing can be suspended until the setting operation ends.

INIST Bit (Initialization Status)

When set to 1, this bit indicates that the SDRAM initialization sequence is in progress. If SDSA is accessed during initialization sequence, the CPU processing can be suspended until the initialization sequence ends.

SRFST Bit (Self-Refresh Transition/Recovery Status)

When set to 1, this bit indicates that a transition to or recovery from self-refresh operation is in progress for SDRAM. “Transition to or recovery from self-refresh operation in progress” refers to the interval from the point at which the bits listed in Table 16.9 are written until the corresponding commands are issued.

Note: Execution of a self-refresh, an initialization sequence, or mode register setting may only be performed when all the status bits are 0. Do not rewrite the registers (bits) listed in Table 16.9 when any of the status bits (SRFST, INIST, MRSST) is set to 1.

Table 16.9 List of Registers and Bits Requiring Checking Status Bits

Function	Register	Bits
Self-refresh	SDSELF	SFEN
Initialization sequence	SDICR	INIRQ
Mode register setting	SDMOD	MR[14:0]

16.3.19 Bus Error Status Clear Register (BERCLR)

Address(es): 0008 1300h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	STSCLR
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	STSCLR	Status Clear	0: Invalid 1: Bus error status register cleared	(W)*1
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only writing 1 is effective; i.e. writing 0 has no effect.

STSCLR Bit (Status Clear)

Writing 1 to this bit clears the bus error status registers 1 and 2 (BERSR1 and BERSR2).

Writing 0 has no effect. It is read as 0.

16.3.20 Bus Error Monitoring Enable Register (BEREN)

Address(es): 0008 1304h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	TOEN	IGAEN
0	0	0	0	0	0	0	0

Value after reset:

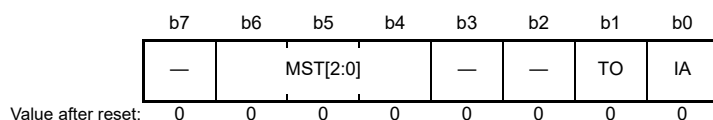
Bit	Symbol	Bit Name	Description	R/W
b0	IGAEN	Illegal Address Access Detection Enable	0: Illegal address access detection is disabled. 1: Illegal address access detection is enabled.	R/W
b1	TOEN	Timeout Detection Enable*1, *2	0: Bus timeout detection is disabled. 1: Bus timeout detection is enabled.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. When detection is disabled (the TOEN bit is set to 0), bus access can cause the bus to freeze.

Note 2. Do not set the TOEN bit to 0 (bus timeout detection disabled) while timeout errors are being detected.

16.3.21 Bus Error Status Register 1 (BERSR1)

Address(es): 0008 1308h



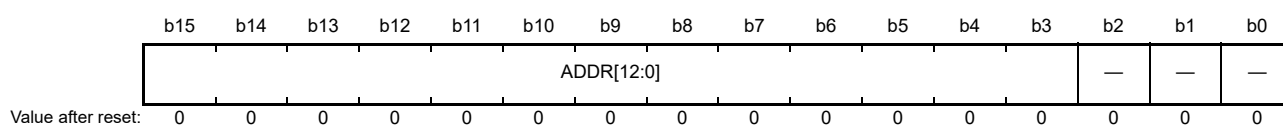
Bit	Symbol	Bit Name	Description	R/W
b0	IA	Illegal Address Access	0: Illegal address access not made 1: Illegal address access made	R
b1	TO	Timeout	0: Timeout not generated 1: Timeout generated	R
b3, b2	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b6 to b4	MST[2:0]	Bus Master Code	b6 b4 0 0 0: CPU 0 0 1: Reserved 0 1 0: Reserved 0 1 1: DTC/DMAC 1 0 0: Reserved 1 0 1: Reserved 1 1 0: Extended bus master 1 1 1: EXDMAC	R
b7	—	Reserved	This bit is read as 0. Writing to this bit has no effect.	R

MST[2:0] Bits (Bus Master Code)

These bits indicate the bus master that accessed a bus when a bus error occurred.

16.3.22 Bus Error Status Register 2 (BERSR2)

Address(es): 0008 130Ah



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b15 to b3	ADDR[12:0]	Bus Error Occurrence Address	The upper 13 bits of an address that was accessed when a bus error occurred (in units of 512 Kbytes).	R

16.3.23 Bus Priority Control Register (BUSPRI)

Address(es): 0008 1310h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	BPEB[1:0]	BPFB[1:0]	BPHB[1:0]	BPGB[1:0]	BPIB[1:0]	BPRO[1:0]	BPRA[1:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	BPRA[1:0]	Memory Bus 1 and 3 (RAM/expansion RAM/ECCRAM) Priority Control	b1 b0 0 0: The order of priority is fixed. 0 1: The order of priority is toggled. 1 0: Setting prohibited 1 1: Setting prohibited	R(W) *1
b3, b2	BPRO[1:0]	Memory Bus 2 (Code Flash Memory) Priority Control	b3 b2 0 0: The order of priority is fixed. 0 1: The order of priority is toggled. 1 0: Setting prohibited 1 1: Setting prohibited	R(W) *1
b5, b4	BPIB[1:0]	Internal Peripheral Bus 1 Priority Control	b5 b4 0 0: The order of priority is fixed. 0 1: The order of priority is toggled. 1 0: Setting prohibited 1 1: Setting prohibited	R(W) *1
b7, b6	BPGB[1:0]	Internal Peripheral Bus 2 and 3 Priority Control	b7 b6 0 0: The order of priority is fixed. 0 1: The order of priority is toggled. 1 0: Setting prohibited 1 1: Setting prohibited	R(W) *1
b9, b8	BPHB[1:0]	Internal Peripheral Bus 4 and 5 Priority Control	b9 b8 0 0: The order of priority is fixed. 0 1: The order of priority is toggled. 1 0: Setting prohibited 1 1: Setting prohibited	R(W) *1
b11, b10	BPFB[1:0]	Internal Peripheral Bus 6 Priority Control	b11b10 0 0: The order of priority is fixed. 0 1: The order of priority is toggled. 1 0: Setting prohibited 1 1: Setting prohibited	R(W) *1
b13, b12	BPEB[1:0]	External Bus Priority Control	b13b12 0 0: The order of priority is fixed. 0 1: The order of priority is toggled. 1 0: Setting prohibited 1 1: Setting prohibited	R(W) *1
b15, b14	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Writing to these bits is only possible once, and this must be done while the DTC, DMAC, EXDMAC, and extended bus master are all stopped. If the value of these bits is changed more than once, further operation is not guaranteed.

BPRA[1:0] Bits (Memory Bus 1 and 3 (RAM/expansion RAM/ECCRAM) Priority Control)

These bits specify the priority order for memory bus 1 (RAM) and memory bus 3 (expansion RAM/ECCRAM).
When the priority order is fixed, internal main bus 2 has priority over CPU buses (instruction and operand buses).
When the priority order is toggled, the bus from which a request has currently been accepted has the lower priority.

BPRO[1:0] Bits (Memory Bus 2 (Code Flash Memory) Priority Control)

These bits specify the priority order for memory bus 2 (code flash memory).
When the priority order is fixed, internal main bus 2 has priority over CPU buses (instruction and operand buses).
When the priority order is toggled, the bus from which a request has currently been accepted has the lower priority.

BPIB[1:0] Bits (Internal Peripheral Bus 1 Priority Control)

These bits specify the priority order for internal peripheral bus 1.

When the priority order is fixed, internal main bus 2 has priority over internal main bus 1.

When the priority order is toggled, a bus has a lower priority when the request of that bus is accepted.

BPGB[1:0] Bits (Internal Peripheral Bus 2 and 3 Priority Control)

These bits specify the priority order for internal peripheral buses 2 and 3.

When the priority order is fixed, internal main bus 2 has priority over internal main bus 1.

When the priority order is toggled, a bus has a lower priority when the request of that bus is accepted.

BPHB[1:0] Bits (Internal Peripheral Bus 4 and 5 Priority Control)

These bits specify the priority order for internal peripheral buses 4 and 5.

When the priority order is fixed, internal main bus 2 has priority over internal main bus 1.

When the priority order is toggled, a bus has a lower priority when the request of that bus is accepted.

BPFB[1:0] Bits (Internal Peripheral Bus 6 Priority Control)

These bits specify the priority order for internal peripheral bus 6.

When the priority order is fixed, internal main bus 2 has priority over internal main bus 1.

When the priority order is toggled, a bus has a lower priority when the request of that bus is accepted.

BPEB[1:0] Bits (External Bus Priority Control)

These bits specify the priority order for the external bus.

When the priority order is fixed, the order is EXDMAC, internal main bus 2, and then internal main bus 1.

When the priority order is toggled, a bus has a lower priority when the request of that bus is accepted, between internal main bus 1 and other buses (internal main bus 2 and EXDMAC). However, the order of priority is EXDMAC and then internal main bus 2, regardless of the BPEB[1:0] bits settings.

16.3.24 Extended Bus Master Priority Control Register (EBMAPCR)

Address(es): 000C 5800h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	PRERR	—	—	—	—	—	—	—	—	—	—	—	—	PR5SEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	PR4SEL[2:0]			—	PR3SEL[2:0]			—	PR2SEL[2:0]			—	PR1SEL[2:0]		
Value after reset:	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	PR1SEL[2:0]	1st Priority Selection	b2 b0 0 0 0: M0 (GLCDC-GRA1) is selected as having the highest priority in arbitration (initial value) 0 0 1: M1 (DRW2D-TX) is selected as having the highest priority in arbitration 0 1 0: M2 (DRW2D-FB) is selected as having the highest priority in arbitration 0 1 1: M3 (GLCDC-GRA2) is selected as having the highest priority in arbitration 1 0 0: M4 (EDMAC) is selected as having the highest priority in arbitration Settings other than above are prohibited.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6 to b4	PR2SEL[2:0]	2nd Priority Selection	b6 b4 0 0 0: M0 (GLCDC-GRA1) is selected as having the second priority in arbitration 0 0 1: M1 (DRW2D-TX) is selected as having the second priority in arbitration (initial value) 0 1 0: M2 (DRW2D-FB) is selected as having the second priority in arbitration 0 1 1: M3 (GLCDC-GRA2) is selected as having the second priority in arbitration 1 0 0: M4 (EDMAC) is selected as having the second priority in arbitration Settings other than above are prohibited.	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b10 to b8	PR3SEL[2:0]	3rd Priority Selection	b10 b8 0 0 0: M0 (GLCDC-GRA1) is selected as having the third priority in arbitration 0 0 1: M1 (DRW2D-TX) is selected as having the third priority in arbitration 0 1 0: M2 (DRW2D-FB) is selected as having the third priority in arbitration (initial value) 0 1 1: M3 (GLCDC-GRA2) is selected as having the third priority in arbitration 1 0 0: M4 (EDMAC) is selected as having the third priority in arbitration Settings other than above are prohibited.	R/W
b11	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Bit Name	Description	R/W
b14 to b12	PR4SEL[2:0]	4th Priority Selection	^{b14} ^{b12} 0 0 0: M0 (GLCDC-GRA1) is selected as having the fourth priority in arbitration 0 0 1: M1 (DRW2D-TX) is selected as having the fourth priority in arbitration 0 1 0: M2 (DRW2D-FB) is selected as having the fourth priority in arbitration 0 1 1: M3 (GLCDC-GRA2) is selected as having the fourth priority in arbitration (initial value) 1 0 0: M4 (EDMAC) is selected as having the fourth priority in arbitration Settings other than above are prohibited.	R/W
b15	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b18 to b16	PR5SEL[2:0]	5th Priority Selection	^{b18} ^{b16} 0 0 0: M0 (GLCDC-GRA1) is selected as having the fifth priority in arbitration 0 0 1: M1 (DRW2D-TX) is selected as having the fifth priority in arbitration 0 1 0: M2 (DRW2D-FB) is selected as having the fifth priority in arbitration 0 1 1: M3 (GLCDC-GRA2) is selected as having the fifth priority in arbitration 1 0 0: M4 (EDMAC) is selected as having the fifth priority in arbitration (initial value) Settings other than above are prohibited.	R/W
b30 to b19	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b31	PRERR	Priority Setting Error Flag	0: The setting of the register is valid. 1: The setting of the register is invalid and operation of priority control is not guaranteed.	R

This register is used to select the priority of bus mastership requests from the extended bus masters in case of contention between them and is only available. The register must be accessed in 32-bit units. The setting of the register must be set while there is no request for bus mastership.

The extended bus master has five transfer sources: EDMAC, GLCDC-GRA1 (GLCDC graphics 1 data read), GLCDC-GRA2 (GLCDC graphics 2 data read), DRW2D-TX (DRW2D texture data read), and DRW2D-FB (DRW2D frame buffer data read write and display list data read). The priority order in cases of conflict of requests for bus mastership is in the order GLCDC-GRA1 > DRW2D-TX > DRW2D-FB > GLCDC-GRA2 > EDMAC. The priority order can be changed by setting the PRnSEL[2:0] bits of the register. The priority order of the transfer source selected by the PR1SEL[2:0] bits is the highest and the one selected by the PR5SEL[2:0] bits becomes the lowest.

PRERR Flag (Priority Setting Error Flag)

When this flag is 1, it indicates that the setting of the EBMAPCR register is invalid. This flag is set to 1 when even two settings among those for the PR1SEL[2:0], PR2SEL[2:0], PR3SEL[2:0], PR4SEL[2:0] and PR5SEL[2:0] bits are the same, or when prohibited settings (101b to 111b) are mode for these bits.

When this flag is 1, the operation of priority control is not guaranteed. Check the value of the flag and make the appropriate setting to the flag.

16.4 Endian and Data Alignment

The external bus has a data-alignment function to control which byte of the data bus (D31 to D24, D23 to D16, D15 to D8, or D7 to D0) is used according to the bus specifications of the area to be accessed (8-bit, 16-bit, or 32-bit bus space), data size, and endian format when accessing the external address space (the CS and SDRAM areas).

16.4.1 Data Alignment Control for CS Area

(1) 32-Bit Bus Space

When a 32-bit width is selected for a bus space by the BSIZE[1:0] bits in CSnCR, the address buses A23 to A2 are enabled to output address signals in units of 32 bits, and the address buses A1 and A0 are disabled (always output the low level).

When byte strobe mode is selected (the WRMOD bit = 0 in CSnMOD), the WR0# to WR3# pins are enabled. The BC0# to BC3# pins are not used.

When single write strobe mode is selected (the WRMOD bit = 1 in CSnMOD), only the WR0# pin is enabled and always outputs the low level during write access, regardless of the data size. Here, the WR1# to WR3# pins are invalid (always output the high level). The valid byte position is indicated by the BC0# to BC3# pins.

In 32-bit bus space, the valid positions of data external to the chip and of control signals differ according to whether the endian is big or little.

Data Size	Access Address	Number of Access	Bus Cycle	Unit of Data	Address	WR3#/BC3# WR2#/BC2# WR1#/BC1# WR0#/BC0# RD# Data Bus D31 D24 D23 D16 D15 D8 D7 D0							
8 bits	4n	One	First	8 bits	4n	[7] 0							
	4n+1	One	First	8 bits	4n	[7] 0							
	4n+2	One	First	8 bits	4n	[7] 0							
	4n+3	One	First	8 bits	4n	[7] 0							
16 bits	4n	One	First	16 bits	4n	[15] 8 [7] 0							
	4n+1	Two	First	8 bits	4n	[7] 0							
			Second	8 bits	4n	[15] 8							
	4n+2	One	First	16 bits	4n	[15] 8 [7] 0							
	4n+3	Two	First	8 bits	4n	[7] 0							
			Second	8 bits	4n+4	[15] 8							
32 bits	4n	One	First	32 bits	4n	[31] 24 [23] 16 [15] 8 [7] 0							
	4n+1	Three	First	8 bits	4n	[7] 0							
			Second	16 bits	4n	[23] 16 [15] 8							
			Third	8 bits	4n+4	[31] 24							
	4n+2	Two	First	16 bits	4n	[15] 8 [7] 0							
			Second	16 bits	4n+4	[31] 24 [23] 16							
	4n+3	Three	First	8 bits	4n	[7] 0							
			Second	16 bits	4n+4	[23] 16 [15] 8							
			Third	8 bits	4n+4	[31] 24							
64 bits	4n	Two	First	32 bits	4n	[31] 24 [23] 16 [15] 8 [7] 0							
			Second	32 bits	4n+4	[63] 56 [55] 48 [47] 40 [39] 32							

Figure 16.6 Data Alignment (Little Endian) in 32-Bit Bus Space

Data Size	Access Address	Number of Access	Data Bus													
			Bus Cycle	Unit of Data	Address	RD#										
						Data Bus										
						D31	D24 D23			D16 D15		D8 D7		D0		
8 bits	4n	One	First	8 bits	4n	7 0										
	4n+1	One	First	8 bits	4n	7 0										
	4n+2	One	First	8 bits	4n	7 0										
	4n+3	One	First	8 bits	4n	7 0										
16 bits	4n	One	First	16 bits	4n	15 8 7 0										
	4n+1	Two	First	8 bits	4n	15 8										
			Second	8 bits	4n	7 0										
	4n+2	One	First	16 bits	4n	15 8 7 0										
	4n+3	Two	First	8 bits	4n	15 8										
			Second	8 bits	4n+4	7 0										
32 bits	4n	One	First	32 bits	4n	31 24 23 16 15 8 7 0										
	4n+1	Three	First	8 bits	4n	31 24										
			Second	16 bits	4n	23 16 15 8										
			Third	8 bits	4n+4	7 0										
	4n+2	Two	First	16 bits	4n	31 24 23 16										
			Second	16 bits	4n+4	15 8 7 0										
	4n+3	Three	First	8 bits	4n	31 24										
			Second	16 bits	4n+4	23 16 15 8										
			Third	8 bits	4n+4	7 0										
64 bits	4n	Two	First	32 bits	4n	63 56 55 48 47 40 39 32										
			Second	32 bits	4n+4	31 24 23 16 15 8 7 0										

Figure 16.7 Data Alignment (Big Endian) in 32-Bit Bus Space

(2) 16-Bit Bus Space

When a 16-bit width is selected for a bus space by the BSIZE[1:0] bits in CSnCR, the address buses A23 to A1 are enabled to output address signals in units of 16 bits, and the address bus A0 is disabled (always output the low level). When byte strobe mode is selected (the WRMOD bit = 0 in CSnMOD), the WR0# and WR1# pins are enabled, and the WR2# and WR3# pins are disabled (fixed high). The BC0# to BC3# pins are not used.

When single write strobe mode is selected (the WRMOD bit = 1 in CSnMOD), only the WR0# pin is enabled and always outputs the low level during write access, regardless of the data size. Here, the WR1# to WR3# pins are invalid (always output the high level). The valid byte position is indicated by the BC0# and BC1# pins. The BC2# and BC3# pins are not used.

In 16-bit bus space, page access can occur in access to data in 32- or 64-bit units. Specifically, page access can occur when an access does not spread over a 32-bit boundary and causes no change in BC0# and BC1# signals. The situations in which page access occurs are indicated by the letter (p) in Figure 16.8 and Figure 16.9.

In 16-bit bus space, the valid positions of data external to the chip and of control signals differ according to whether the endian is big or little.

Data Size	Access Address	Number of Access	Bus Cycle	Unit of Data	Address	WR1#/BC1# WR0#/BC0# RD# Data Bus D15 D8 D7 D0			
8 bits	4n	One	First	8 bits	4n	7 0			
	4n+1	One	First	8 bits	4n	7 0			
	4n+2	One	First	8 bits	4n+2	7 0			
	4n+3	One	First	8 bits	4n+2	7 0			
16 bits	4n	One	First	16 bits	4n	15 8 7 0			
	4n+1	Two	First	8 bits	4n	7 0			
			Second	8 bits	4n+2	15 8			
	4n+2	One	First	16 bits	4n+2	15 8 7 0			
	4n+3	Two	First	8 bits	4n+2	7 0			
			Second	8 bits	4n+4	15 8			
32 bits	4n	Two	First	16 bits	4n	15 8 7 0			
			Second	16 bits	4n+2 (p)	31 24 23 16			
	4n+1	Three	First	8 bits	4n	7 0			
			Second	16 bits	4n+2	23 16 15 8			
			Third	8 bits	4n+4	31 24			
	4n+2	Two	First	16 bits	4n+2	15 8 7 0			
			Second	16 bits	4n+4	31 24 23 16			
	4n+3	Three	First	8 bits	4n+2	7 0			
			Second	16 bits	4n+4	23 16 15 8			
			Third	8 bits	4n+6	31 24			
64 bits	4n	Four	First	16 bits	4n	15 8 7 0			
			Second	16 bits	4n+2 (p)	31 24 23 16			
			Third	16 bits	4n+4	47 40 39 32			
			Fourth	16 bits	4n+6 (p)	63 56 55 48			

(p): Page access (only when page access is enabled with the PRENB and PWENB bits in CSnMOD)

Figure 16.8 Data Alignment (Little Endian) in 16-Bit Bus Space

Data Size	Access Address	Number of Access	Bus Cycle	Unit of Data	Address	WR1#/BC1#	WR0#/BC0#
						RD#	
						Data Bus	
						D15	D8 D7
8 bits	4n	One	First	8 bits	4n	7	0
	4n+1	One	First	8 bits	4n		7 0
	4n+2	One	First	8 bits	4n+2	7	0
	4n+3	One	First	8 bits	4n+2		7 0
16 bits	4n	One	First	16 bits	4n	15	8 7 0
	4n+1	Two	First	8 bits	4n		15 8
			Second	8 bits	4n+2	7	0
	4n+2	One	First	16 bits	4n+2	15	8 7 0
	4n+3	Two	First	8 bits	4n+2		15 8
			Second	8 bits	4n+4	7	0
32 bits	4n	Two	First	16 bits	4n	31	24 23 16
			Second	16 bits	4n+2 (p)	15	8 7 0
	4n+1	Three	First	8 bits	4n		31 24
			Second	16 bits	4n+2	23	16 15 8
			Third	8 bits	4n+4	7	0
	4n+2	Two	First	16 bits	4n+2	31	24 23 16
			Second	16 bits	4n+4	15	8 7 0
	4n+3	Three	First	8 bits	4n+2		31 24
			Second	16 bits	4n+4	23	16 15 8
			Third	8 bits	4n+6	7	0
64 bits	4n	Four	First	16 bits	4n	63	56 55 48
			Second	16 bits	4n+2 (p)	47	40 39 32
			Third	16 bits	4n+4	31	24 23 16
			Fourth	16 bits	4n+6 (p)	15	8 7 0

(p): Page access (only when page access is enabled with the PRENB and PWENB bits in CSnMOD)

Figure 16.9 Data Alignment (Big Endian) in 16-Bit Bus Space

(3) 8-Bit Bus Space

When an 8-bit bus space is selected by the BSIZE[1:0] bits in CSnCR, the address buses A23 to A0 are enabled to output address signals in byte units.

In 8-bit bus space, only the WR0# pin is valid regardless of write access mode, and always outputs the low level during write access. The WR1# to WR3# pins and the BC0# to BC3# pins are not used.

Page access can occur in access to data in 16-, 32- or 64-bit units. Specifically, page access can occur when an access does not spread over a 32-bit boundary. The situations in which page access occurs are indicated by the letter (p) in Figure 16.10 and Figure 16.11.

In 8-bit bus space, the valid positions of data external to the chip are D7 to D0 and WR0# is used as the control signal, regardless of the endian mode.

Data Size	Access Address	Number of Access	Bus Cycle	Unit of Data	Address	Data Bus			
						D15	D8	D7	D0
8 bits	4n	One	First	8 bits	4n	7	0		
	4n+1	One	First	8 bits	4n+1	7	0		
	4n+2	One	First	8 bits	4n+2	7	0		
	4n+3	One	First	8 bits	4n+3	7	0		
16 bits	4n	Two	First	8 bits	4n	7	0		
			Second	8 bits	4n+1 (p)	15	8		
	4n+1	Two	First	8 bits	4n+1	7	0		
			Second	8 bits	4n+2 (p)	15	8		
	4n+2	Two	First	8 bits	4n+2	7	0		
			Second	8 bits	4n+3 (p)	15	8		
	4n+3	Two	First	8 bits	4n+3	7	0		
			Second	8 bits	4n+4	15	8		
32 bits	4n	Four	First	8 bits	4n	7	0		
			Second	8 bits	4n+1 (p)	15	8		
			Third	8 bits	4n+2 (p)	23	16		
			Fourth	8 bits	4n+3 (p)	31	24		
	4n+1	Four	First	8 bits	4n+1	7	0		
			Second	8 bits	4n+2 (p)	15	8		
			Third	8 bits	4n+3 (p)	23	16		
			Fourth	8 bits	4n+4	31	24		
	4n+2	Four	First	8 bits	4n+2	7	0		
			Second	8 bits	4n+3 (p)	15	8		
			Third	8 bits	4n+4	23	16		
			Fourth	8 bits	4n+5 (p)	31	24		
	4n+3	Four	First	8 bits	4n+3	7	0		
			Second	8 bits	4n+4	15	8		
			Third	8 bits	4n+5 (p)	23	16		
			Fourth	8 bits	4n+6 (p)	31	24		
64 bits	4n	Eight	First	8 bits	4n	7	0		
			Second	8 bits	4n+1 (p)	15	8		
			Third	8 bits	4n+2 (p)	23	16		
			Fourth	8 bits	4n+3 (p)	31	24		
			Fifth	8 bits	4n+4	39	32		
			Sixth	8 bits	4n+5 (p)	47	40		
			Seventh	8 bits	4n+6 (p)	55	48		
			Eighth	8 bits	4n+7 (p)	63	56		

(p): Page access (only when page access is enabled with the PRENB and PWENB bits in CSnMOD)

Figure 16.10 Data Alignment (Little Endian) in 8-Bit Bus Space

Data Size	Access Address	Number of Access	Bus Cycle	Unit of Data	Address	Data Bus			
						RD#			
						D15	D8	D7	D0
8 bits	4n	One	First	8 bits	4n	7	0		
	4n+1	One	First	8 bits	4n+1	7	0		
	4n+2	One	First	8 bits	4n+2	7	0		
	4n+3	One	First	8 bits	4n+3	7	0		
16 bits	4n	Two	First	8 bits	4n	15	8		
			Second	8 bits	4n+1 (p)	7	0		
	4n+1	Two	First	8 bits	4n+1	15	8		
			Second	8 bits	4n+2 (p)	7	0		
	4n+2	Two	First	8 bits	4n+2	15	8		
			Second	8 bits	4n+3 (p)	7	0		
	4n+3	Two	First	8 bits	4n+3	15	8		
			Second	8 bits	4n+4	7	0		
32 bits	4n	Four	First	8 bits	4n	31	24		
			Second	8 bits	4n+1 (p)	23	16		
			Third	8 bits	4n+2 (p)	15	8		
			Fourth	8 bits	4n+3 (p)	7	0		
	4n+1	Four	First	8 bits	4n+1	31	24		
			Second	8 bits	4n+2 (p)	23	16		
			Third	8 bits	4n+3 (p)	15	8		
			Fourth	8 bits	4n+4	7	0		
	4n+2	Four	First	8 bits	4n+2	31	24		
			Second	8 bits	4n+3 (p)	23	16		
			Third	8 bits	4n+4	15	8		
			Fourth	8 bits	4n+5 (p)	7	0		
	4n+3	Four	First	8 bits	4n+3	31	24		
			Second	8 bits	4n+4	23	16		
			Third	8 bits	4n+5 (p)	15	8		
			Fourth	8 bits	4n+6 (p)	7	0		
64 bits	4n	Eight	First	8 bits	4n	63	56		
			Second	8 bits	4n+1 (p)	55	48		
			Third	8 bits	4n+2 (p)	47	40		
			Fourth	8 bits	4n+3 (p)	39	32		
			Fifth	8 bits	4n+4	31	24		
			Sixth	8 bits	4n+5 (p)	23	16		
			Seventh	8 bits	4n+6 (p)	15	8		
			Eighth	8 bits	4n+7 (p)	7	0		

(p): Page access (only when page access is enabled with the PRENB and PWENB bits in CSnMOD)

Figure 16.11 Data Alignment (Big Endian) in 8-Bit Bus Space

16.4.2 Data Alignment Control for SDRAM Area

(1) 32-Bit Bus Space

When a 32-bit width is selected for a bus space by the BSIZE[1:0] bits in SDCCR, the address buses A26 to A2 are enabled to output address signals in units of 32 bits, and the address buses A1 and A0 are disabled (always output the low level).

The external data is accessed using the D31 to D24, D23 to D16, D15 to D8, and D7 to D0 pins. Data widths that can be handled in a single cycle of access are 8, 16 and 32 bits. 64-bit access is executed as two cycles of 32-bit access. The valid byte position is indicated by DQM0 to DQM3 signals.

In 32-bit bus space, the valid positions of data external to the chip and of SDRAM control signals (DQM0 to DQM3) differ according to whether the endian is big or little. Figure 16.12 and Figure 16.13 show data alignment control when the endian is little and big, respectively.

In 32-bit bus space, consecutive access can occur in access to data in 8-, 16-, or 32-bit units. Specifically, consecutive access can occur when a single round of bus access is generated in response to a single transfer request. The situations in which consecutive access occurs are indicated by the letter “(r1)” in Figure 16.12 and Figure 16.13. Figure 16.18 shows a consecutive access example.

Data Size	Access Address	Access Address	Bus Cycle	Unit of Data	Address	Data Bus																	
						DQM3				DQM2				DQM1				DQM0					
						WE#																	
						D31	D24	D23	D16				D15	D8				D7	D0				
8 bits	4n	One	First	8 bits	4n (r1)	7 0																	
	4n+1	One	First	8 bits	4n (r1)	7 0																	
	4n+2	One	First	8 bits	4n (r1)	7 0																	
	4n+3	One	First	8 bits	4n (r1)	7 0																	
16 bits	4n	One	First	16 bits	4n (r1)	15 8 7 0																	
	4n+1	Two	First	8 bits	4n	7 0																	
			Second	8 bits	4n	15 8																	
	4n+2	One	First	16 bits	4n (r1)	15 8 7 0																	
4n+3	Two	First	8 bits	4n	7 0																		
32 bits	4n	One	First	32 bits	4n (r1)	31 24 23 16 15 8 7 0																	
			4n+1	Three	First	8 bits	4n	7 0															
					Second	16 bits	4n	23 16 15 8															
	Third	8 bits	4n+4	31 24																			
	4n+2	Two	First	16 bits	4n	15 8 7 0																	
			Second	16 bits	4n+4	31 24 23 16																	
	4n+3	Three	First	8 bits	4n	7 0																	
			Second	16 bits	4n+4	23 16 15 8																	
Third			8 bits	4n+4	31 24																		
64 bits	4n	Two	First	32 bits	4n	31 24 23 16 15 8 7 0																	
			Second	32 bits	4n+4	63 56 55 48 47 40 39 32																	

(r1): Consecutive access (only when consecutive access is enabled by BE = 1 in SDAMOD during the EXDMAC block or cluster transfer in single address mode)

Figure 16.12 Data Alignment (Little Endian) in 32-Bit Bus Space

(r1): Consecutive access (only when consecutive access is enabled by BE = 1 in SDAMOD during the EXDMAC block or cluster transfer in single address mode)

Figure 16.13 Data Alignment (Big Endian) in 32-Bit Bus Space

(2) 16-Bit Bus Space

When a 16-bit width is selected for a bus space by the BSIZE[1:0] bits in SDCCR, the address buses A26 to A1 are enabled to output address signals in units of 16 bits, and the address buses A0 is disabled (always output the low level). The valid byte position is indicated by DQM0 and DQM1 signals. DQM2 and DQM3 signals are not used.

In 16-bit bus space, the external data is accessed using the D15 to D8 and D7 to D0 pins and DQM0 and DQM1 control signals. Data widths that can be handled in a single cycle of access are 8 and 16 bits. 32-bit access is executed as two cycles of 16-bit access or three cycles in order of 8-, 16-, and 8-bit access (depending on the alignment of the address), and 64-bit access is executed as four cycles of 16-bit access.

In 16-bit bus space, the valid positions of data external to the chip and of control signals differ according to whether the endian is big or little. Figure 16.14 and Figure 16.15 show data alignment control when the endian is little and big, respectively.

In 16-bit bus space, consecutive access can occur in access to data in 8- or 16-bit units. Specifically, consecutive access can occur when a single round of bus access is generated in response to a single transfer request. The situations in which consecutive access occurs are indicated by the letter “(r1)” in Figure 16.14 and Figure 16.15. Figure 16.18 shows a consecutive access example.

Data Size	Access Address	Number of Access	Bus Cycle	Unit of Data	Address	(r1)	Data Bus			
							D15	D8	D7	D0
8 bits	4n	One	First	8 bits	4n	(r1)	7	0		
	4n+1	One	First	8 bits	4n	(r1)	7	0		
	4n+2	One	First	8 bits	4n+2	(r1)	7	0		
	4n+3	One	First	8 bits	4n+2	(r1)	7	0		
16 bit	4n	One	First	16 bits	4n	(r1)	15	8	7	0
	4n+1	Two	First	8 bits	4n		7	0		
			Second	8 bits	4n+2			15	8	
	4n+2	One	First	16 bits	4n+2	(r1)	15	8	7	0
	4n+3	Two	First	8 bits	4n+2		7	0		
			Second	8 bits	4n+4			15	8	
32 bits	4n	Two	First	16 bits	4n		15	8	7	0
			Second	16 bits	4n+2		31	24	23	16
	4n+1	Three	First	8 bits	4n		7	0		
			Second	16 bits	4n+2		23	16	15	8
			Third	8 bits	4n+4			31	24	
	4n+2	Two	First	16 bits	4n+2		15	8	7	0
			Second	16 bits	4n+4		31	24	23	16
	4n+3	Three	First	8 bits	4n+2		7	0		
			Second	16 bits	4n+4		23	16	15	8
64 bits			Third	8 bits	4n+6			31	24	
	4n	Four	First	16 bits	4n		15	8	7	0
			Second	16 bits	4n+2		31	24	23	16
			Third	16 bits	4n+4		47	40	39	32
			Fourth	16 bits	4n+6		63	56	55	48

(r1): Consecutive access (only when consecutive access is enabled by BE = 1 in SDAMOD during the EXDMAC block or cluster transfer in single address mode)

Figure 16.14 Data Alignment (Little Endian) in 16-Bit Bus Space

Data Size	Access Address	Number of Access	Bus Cycle	Unit of Data	Address		DQM1 DQM0 WE# Data Bus D15 D8 D7 D0			
8 bits	4n	One	First	8 bits	4n	(r1)	[7 0]			
	4n+1	One	First	8 bits	4n	(r1)	[7 0]			
	4n+2	One	First	8 bits	4n+2	(r1)	[7 0]			
	4n+3	One	First	8 bits	4n+2	(r1)	[7 0]			
16 bits	4n	One	First	16 bits	4n	(r1)	[15 8 7 0]			
	4n+1	Two	First	8 bits	4n		[15 8]			
			Second	8 bits	4n+2		[7 0]			
	4n+2	One	First	16 bits	4n+2	(r1)	[15 8 7 0]			
	4n+3	Two	First	8 bits	4n+2		[15 8]			
			Second	8 bits	4n+4		[7 0]			
32 bits	4n	Two	First	16 bits	4n		[31 24 23 16]			
			Second	16 bits	4n+2		[15 8 7 0]			
	4n+1	Three	First	8 bits	4n		[31 24]			
			Second	16 bits	4n+2		[23 16 15 8]			
			Third	8 bits	4n+4		[7 0]			
	4n+2	Two	First	16 bits	4n+2		[31 24 23 16]			
			Second	16 bits	4n+4		[15 8 7 0]			
	4n+3	Three	First	8 bits	4n+2		[31 24]			
			Second	16 bits	4n+4		[23 16 15 8]			
			Third	8 bits	4n+6		[7 0]			
64 bits	4n	Four	First	16 bits	4n		[63 56 55 48]			
			Second	16 bits	4n+2		[47 40 39 32]			
			Third	16 bits	4n+4		[31 24 23 16]			
			Fourth	16 bits	4n+6		[15 8 7 0]			

(r1): Consecutive access (only when consecutive access is enabled by BE = 1 in SDAMOD during the EXDMAC block or cluster transfer in single address mode)

Figure 16.15 Data Alignment (Big Endian) in 16-Bit Bus Space

(3) 8-Bit Bus Space

When an 8-bit width is selected for a bus space by the BSIZE[1:0] bits in SDCCR, the address buses A26 to A0 are enabled to output address signals in units of 8 bits.

In 8-bit bus space, the external data is accessed using the D7 to D0 pins and DQM0 control signal. Eight-bit data can be accessed at a time; 16-bit data is accessed with two 8-bit accesses, 32-bit data is accessed with four 8-bit accesses, and 64-bit data is accessed with eight 8-bit accesses.

Figure 16.16 and Figure 16.17 show data alignment control when the endian is little and big, respectively.

In 8-bit bus space, consecutive access can occur in access to data in 8-bit units. Specifically, consecutive access can occur when a single round of bus access is generated in response to a single transfer request. The situations in which consecutive access occurs are indicated by the letter “(r1)” in Figure 16.16 and Figure 16.17. Figure 16.18 shows a consecutive access example.

Data Size	Access Address	Number of Access	Bus Cycle	Unit of Data	Address	Data Bus			
						D15	D8	D7	D0
8 bits	4n	One	First	8 bits	4n (r1)	7		0	
	4n+1	One	First	8 bits	4n+1 (r1)	7		0	
	4n+2	One	First	8 bits	4n+2 (r1)	7		0	
	4n+3	One	First	8 bits	4n+3 (r1)	7		0	
16 bits	4n	Two	First	8 bits	4n	7		0	
			Second	8 bits	4n+1	15		8	
	4n+1	Two	First	8 bits	4n+1	7		0	
			Second	8 bits	4n+2	15		8	
	4n+2	Two	First	8 bits	4n+2	7		0	
			Second	8 bits	4n+3	15		8	
	4n+3	Two	First	8 bits	4n+3	7		0	
			Second	8 bits	4n+4	15		8	
32 bits	4n	Four	First	8 bits	4n	7		0	
			Second	8 bits	4n+1	15		8	
			Third	8 bits	4n+2	23		16	
			Fourth	8 bits	4n+3	31		24	
	4n+1	Four	First	8 bits	4n+1	7		0	
			Second	8 bits	4n+2	15		8	
			Third	8 bits	4n+3	23		16	
			Fourth	8 bits	4n+4	31		24	
	4n+2	Four	First	8 bits	4n+2	7		0	
			Second	8 bits	4n+3	15		8	
			Third	8 bits	4n+4	23		16	
			Fourth	8 bits	4n+5	31		24	
	4n+3	Four	First	8 bits	4n+3	7		0	
			Second	8 bits	4n+4	15		8	
			Third	8 bits	4n+5	23		16	
			Fourth	8 bits	4n+6	31		24	
64 bits	4n	Eight	First	8 bits	4n	7		0	
			Second	8 bits	4n+1	15		8	
			Third	8 bits	4n+2	23		16	
			Fourth	8 bits	4n+3	31		24	
			Fifth	8 bits	4n+4	39		32	
			Sixth	8 bits	4n+5	47		40	
			Seventh	8 bits	4n+6	55		48	
			Eighth	8 bits	4n+7	63		56	

(r1): Consecutive access (only when consecutive access is enabled by BE = 1 in SDAMOD during the EXDMAC block or cluster transfer in single address mode)

Figure 16.16 Data Alignment (Little Endian) in 8-Bit Bus Space

Data Size	Access Address	Number of Access	Bus Cycle	Unit of Data	Address	Data Bus			
						D15	D8	D7	D0
8 bits	4n	One	First	8 bits	4n (r1)	7		0	
	4n+1	One	First	8 bits	4n+1 (r1)	7		0	
	4n+2	One	First	8 bits	4n+2 (r1)	7		0	
	4n+3	One	First	8 bits	4n+3 (r1)	7		0	
16 bits	4n	Two	First	8 bits	4n	15		8	
			Second	8 bits	4n+1	7		0	
	4n+1	Two	First	8 bits	4n+1	15		8	
			Second	8 bits	4n+2	7		0	
	4n+2	Two	First	8 bits	4n+2	15		8	
			Second	8 bits	4n+3	7		0	
	4n+3	Two	First	8 bits	4n+3	15		8	
			Second	8 bits	4n+4	7		0	
32 bits	4n	Four	First	8 bits	4n	31		24	
			Second	8 bits	4n+1	23		16	
			Third	8 bits	4n+2	15		8	
			Fourth	8 bits	4n+3	7		0	
	4n+1	Four	First	8 bits	4n+1	31		24	
			Second	8 bits	4n+2	23		16	
			Third	8 bits	4n+3	15		8	
			Fourth	8 bits	4n+4	7		0	
	4n+2	Four	First	8 bits	4n+2	31		24	
			Second	8 bits	4n+3	23		16	
			Third	8 bits	4n+4	15		8	
			Fourth	8 bits	4n+5	7		0	
	4n+3	Four	First	8 bits	4n+3	31		24	
			Second	8 bits	4n+4	23		16	
			Third	8 bits	4n+5	15		8	
			Fourth	8 bits	4n+6	7		0	
64 bits	4n	Eight	First	8 bits	4n	63		56	
			Second	8 bits	4n+1	55		48	
			Third	8 bits	4n+2	47		40	
			Fourth	8 bits	4n+3	39		32	
			Fifth	8 bits	4n+4	31		24	
			Sixth	8 bits	4n+5	23		16	
			Seventh	8 bits	4n+6	15		8	
			Eighth	8 bits	4n+7	7		0	

(r1): Consecutive access (only when consecutive access is enabled by BE = 1 in SDAMOD during the EXDMAC block or cluster transfer in single address mode)

Figure 16.17 Data Alignment (Big Endian) in 8-Bit Bus Space

EXDMAC cluster transfer mode or single address mode with consecutive access mode enabled (SDAMOD.BE = 1)

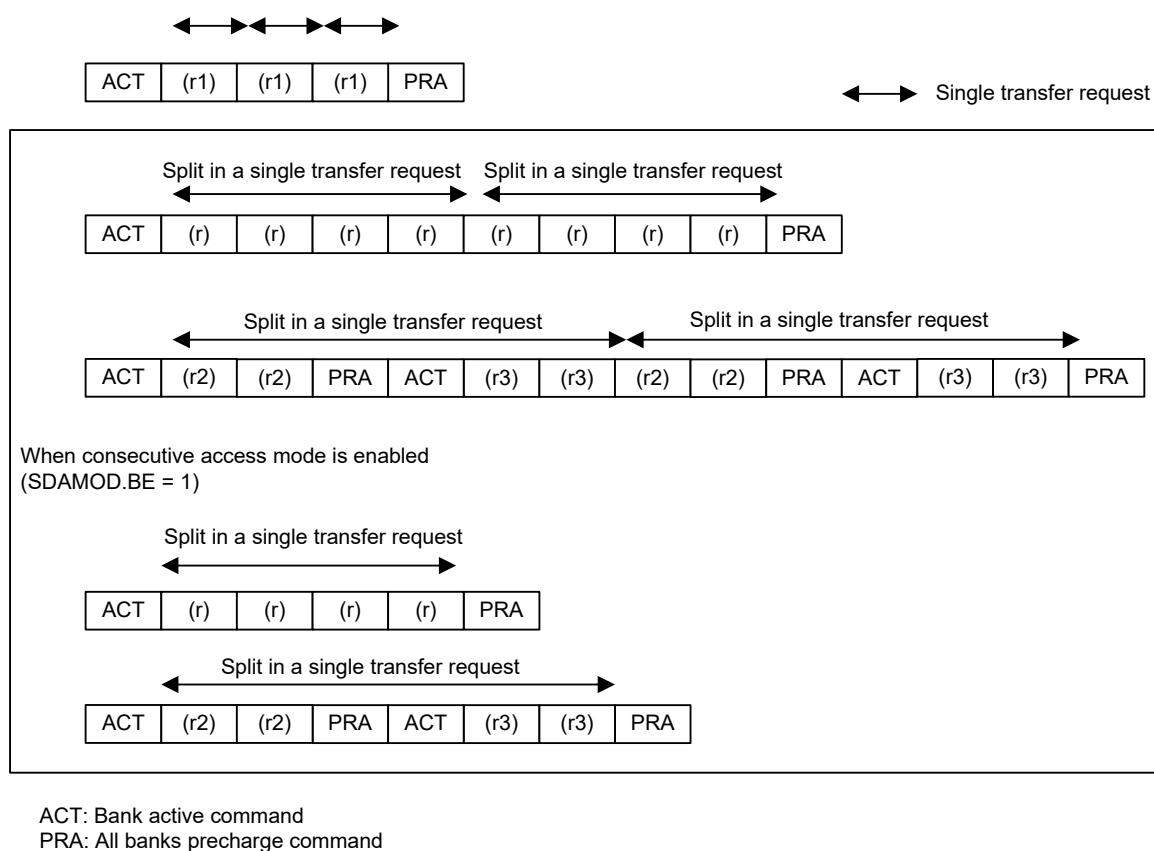


Figure 16.18 Consecutive Access Example

16.5 Operation of CS Area Controller

16.5.1 Separate Bus

The various periods in the timing charts are described below.

The CS area controller (CSC) operates in synchronization with the external bus clock (BCLK). The operation cycles, such as wait cycles specified with the CSC register, are counted on BCLK. In the following description, frequencies of BCLK and BCLK pin output are the same, unless otherwise noted.

Access via the external bus starts at the same point as the output of a rising edge on the BCLK pin. However, if the external bus clock (BCLK) and the output on the BCLK pin are at different frequencies so that a single request from a bus master for transfer leads to two or more rounds of access via the external bus, the wait settings may cause the start of access for the second and subsequent rounds to coincide with the falling edge of the output on the BCLK pin (refer to Figure 16.26 to Figure 16.30). If recovery cycles are inserted for bus access, the setting for the number of recovery cycles may also cause the start of access for the second and subsequent rounds to coincide with the falling edge of the output on the BCLK pin (refer to Figure 16.46).

(a) Tw1 to Twn (Clock Cycles of Waiting for a Normal Read Cycle or Normal Write Cycle)

The period Tw1 to Twn is made up of the number of clock cycles between the start of access via the external bus clock and one cycle before the strobe signal is valid. The number of cycles are selectable within the range from zero to 31. Within this period, the timing of CSn#, RD#, and WRn# assertion (placing the signals at the low level) is determined by the respective wait settings. Specifically, the periods of waiting are controlled by the CS assert wait select bits (CSON[2:0]), the RD assert wait select bits (RDON[2:0]), the WR assert wait select bits (WRON[2:0]), and the write data output wait select bits (WDON[2:0]) in CSn wait control register 2 (CSnWCR2). The number of clock cycles for each of these periods of waiting is selectable as a value from zero to seven counted from the start of external bus access. Selectable numbers of cycles are also within the overall number of clock cycles of waiting for reading or writing.

(b) Tend (Clock Cycle where the Strobe Signal is Valid)

Tend is the next clock cycle after completion of the period of waiting for a normal cycle of reading or writing or for a cycle of page reading or page writing. If each wait select bit for a normal cycle of reading or writing or for a cycle of page reading or page writing is zero, the clock cycle where bus access starts is the clock cycle where the strobe signal is valid. The RD# and WRn# signals are negated in the next clock cycle after the cycle where the strobe signal is valid. In the case of read access, the clock cycle where the strobe signal is valid becomes the clock cycle where the data to be read are sampled.

If an external wait is enabled, the wait signal is sampled at the time of the cycle where the strobe signal is valid. The bus cycle is extended if the wait signal is at the low level. The bus cycle is completed in the next clock cycle if the wait signal is at the high level. Tend indicates the cycle where sampling of the wait signal starts.

After the first cycle where the strobe signal is valid during page access, second and subsequent page access operations (point (e) below) start in the next cycle except in cases of write access where a setting (other than zero) for write-data output extension clock cycles (point (d) below) has been made. If the setting for the RD or WR assertion wait is a value other than zero, the RD# and WRn# signals are negated in the next clock cycle. If the setting is zero, assertion continues. Furthermore, the CSn# signal continues to be asserted rather than being negated.

(c) Tn1 to Tnm (Clock Cycles of CS Extension)

In the case of normal access, Tn1 to Tnm represent the clock cycles of the period following the cycle where the strobe signal is valid (Tend) up to negation of the CSn# signal. For read or write access, the timing of negation can be controlled by the read-access CS extension cycle select bits (CSROFF[2:0]) and the write-access CS extension cycle select bits (CSWOFF[2:0]) in the CSn wait control register 2 (CSnWCR2), respectively.

The number of cycles are counted from the cycle following the cycle where the strobe signal is valid.

In the case of page access, Tn1 to Tnm represent the clock cycles of the period for the cycle following the last cycle where the strobe signal is valid up to negation of the CSn# signal.

For write access, setting the write data output extension cycle select bits (WDOFF) controls extension of the period where the address and output data are valid.

(d) Tdw1 to Tdwn (Write-Data Output Extension Clock Cycles)

For write access, if the setting for write-data output extension wait is a value other than zero, clock cycles of write-data output extension are inserted from the cycle that follows the cycle where the strobe signal is valid (Tend).

In the case of normal access, this is inserted within the period of clock cycles of CS extension (point (c) above).

In the case of page access, this is inserted within the period of the cycle where the strobe signal is valid and subsequent page access or within the period of clock cycles of CS extension (point (c) above). Valid address and data output are extended over this period, and the WRn# signal is negated.

(e) Tpw1 to TpwN (Page-Read Cycle Wait or Page-Write Cycle Wait)

For the second and subsequent bus cycles during page access, the values for a page-read cycle wait or page-write cycle wait are used instead of the settings for a normal read or write cycle wait. Setting the WR assert wait select bits becomes enabled in the same way as for the first round of access. How the setting for RD assertion controls operation depends on the setting for page-read access mode (the PRMOD bit in CSnMOD) as described below.

CSnMOD.PRMOD = 0: A wait until RD assertion is inserted in the same way as for the first round of access, and the RD# signal is negated.

CSnMOD.PRMOD = 1: Although a wait until RD assertion is inserted in the same way as for normal-access compatibility mode, the RD# signal continues to be asserted over this period.

(f) Tr1 to Trn (Recovery Cycles)

Recovery cycles can be inserted from the point where a bus cycle is completed (CSn# signal negation). The number of recovery cycles can be controlled by the setting of the read recovery (RRCV) or write recovery (WRCV) bits in the CSn recovery cycle register (CSnREC). Both numbers of recovery cycles are counted from the end of a bus cycle (CSn# negation) and can be selected from 0 to 15 cycles. For details on recovery cycles, refer to section 16.5.4, Insertion of Recovery Cycles.

(1) Normal Access

When the PRENB and PWENB bits in CSnMOD are set to 0 to disable page-read and page-write access, respectively, all bus accesses will take the form of normal read and write operations.

Even when the PRENB and PWENB bits in CSnMOD are set to 1 to enable page-read and page-write access, respectively, bus access other than page access will take the form of normal read and write operations.

Figure 16.19 to Figure 16.21 show the normal access operations.

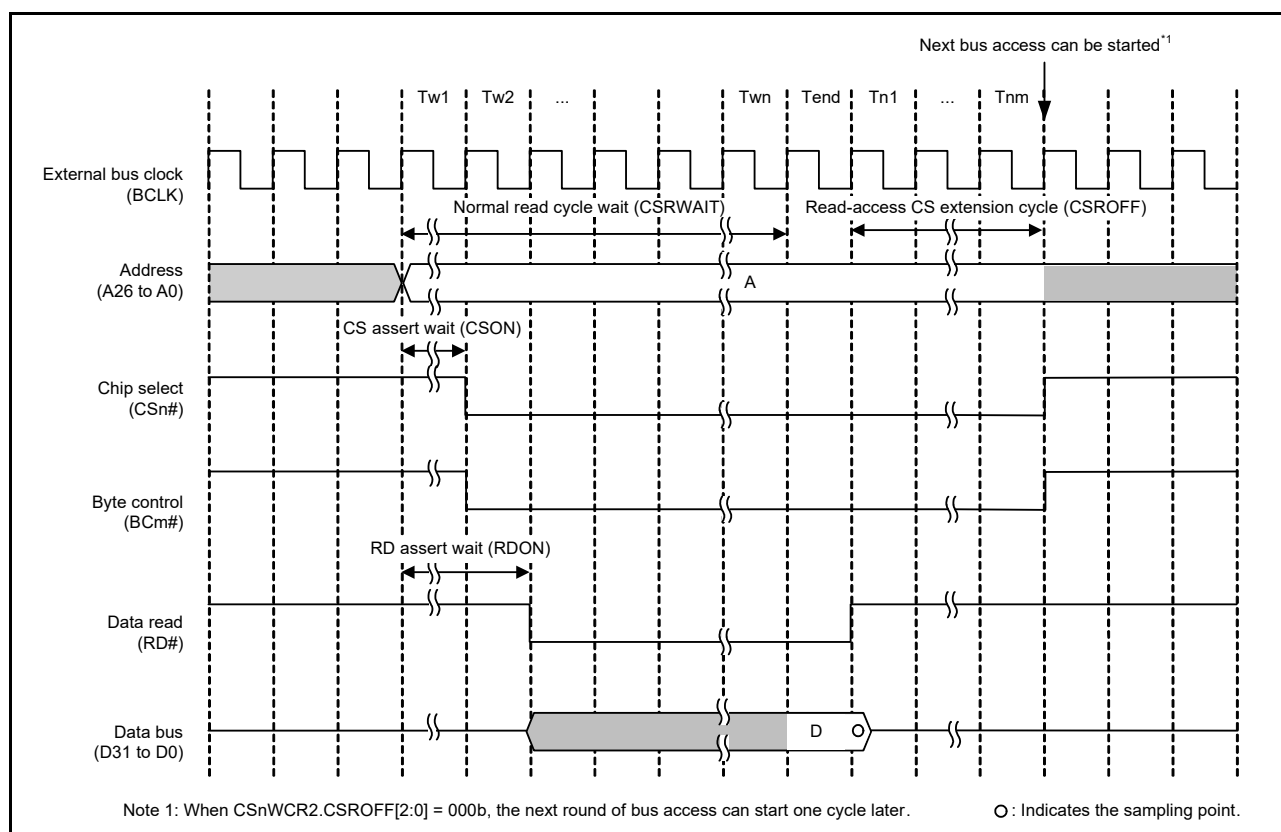


Figure 16.19 Bus Timing (Normal-Read Operation) (n = 0 to 7, m = 0 to 3)

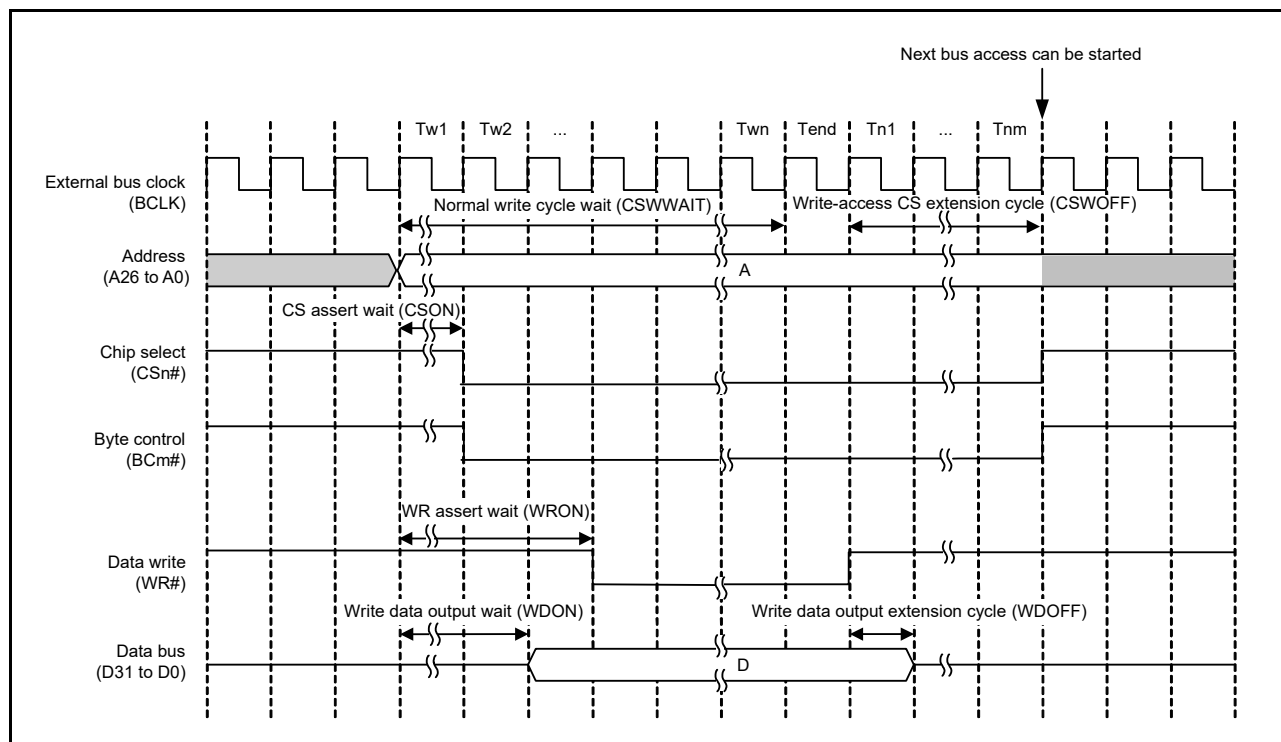


Figure 16.20 Bus Timing (Normal-Write Operation, Single Write Strobe Mode) (n = 0 to 7, m = 0 to 3)

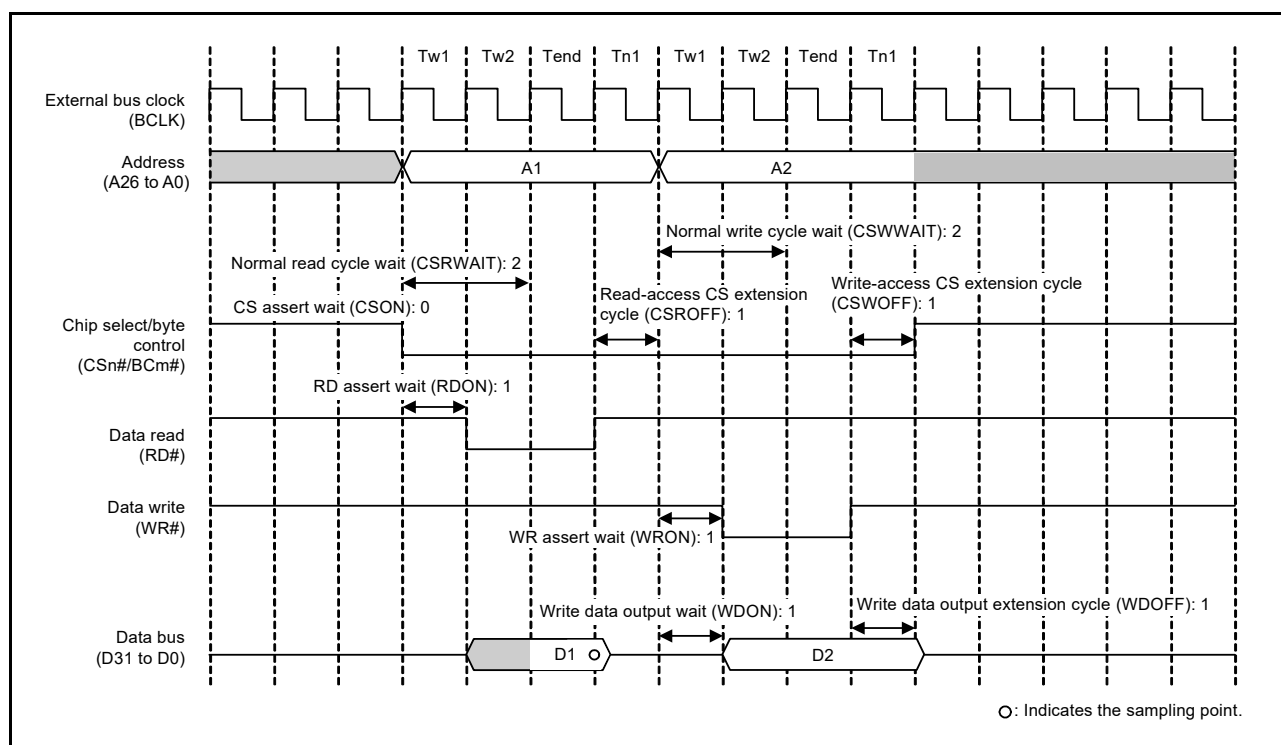


Figure 16.21 Example of Normal Access Operation (Read/Write) ($n = 0$ to 7, $m = 0$ to 3)

When two or more rounds of external bus access are required in response to a single request for transfer from a bus master, normal access operations (steps (a) to (d) above) are repeated. Figure 16.22 and Figure 16.23 show examples of operations when two rounds of bus access are generated in response to a single request for transfer. If the recovery cycle insertion condition is satisfied, recovery cycles (step (f) above) are also inserted in the second and subsequent external bus accesses (refer to Figure 16.44).

The values of the wait control registers are example settings. In practice, set the register bits according to the specifications of connected devices.

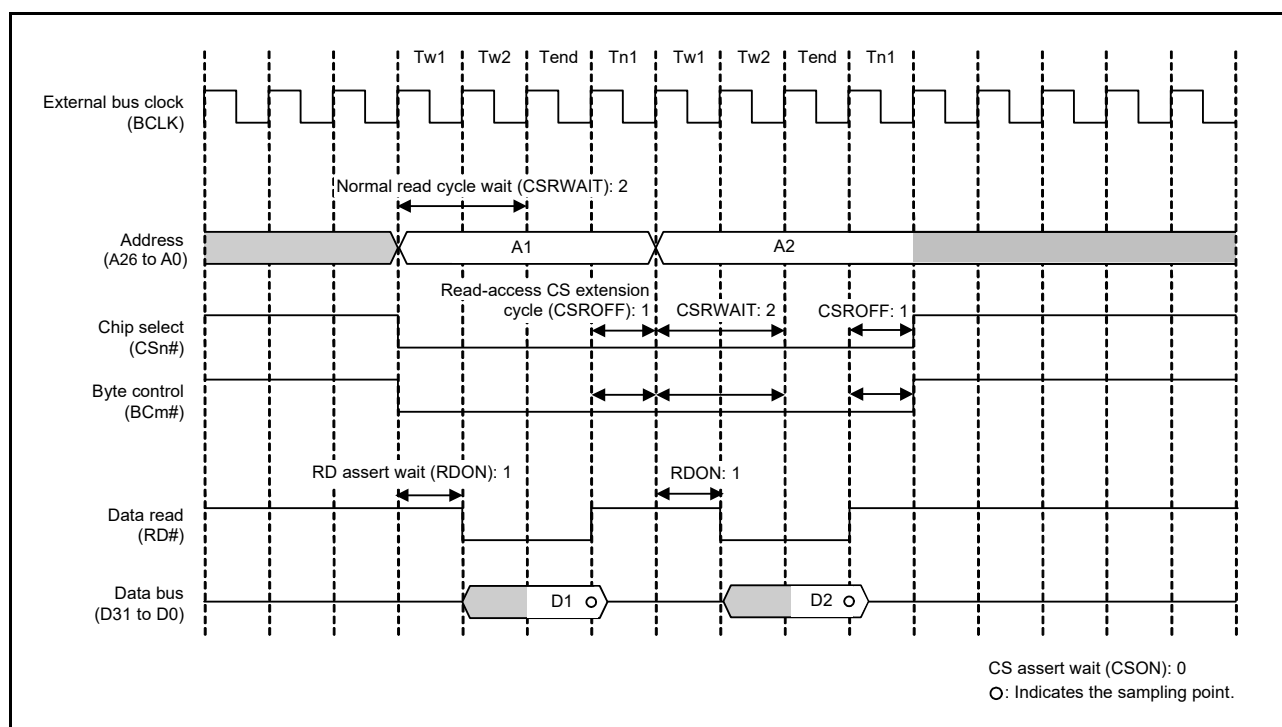


Figure 16.22 Example of Normal-Read Operation
(when Two Rounds of Bus Access are Generated in Response to a Single Request for Transfer)
($n = 0$ to 7, $m = 0$ to 3)

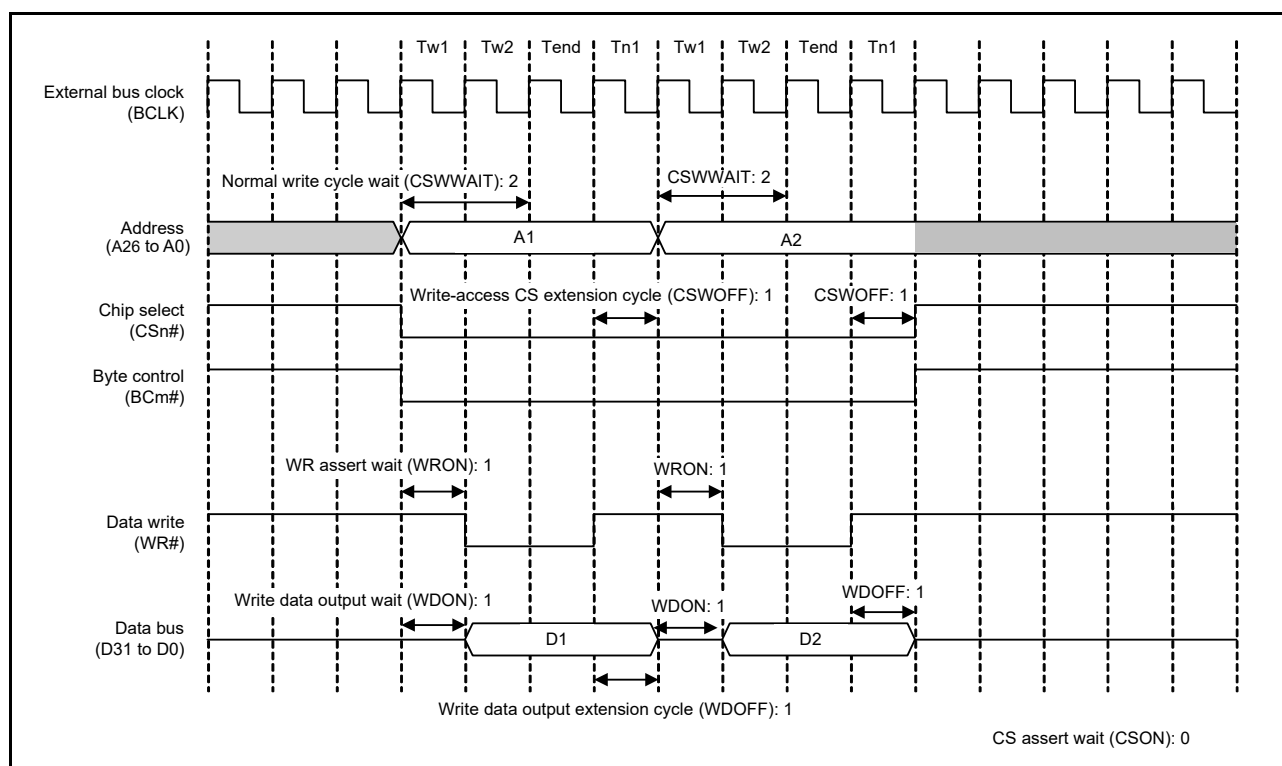


Figure 16.23 Example of Normal-Write Operation
(when Two Rounds of Bus Access are Generated in Response to a Single Request for Transfer, in Single Write Strobe Mode) ($n = 0$ to 7, $m = 0$ to 3)

Figure 16.24 and Figure 16.25 show examples of normal read and write accesses to a 32-bit bus space in 16 bits.

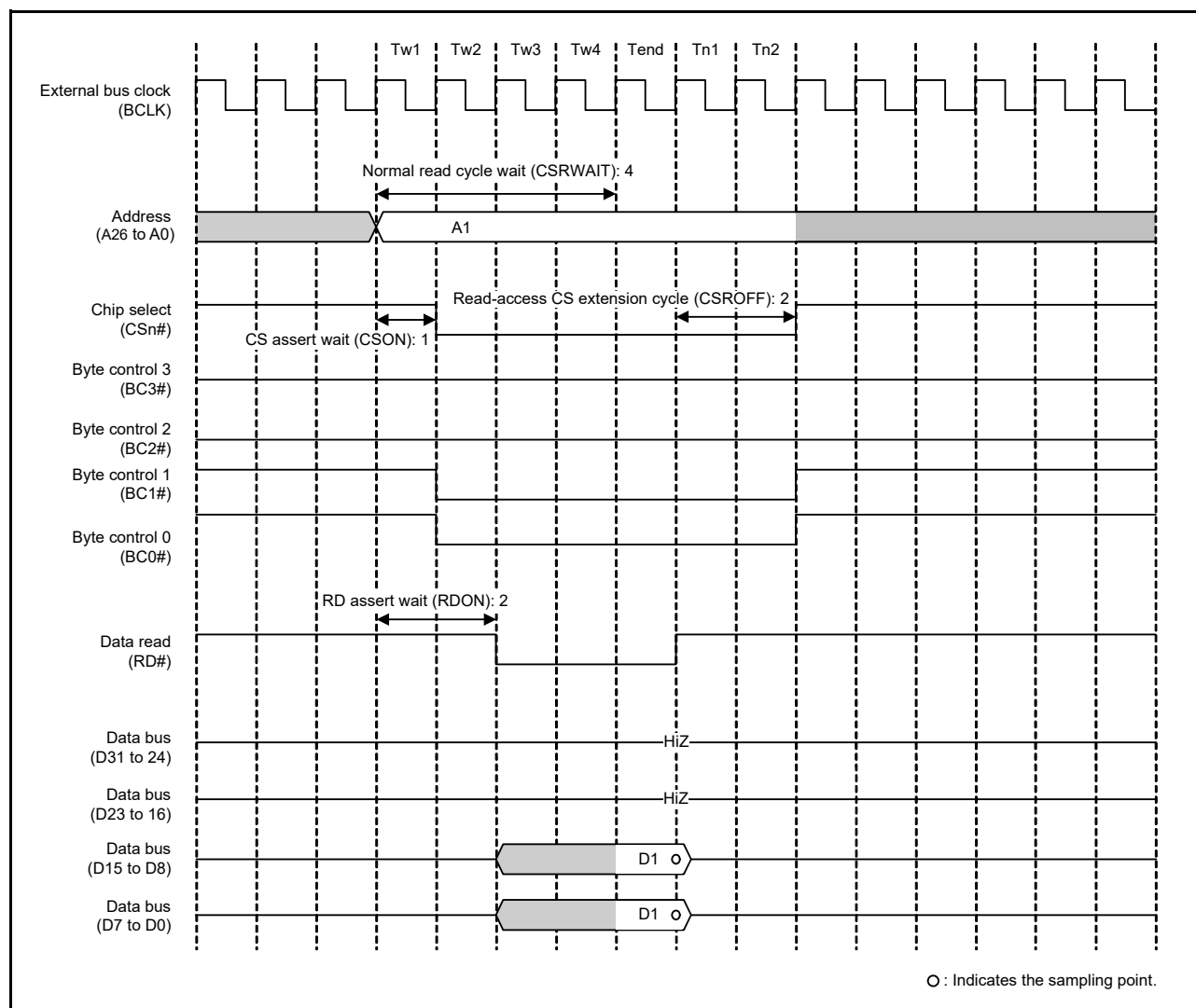


Figure 16.24 Example of Normal-Read Operation (when 32-Bit Bus Space is Accessed in 16 Bits) (n = 0 to 7)

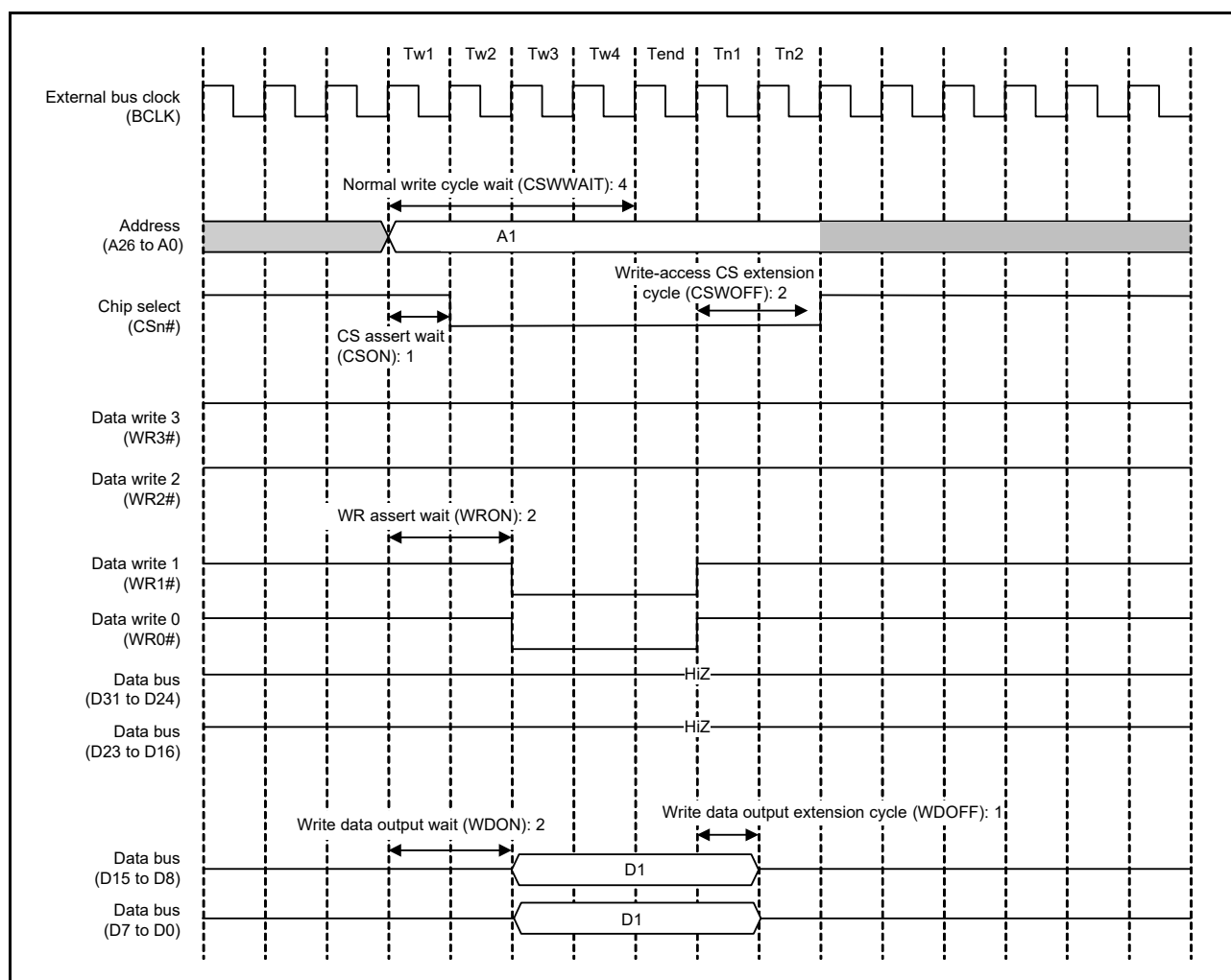


Figure 16.25 Example of Normal-Write Operation
(when 32-Bit Bus Space is Accessed in 16 Bits, in Byte Strobe Mode) ($n = 0$ to 7)

Figure 16.26 to Figure 16.30 show examples of normal accesses made with the 1/2 BCLK selected with the BCLK pin output select bit.

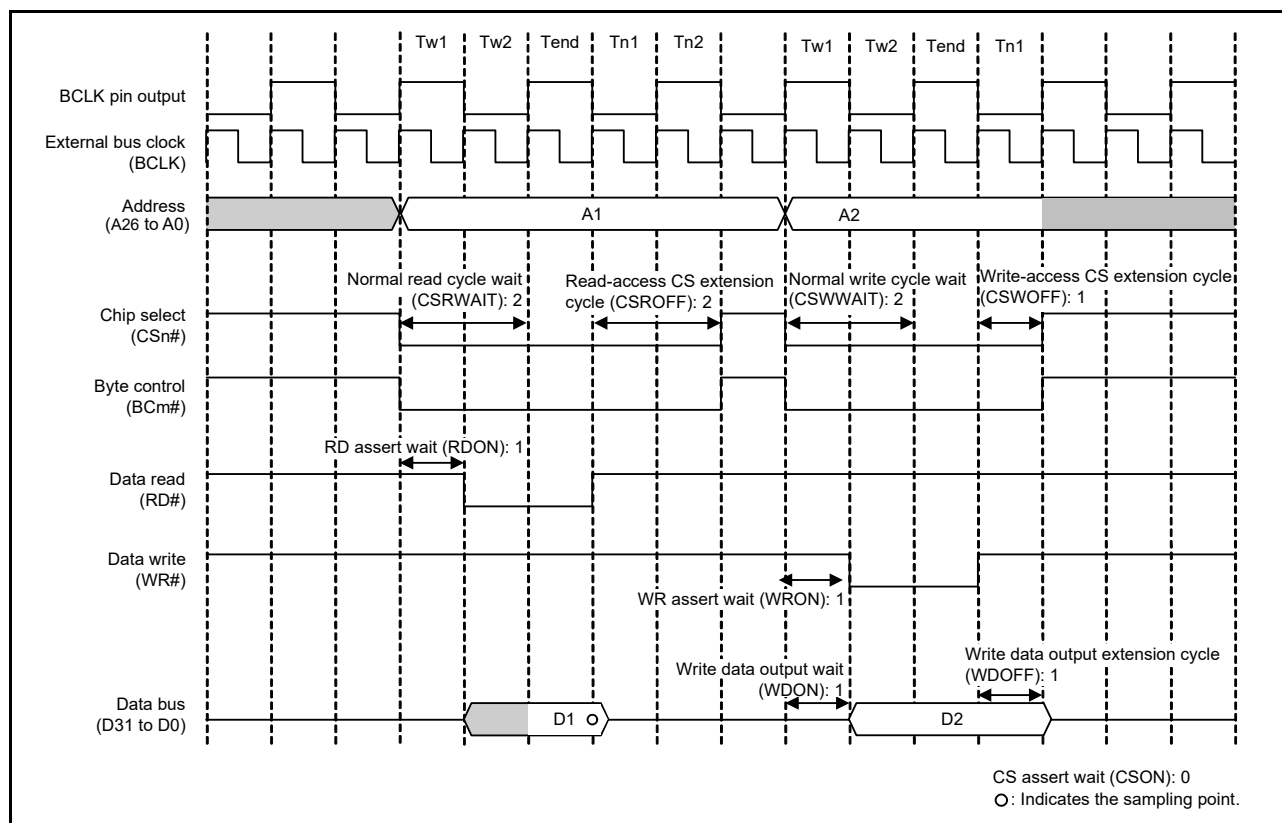


Figure 16.26 Example of Normal Access
(when 1/2 BCLK is Selected with the BCLK Pin Output Select Bit) (n = 0 to 7, m = 0 to 3)

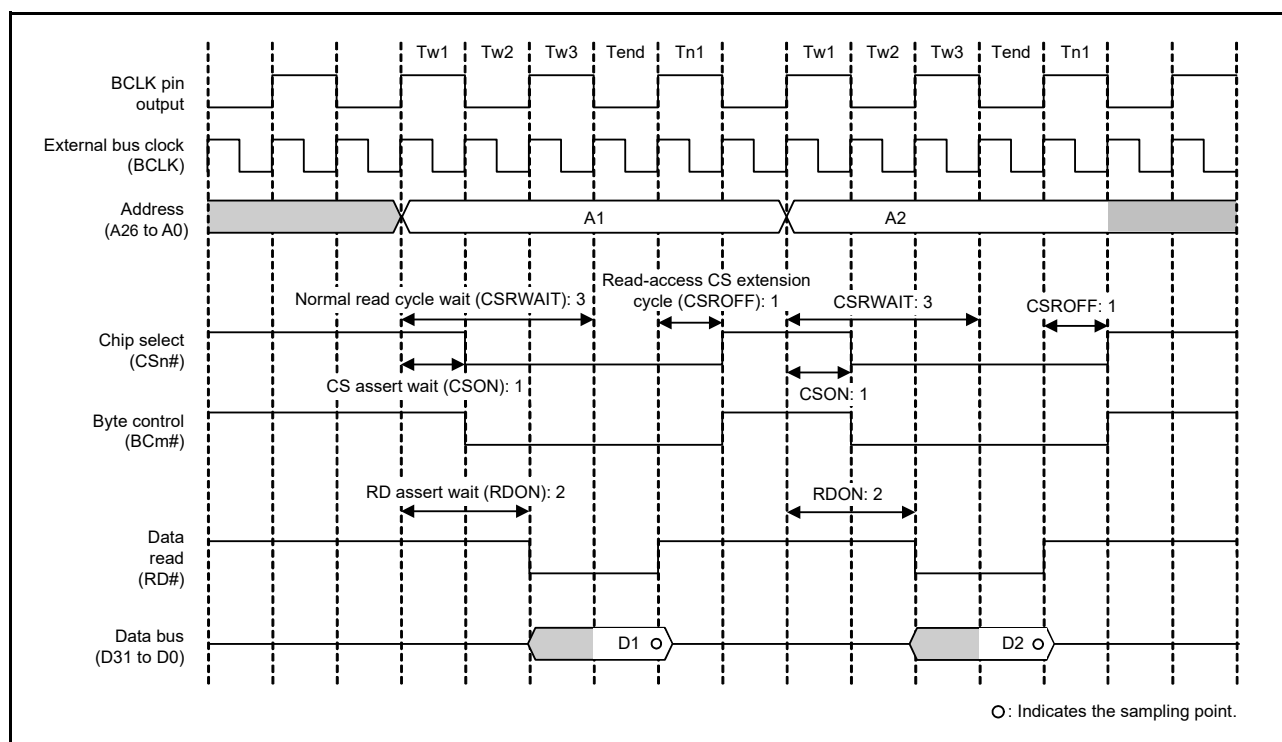


Figure 16.27 Example of Normal-Read Operation
(when 1/2 BCLK is Selected with the BCLK Pin Output Select Bit) (n = 0 to 7, m = 0 to 3)

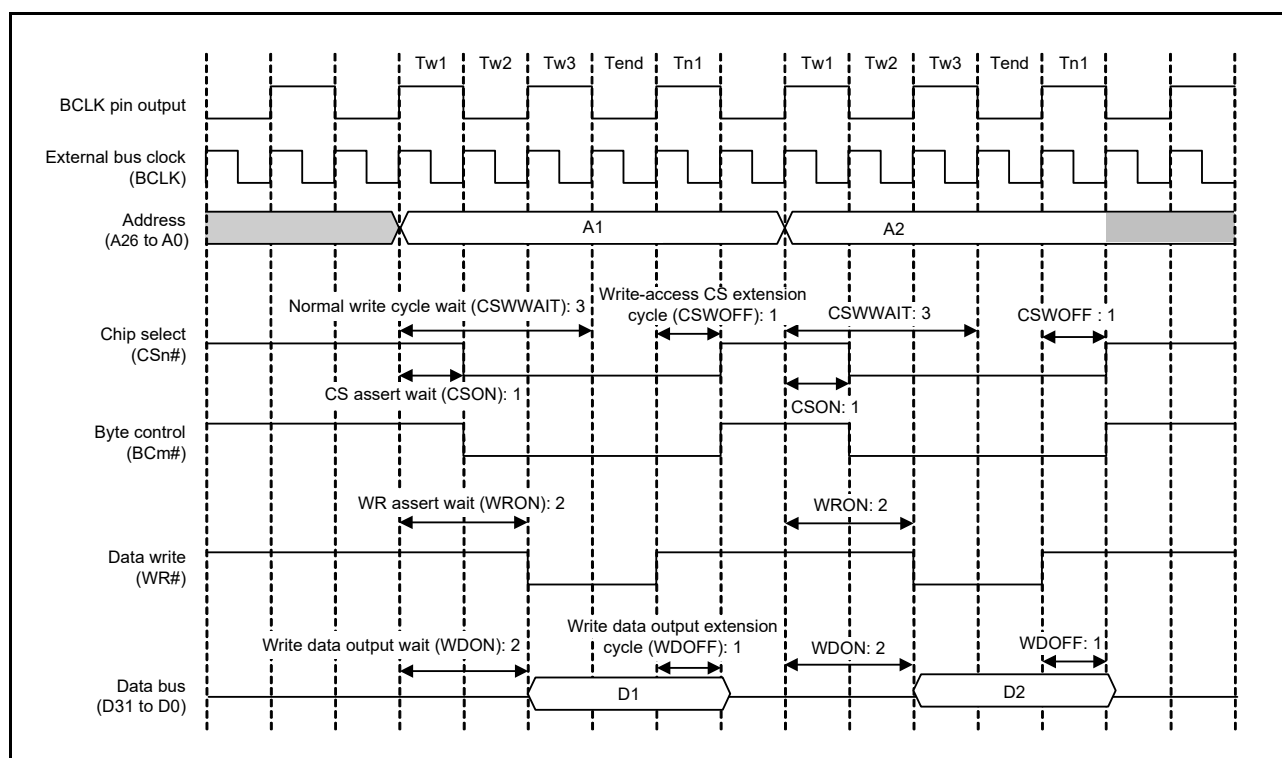


Figure 16.28 Example of Normal-Write Operation
(when 1/2 BCLK is Selected with the BCLK Pin Output Select Bit) ($n = 0$ to 7, $m = 0$ to 3)

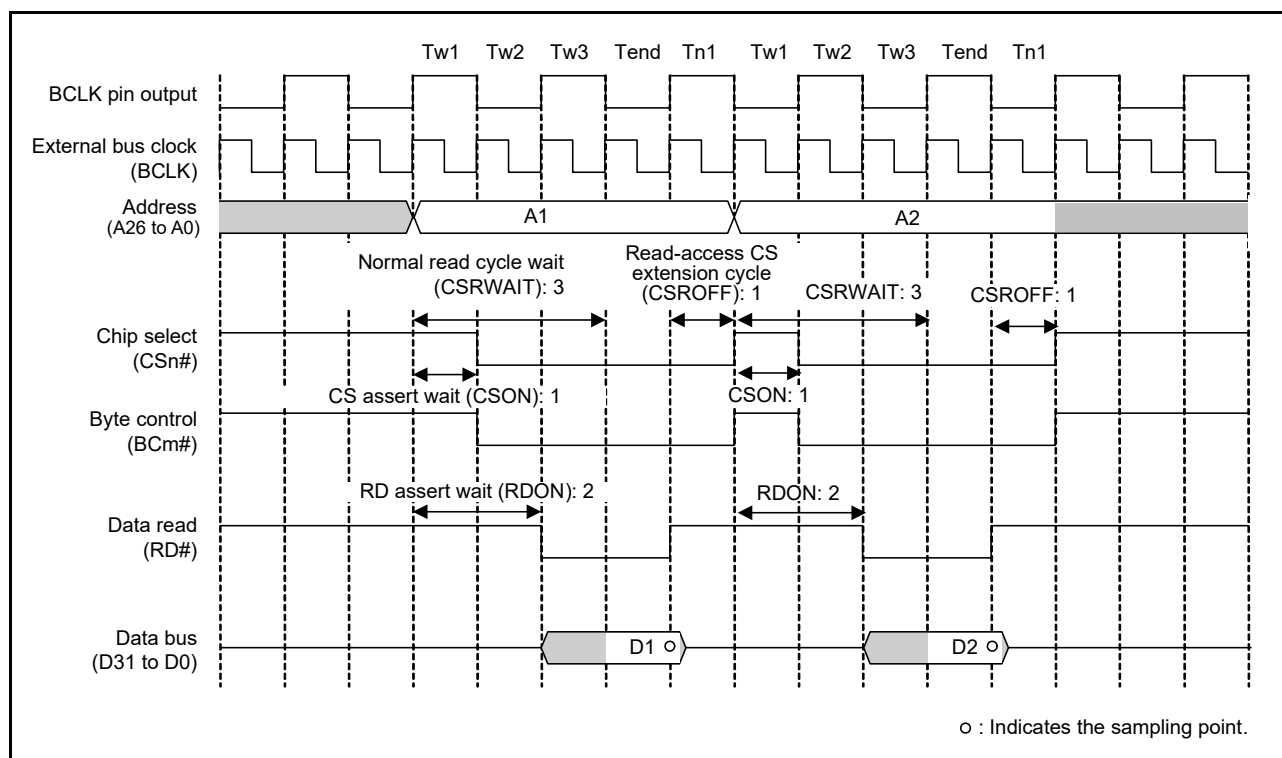


Figure 16.29 Example of Normal-Read Operation
(when 1/2 BCLK is Selected with the BCLK Pin Output Select Bit and Two Rounds of Bus Access are Generated in Response to a Single Request for Transfer) ($n = 0$ to 7, $m = 0$ to 3)

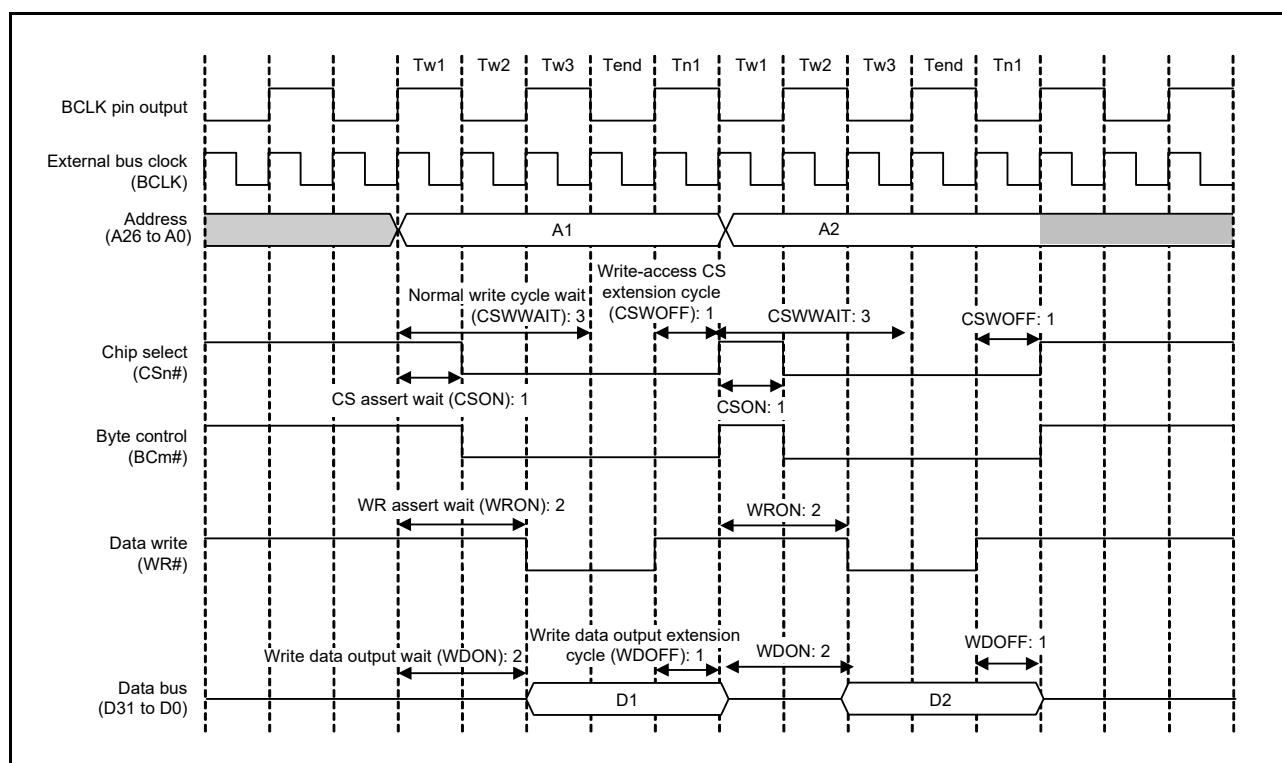


Figure 16.30 Example of Normal-Write Operation
 (when 1/2 BCLK is Selected with the BCLK Pin Output Select Bit and Two Rounds of Bus Access are Generated in Response to a Single Request for Transfer) (n = 0 to 7, m = 0 to 3)

(2) Page Access

When the PRENB and PWENB bits in CSnMOD are set to 1 to enable page-read and page-write access, respectively, the bus access for page access operations becomes page reading and writing. Specifically, page access can occur when two or more rounds of external bus access are required for a single transfer request from the bus master. However, normal access is made when the split accesses are not aligned or the access spreads over the 32-bit boundary. Refer to Figure 16.8 to Figure 16.11 for the conditions under which page access occurs.

Figure 16.31 and Figure 16.32 show examples of page access operations.

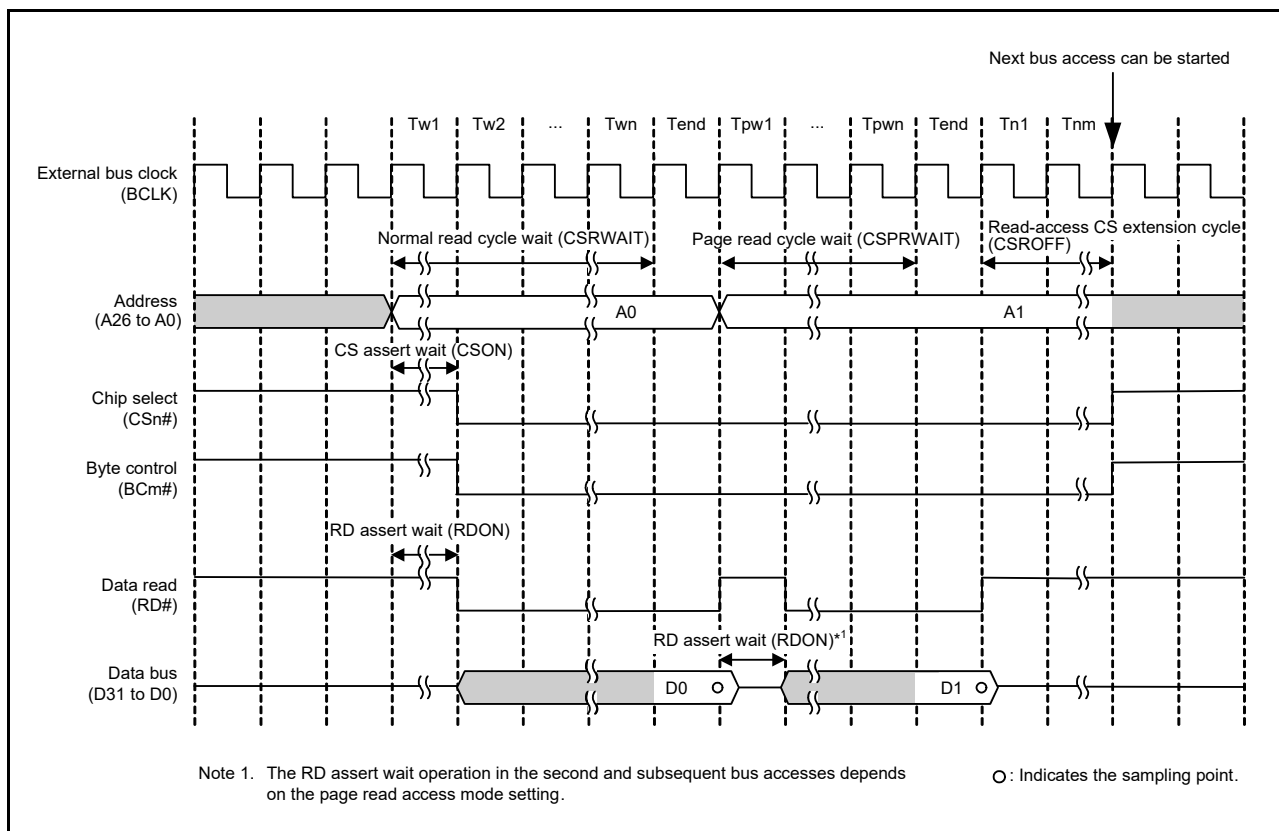


Figure 16.31 Page-Read Access Timing (n = 0 to 7, m = 0 to 3)

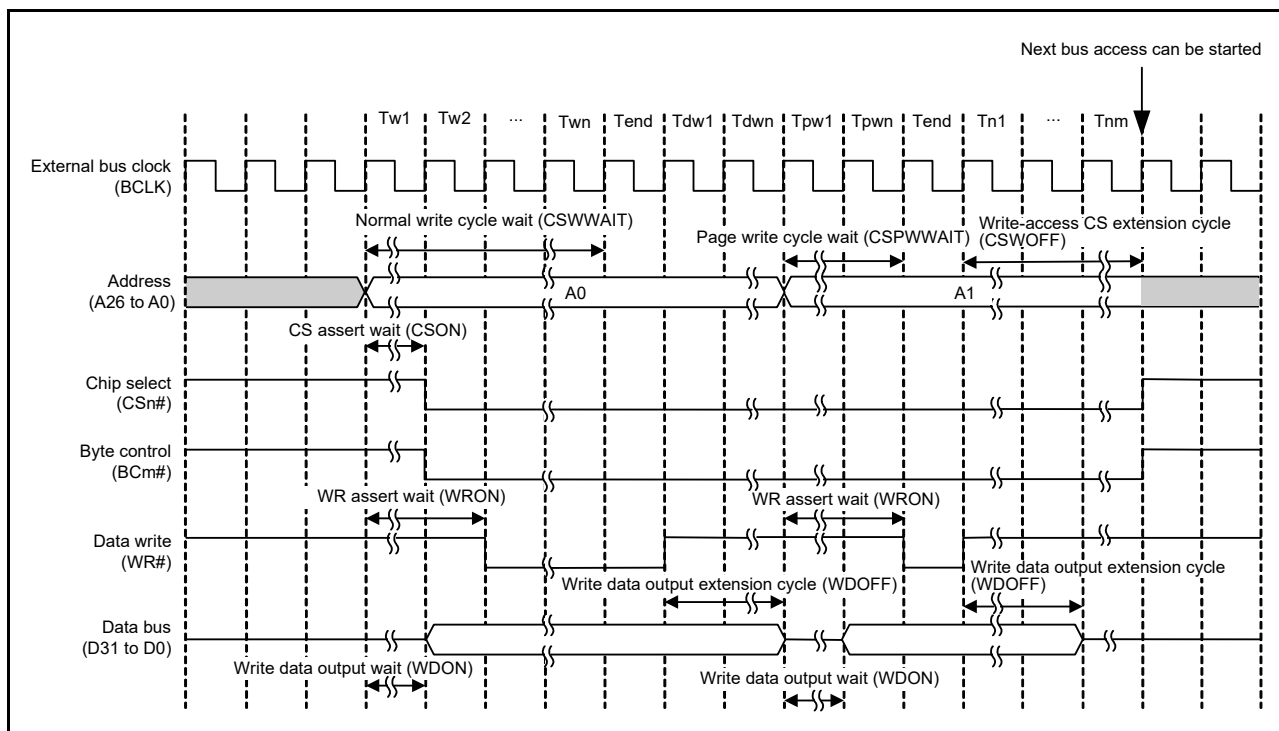


Figure 16.32 Page-Write Access Timing (n = 0 to 7, m = 0 to 3)

Figure 16.33 and Figure 16.34 show examples of operations for access to a 16-bit bus space in 32 bits. The values of the wait control registers are example settings. In practice, the register settings will correspond to the specifications of connected devices.

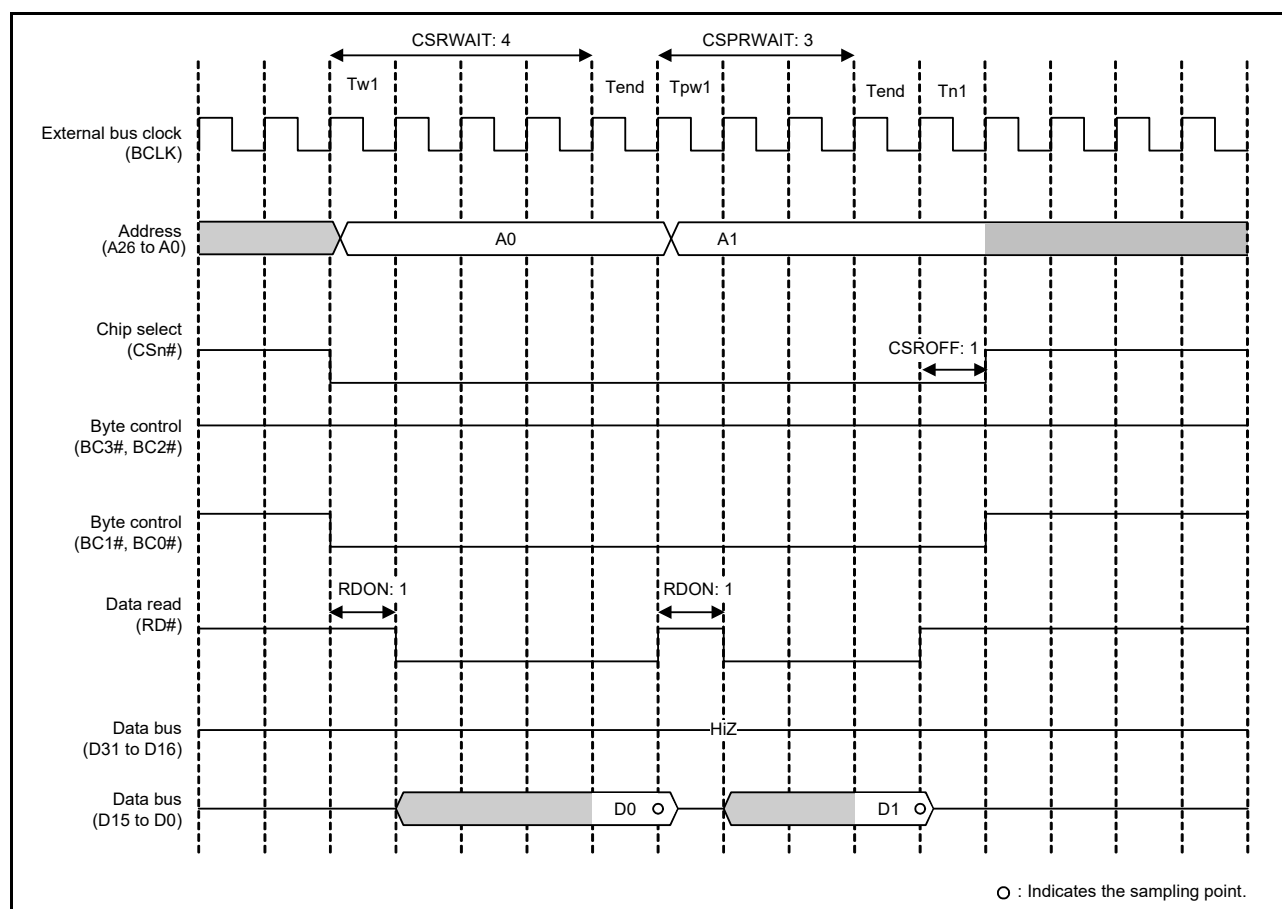


Figure 16.33 Example of Page-Read Access Operation
(when 16-Bit Bus Space is Accessed in 32 Bits) (n = 0 to 7)

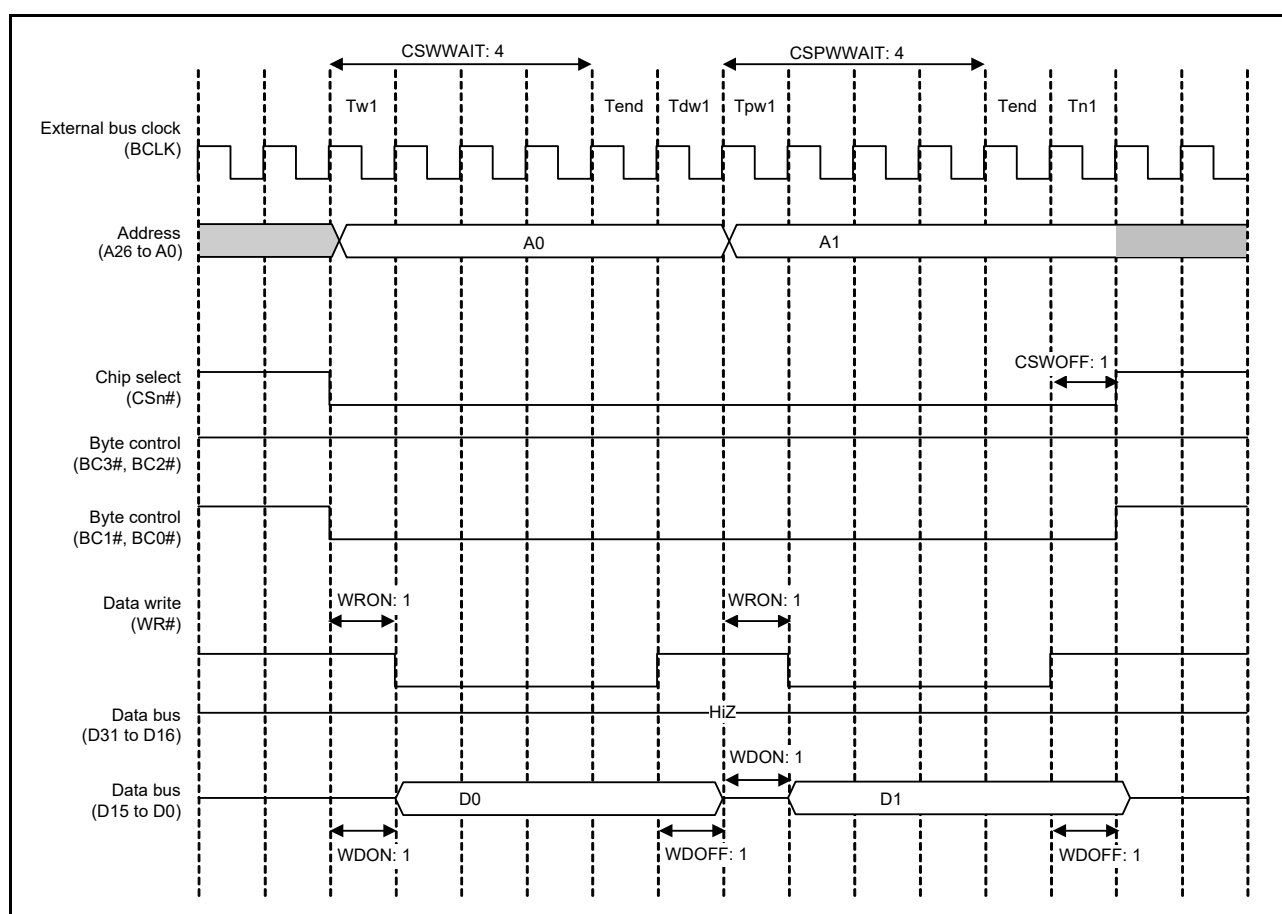


Figure 16.34 Example of Page-Write Access Operation
(when 16-Bit Bus Space is Accessed in 32 Bits, in Single Write Strobe Mode) (n = 0 to 7)

Figure 16.35 and Figure 16.36 show examples of page access operations performed with the 1/2 BCLK selected with the BCLK pin output select bit.

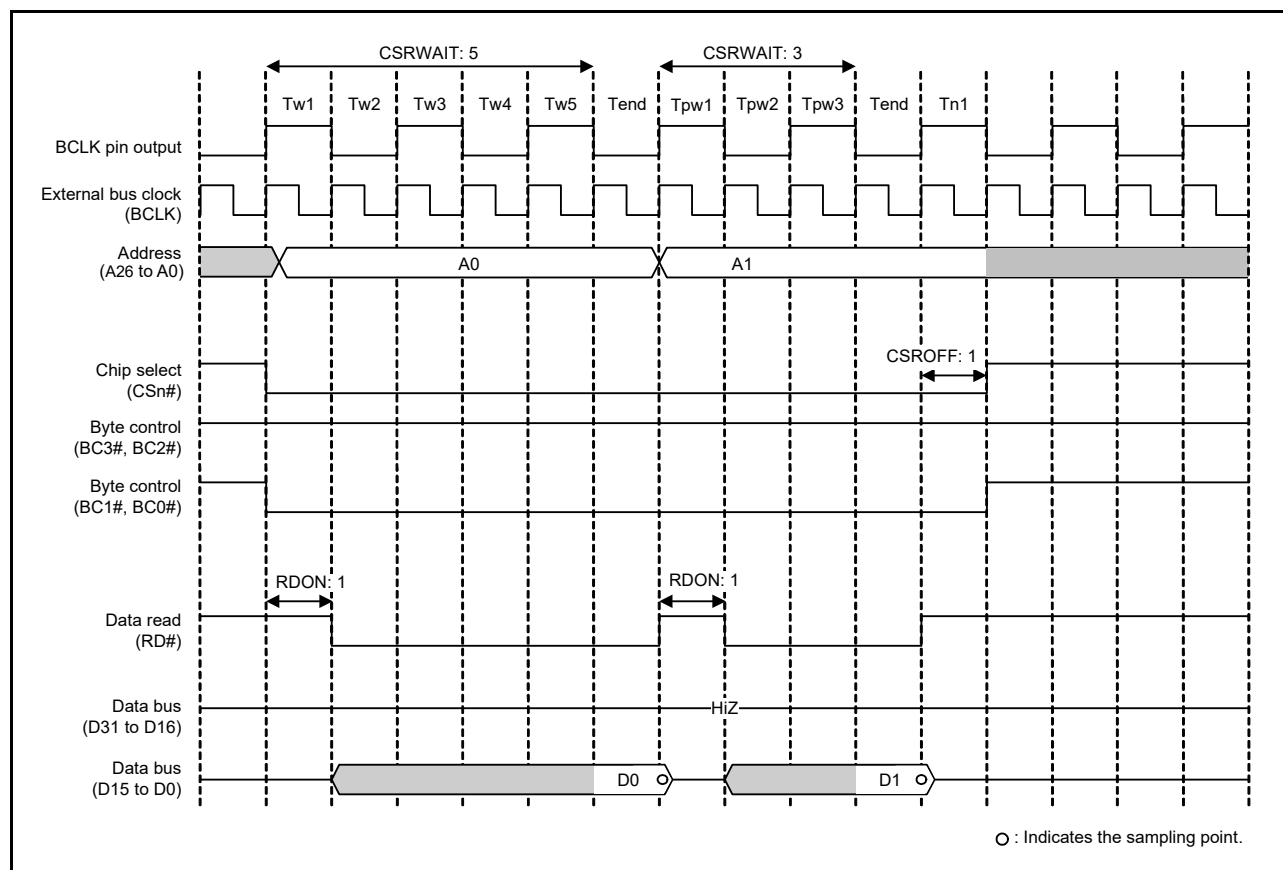


Figure 16.35 Example of Page Read Access Operation
(when 1/2 BCLK is Selected with the BCLK Pin Output Select Bit and Two Rounds of Bus Access are Generated in Response to a Single Request for Transfer) (n = 0 to 7)

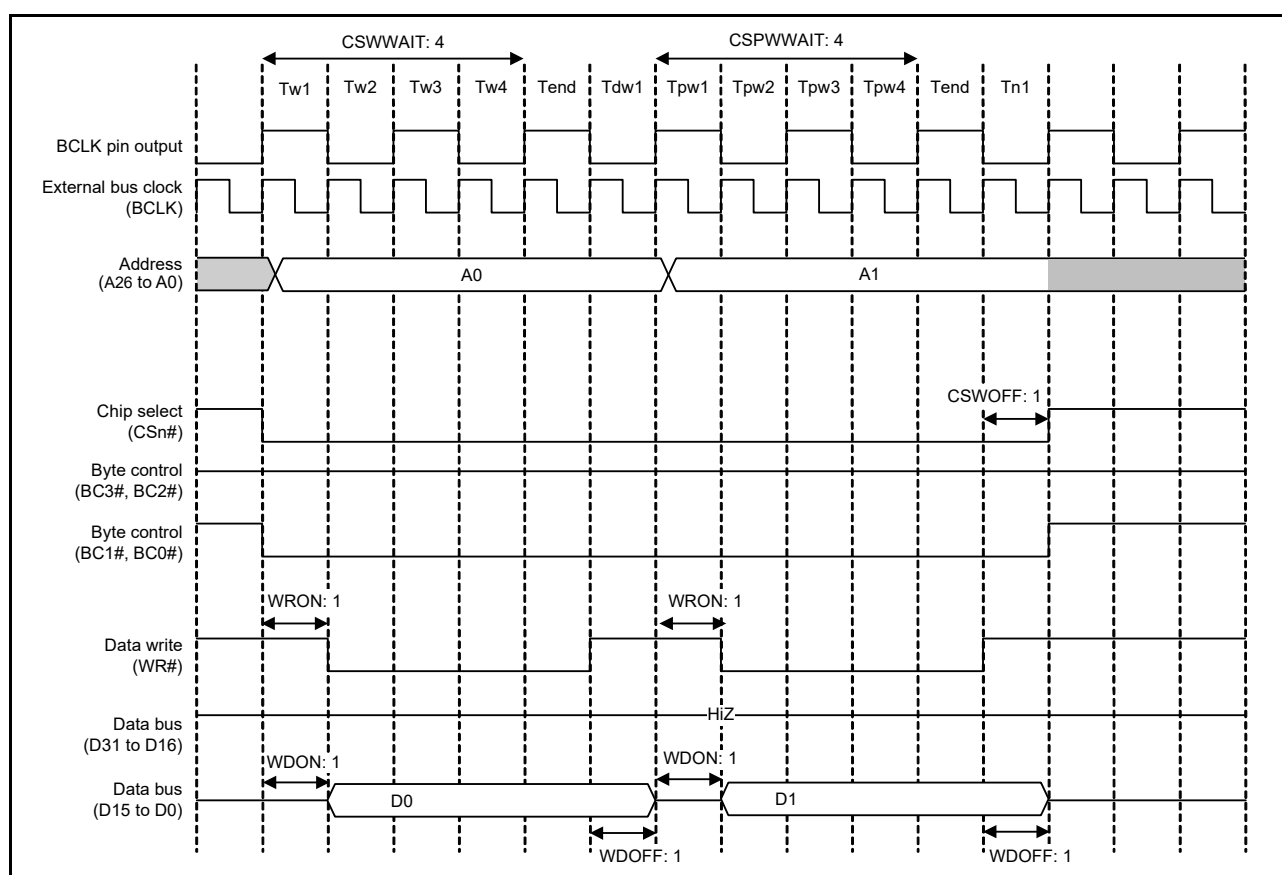


Figure 16.36 Example of Page Write Access Operation
 (when 1/2 BCLK is Selected with the BCLK Pin Output Select Bit and Two Rounds of Bus Access are Generated in Response to a Single Request for Transfer, in Single Write Strobe Mode)
 (n = 0 to 7)

16.5.2 Address/Data Multiplexed Bus

When the address/data multiplexed I/O interface select bit (MPXEN) in CSnCR is set to 1, addresses and data can be multiplexing input/output to/from the D15 to D0 pins in the corresponding area. Using this function enables direct connection of this MCU to peripheral LSIs requiring address/data multiplexing. When 8-bit width is selected with the BSIZE[1:0] bits in CSnCR, D7 to D0 are multiplexed with A7 to A0. When 16-bit width is selected, D15 to D0 are multiplexed with A15 to A0. In the address/data multiplexed I/O space, accesses are controlled with the ALE, RD#, WRn#, and BCn# signals.

Byte strobe mode or single-write strobe mode is selectable in the same way as for a separate bus. However, with regard to the BCn# signals within the address cycle, the byte-control signal is output for the data being read or written.

During the address/data multiplexed I/O space access, after the number of wait cycles specified by the address cycle wait select bits (AWAIT[1:0]) in CSnWCR2 is inserted in the address output cycle, data access is performed.

- Ta1 to Tan (Address Cycle Wait)

The period Ta1 to Tan is valid only when the address/data multiplexed I/O space is specified. This period is made up of the number of clock cycles between the start of external bus access and one cycle before the address latch (ALE) signal is negated. The number of cycles are selectable within the range from zero to three. Addresses are output until the next cycle of ALE signal negation (address cycle). The timing of ALE signal is the same as that of CS# assertion. After the address cycle, a data cycle is started. CSnWCR1 and CSnWCR2 should be set so that an address cycle and a data cycle do not overlap.

Page access to the address/data multiplexed I/O space is invalid. When the PRENB or PWENB bit in CSnMOD is set to 1 to enable page-read or page-write access, these settings are ignored and normal read or write operation is performed.

Figure 16.37 to Figure 16.39 show examples of operations with the address/data multiplexed I/O interface.

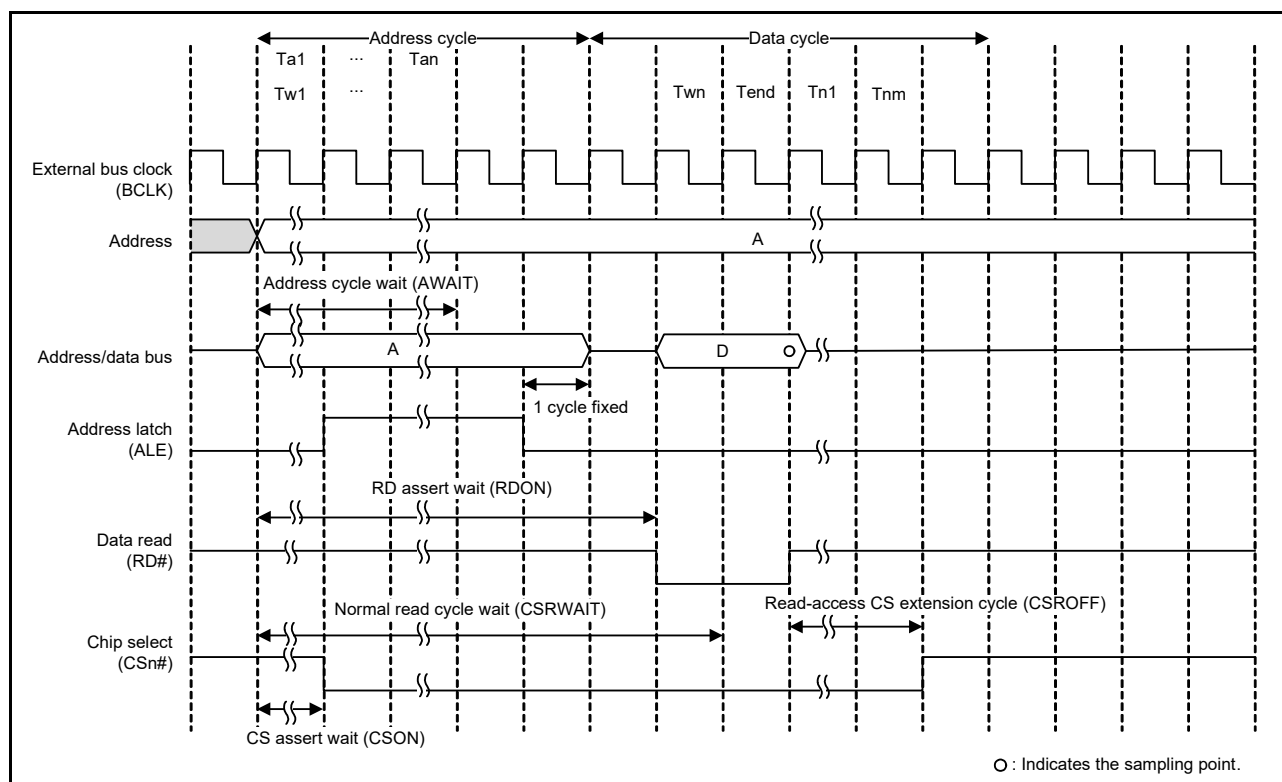


Figure 16.37 Example of Read Access Operation with Address/Data Multiplexed I/O Interface (n = 0 to 7)

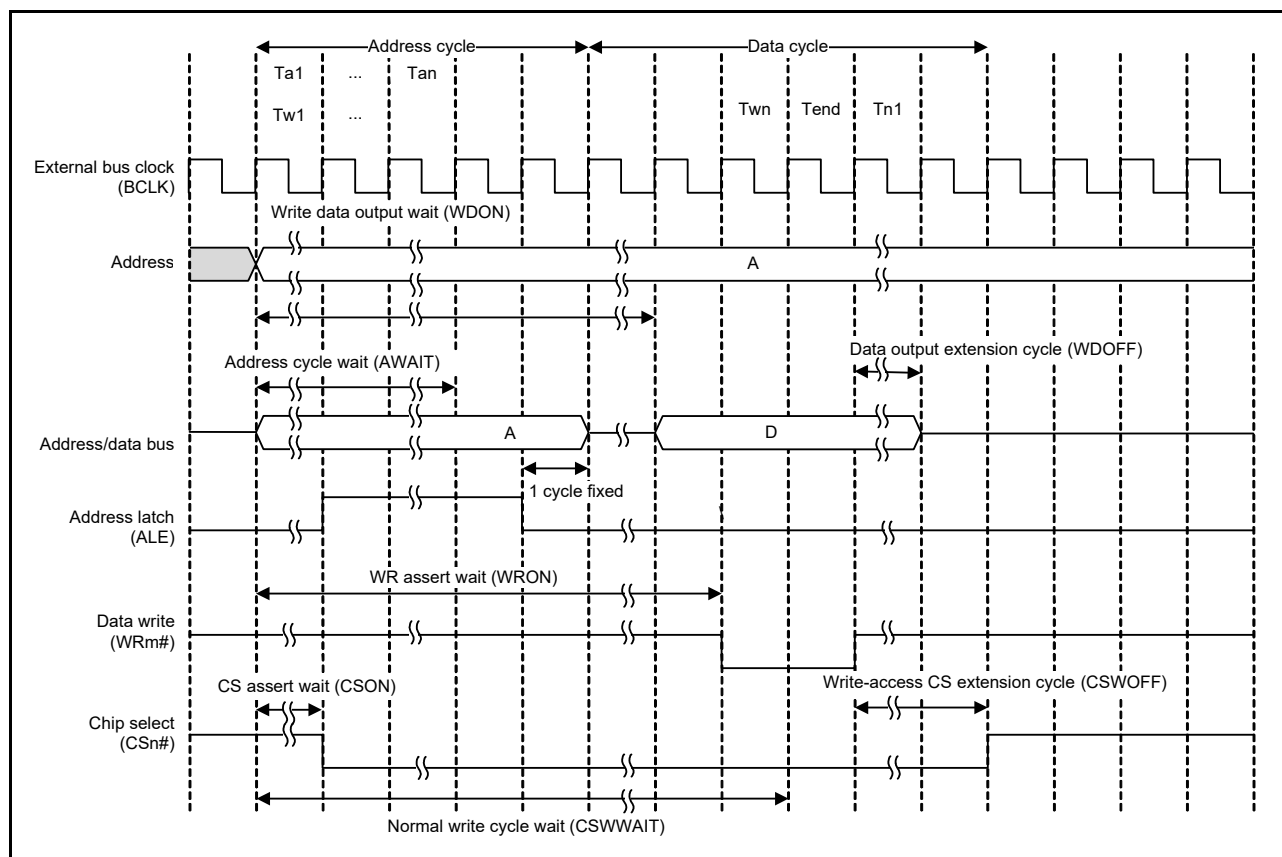


Figure 16.38 Example of Write Access Operation with Address/Data Multiplexed I/O Interface (n = 0 to 7; m = 0, 1)

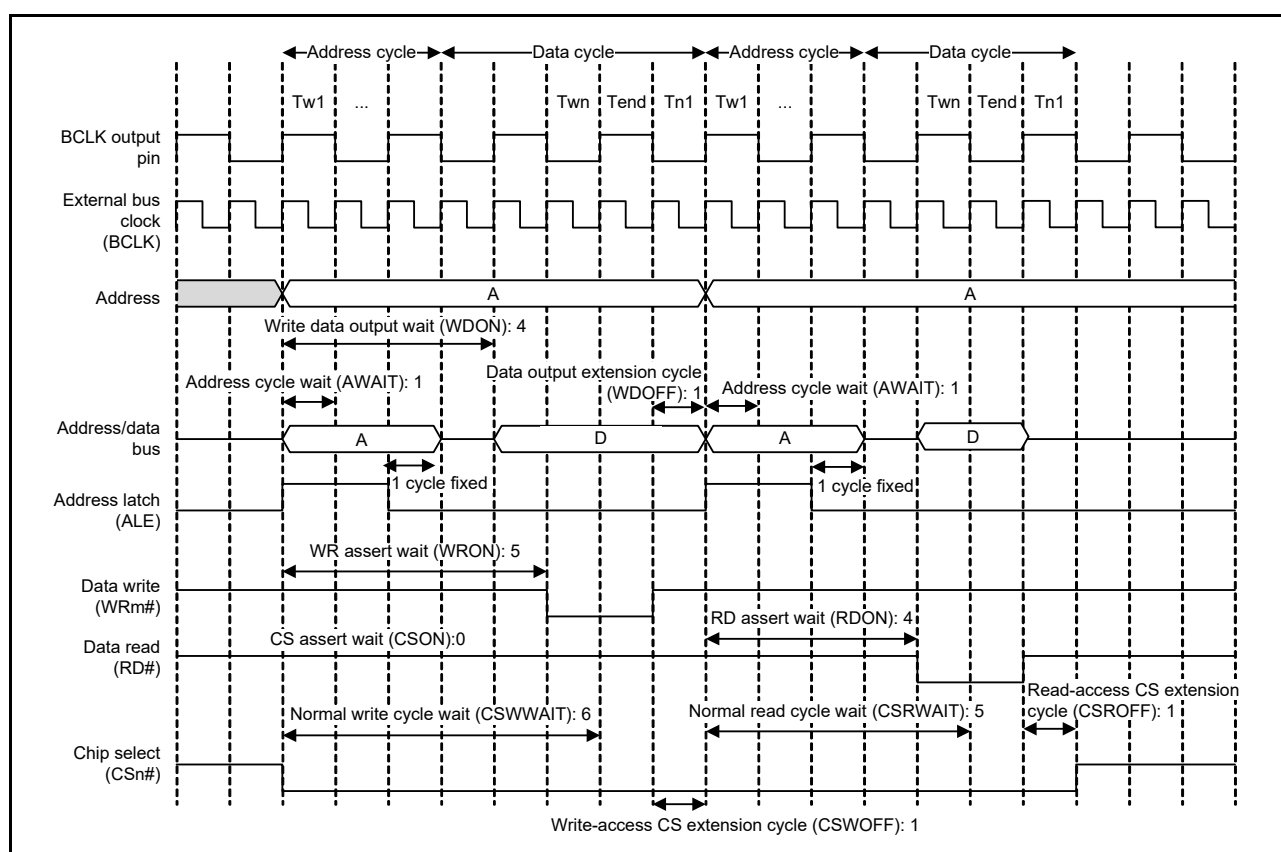


Figure 16.39 Example of Bus Timing with Address/Data Multiplexed I/O Interface ($n = 0$ to 7 ; $m = 0, 1$)

16.5.3 External Wait Function

Wait cycles can be extended by the WAIT# signal over the length of normal access cycle wait (specified by the CSWAIT[4:0] and CSWAIT[4:0] bits in CSnWCR1) and page access cycle wait (specified by the CSPRWAIT[2:0] and CSPRWAIT[2:0] bits in CSnWCR1).

When external wait is enabled (the EWENB bit = 1 in CSnMOD), wait cycles are inserted while the WAIT# signal is held low. When external wait is disabled (the EWENB bit = 0 in CSnMOD), the WAIT# signal has no effect.

All wait cycles specified in CSnWCR1 are inserted independently of the WAIT# signal.

(1) Normal Access

Sampling of the WAIT# signal begins upon completion of the wait cycle (Tend) specified in CSnWCR1. The bus cycle is extended while the WAIT# signal is held low. The wait cycle ends (Tend) at the next cycle after the WAIT# signal becomes high.

(2) Page Access

The first access operation is the same as the normal access operation. Sampling of the WAIT# signal begins upon completion of the wait cycle (Tend) specified in the CSnWCR1 register. The bus cycle is extended while the WAIT# signal is held low. The wait cycle ends (Tend) at the next cycle after the WAIT# signal becomes high.

With respect to the second and subsequent accesses, sampling of the WAIT# signal begins upon completion of the wait cycle of the page access (Tend). The wait cycle of the page access is extended while the WAIT# signal is held low, and ends (Tend) at the next cycle after the WAIT# signal becomes high.

Figure 16.40 and Figure 16.41 show examples of external wait insertion timing with the separate bus interface.

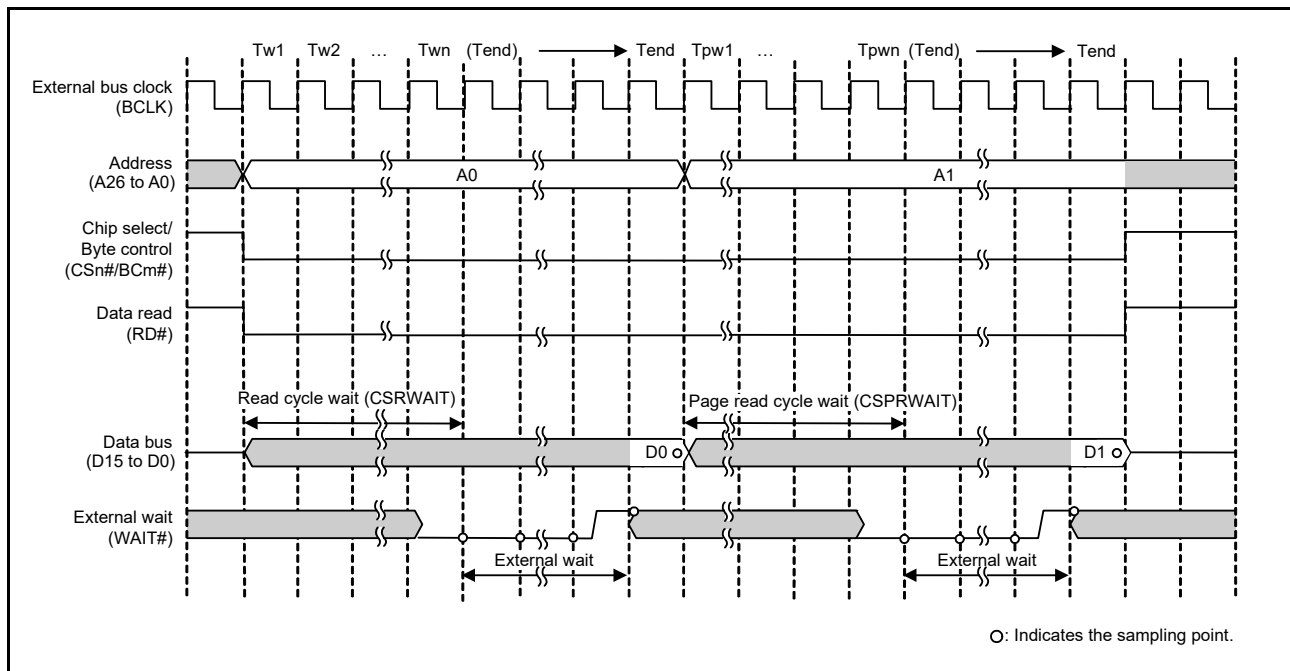


Figure 16.40 Example of External Wait Timing (Page-Read Access to 16-Bit Bus Space) (n = 0 to 7; m = 0, 1)

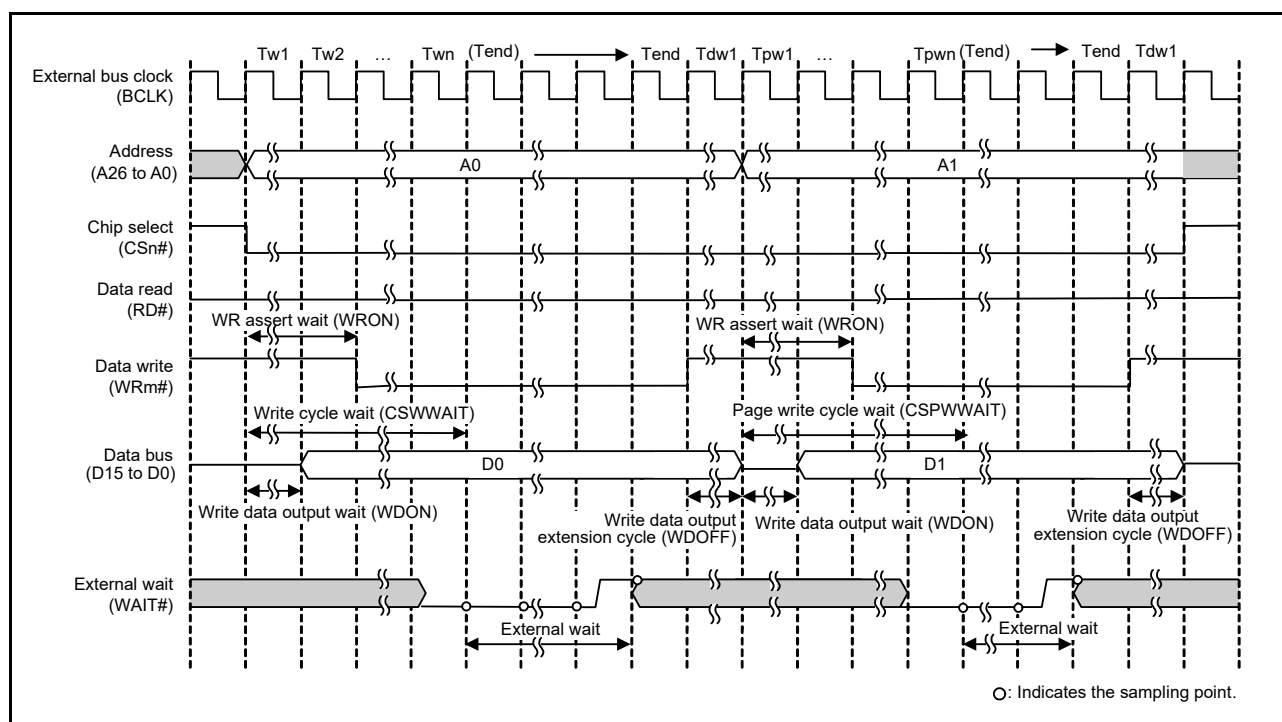


Figure 16.41 Example of External Wait Timing
(Page-Write Access to 16-Bit Bus Space, in Byte Strobe Mode) ($n = 0$ to 7 ; $m = 0, 1$)

(3) Address/Data Multiplexed I/O Interface

In a data cycle with the address/data multiplexed I/O interface, programmed waits and pin waits using the WAIT pin can be inserted in the same way as that with the separate bus interface.

Address cycles are not affected by the wait control settings. Figure 16.42 shows an example of external wait insertion timing with the address/data multiplexed I/O interface.

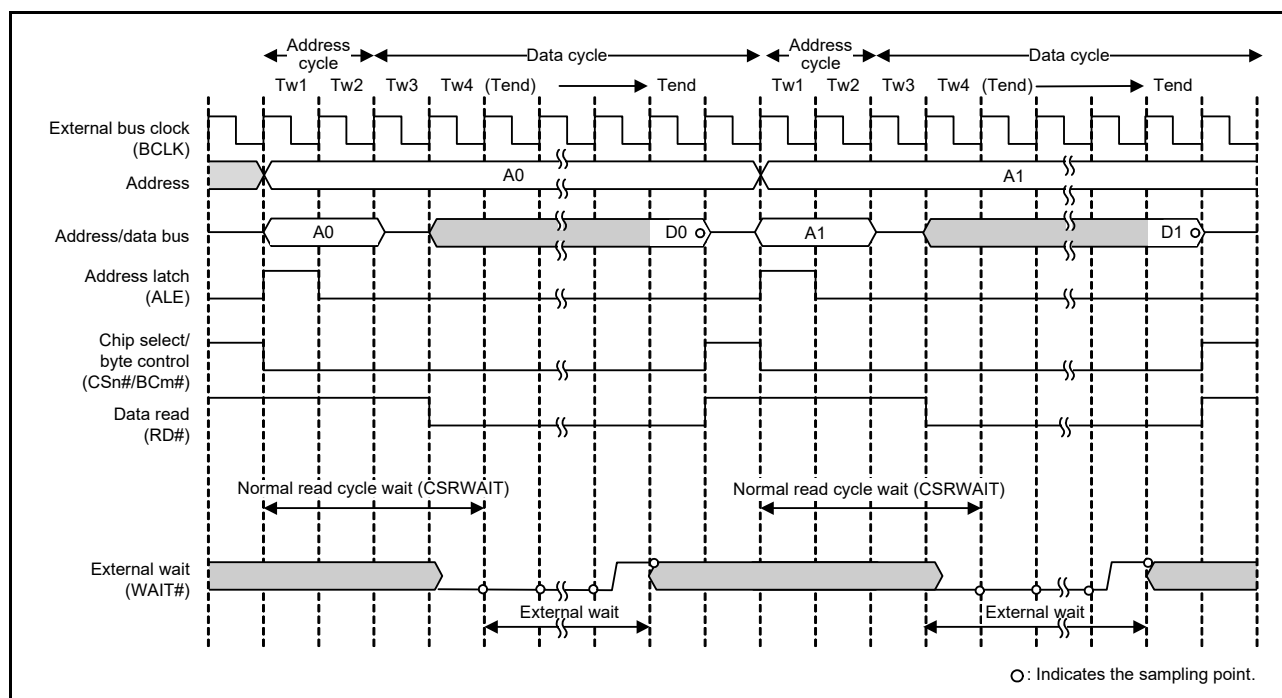


Figure 16.42 Example of External Wait Insertion Timing with Address/Data Multiplexed I/O Interface ($m = 0, 1$)

16.5.4 Insertion of Recovery Cycles

Recovery cycles can be inserted between consecutive rounds of external bus access by setting the recovery cycle insertion enable bit in CSRECEN to 1.

The number of recovery cycles to be inserted after read cycles and write cycles can be separately set for each area using CSnREC. When the preceding bus cycle is a write access, the number of write recovery cycles should be set with the WRCV[3:0] bits for the area. When the preceding bus cycle is a read access, the number of read recovery cycles should be set with the RRCV[3:0] bits for the area. For example, when CS1 read access occurs after CS0 read access, the number of recovery cycles to be inserted between them is set by the RRCV[3:0] bits in CS0REC.

Recovery cycles can be inserted on any of the following eight conditions. The recovery cycle insertion can be enabled or disabled with the RCVENj (j = 0 to 7) in CSRECEN when the preceding bus access is a separate bus access, and with RCVENMj (j = 0 to 7) when the preceding bus access is an address/data multiplexed bus access.

- After a read access to the external bus, a read access is made to the external bus in the same area.
- After a read access to the external bus, a read access is made to the external bus in a different area.
- After a read access to the external bus, a write access is made to the external bus in the same area.
- After a read access to the external bus, a write access is made to the external bus in a different area.
- After a write access to the external bus, a read access is made to the external bus in the same area.
- After a write access to the external bus, a read access is made to the external bus in a different area.
- After a write access to the external bus, a write access is made to the external bus in the same area.
- After a write access to the external bus, a write access is made to the external bus in a different area.

The recovery cycle starts at the end of the preceding bus cycle, i.e. when the CSn# signal (n = 0 to 7) is negated. A high-level period of the CSn# signal is inserted for the specified recovery cycle period starting from this point.

The CSn# signal for the next round of bus access is asserted immediately after the end of recovery cycles in the fastest case. Even if the next request for access to an external address space is generated during the recovery period, the next round of access over the external bus will start immediately after the end of recovery cycles.

When two or more external bus access cycles are required for a single transfer request from a bus master and the recovery cycle insertion condition is satisfied, recovery cycles are also inserted between these bus access cycles.

However, when page read access is enabled (CSnMOD.PRENB = 1) or page write access is enabled (CSnMOD.PWENB = 1), recovery cycles are not inserted except after the last bus access cycle of the transfer even if the recovery cycle insertion condition is satisfied (Figure 16.45).

Similarly, during normal accesses with page access enabled, recovery cycles are not inserted between bus access cycles but inserted only after the last bus access cycle of the transfer.

With the address/data multiplexed I/O interface, when the recovery cycle insertion condition is satisfied, recovery cycles are inserted between bus access cycles regardless of the page access enable setting.

Figure 16.43 to Figure 16.45 show examples of recovery cycle insertion with the separate bus interface.

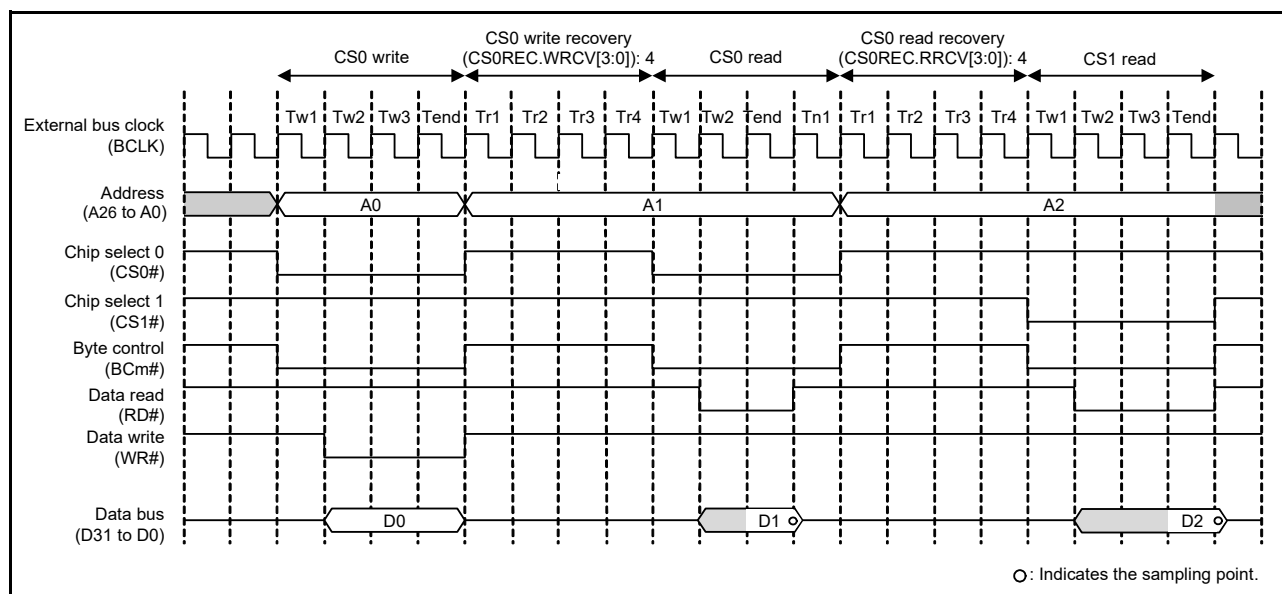


Figure 16.43 Example of Recovery Cycle Insertion with Separate Bus Interface ($m = 0$ to 3)

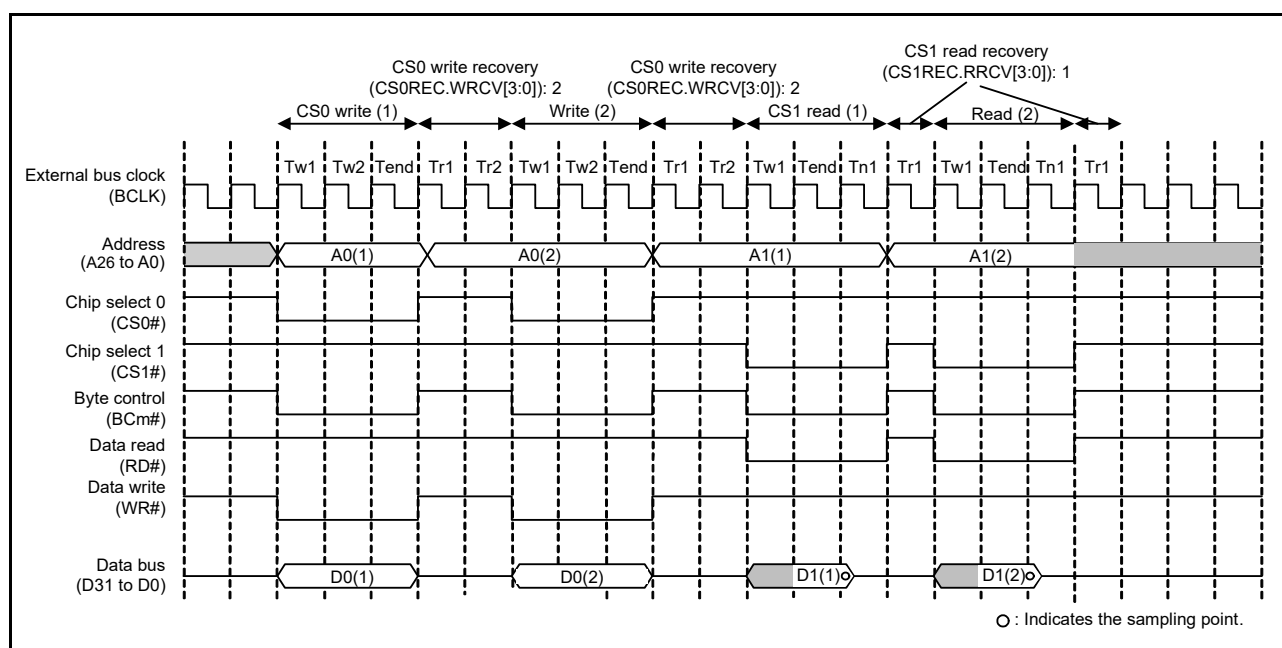


Figure 16.44 Example of Recovery Cycle Insertion When a Bus Access is Split (with Separate Bus Interface, Normal Access) ($m = 0$ to 3)

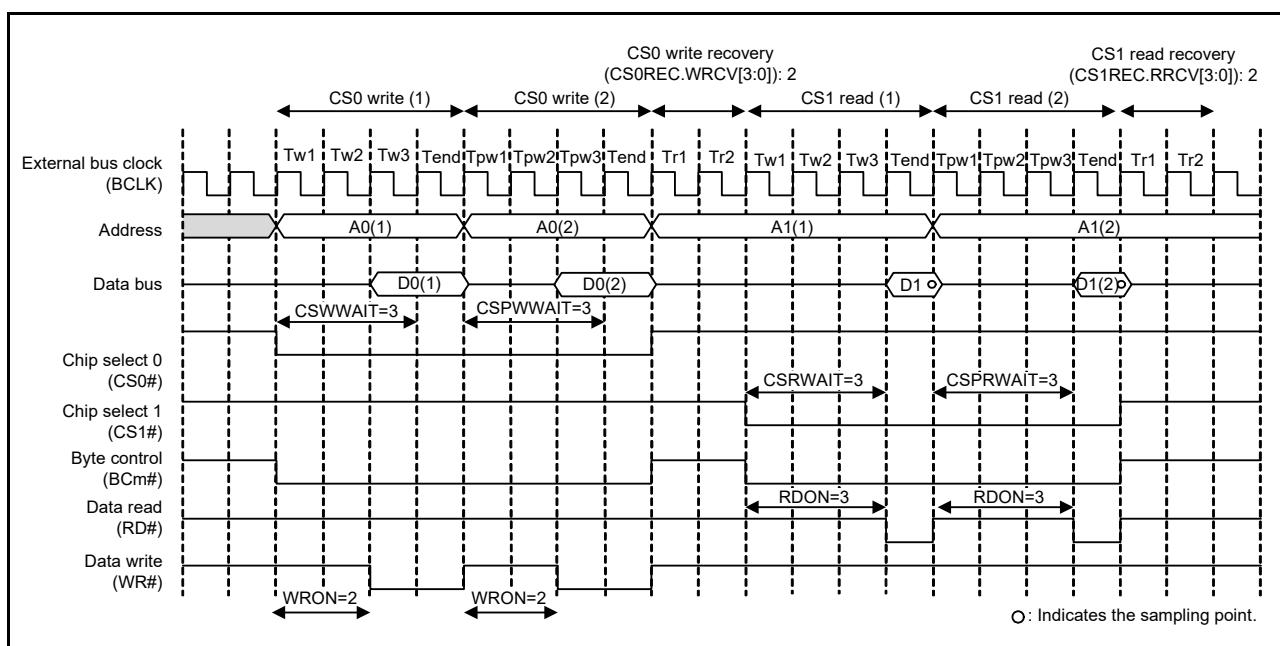


Figure 16.45 Example of Recovery Cycle Insertion When a Bus Access is Split (with Separate Bus Interface, Page Access) ($m = 0$ to 3)

Figure 16.46 shows examples of operations when the BCLK pin output selection bits are set for frequency-division of BCLK by 2.

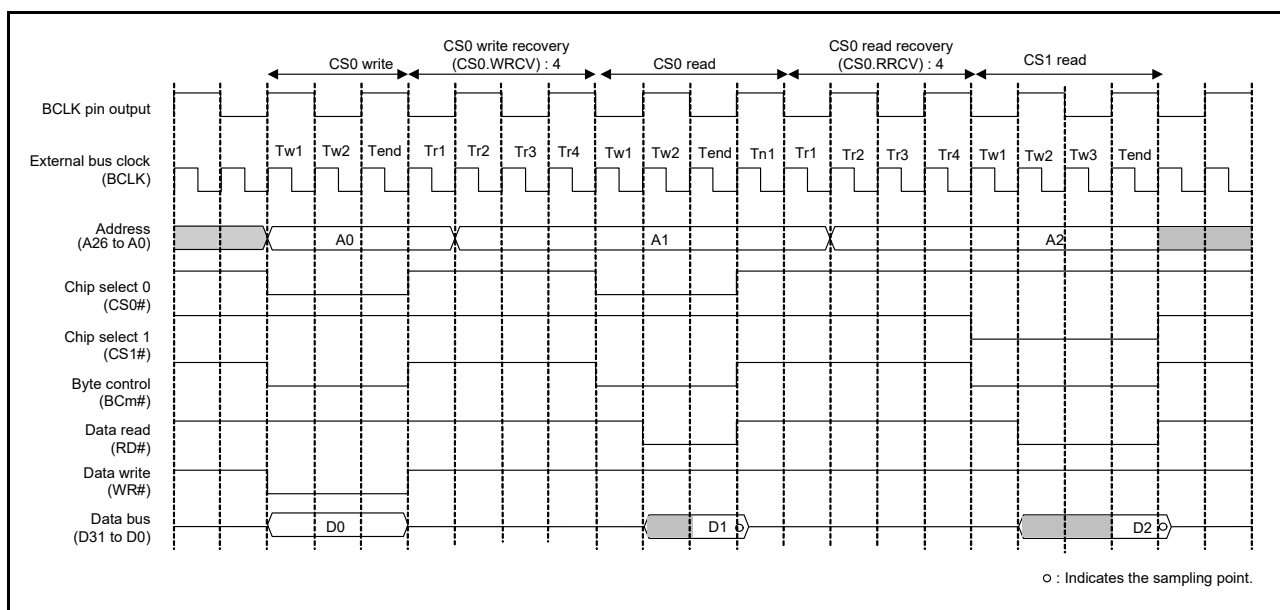


Figure 16.46 Example of Operation for Recovery Cycles when the BCLK Pin Output Selection Bits are Set for Frequency-Division of BCLK by 2 (For the Case of Normal Access through a Separate Bus Interface) ($m = 0$ to 3)

With the address/data multiplexed I/O interface, recovery cycles are inserted in the same way as that with the separate bus interface. Figure 16.47 and Figure 16.48 show examples of recovery cycle insertion with the address/data multiplexed I/O interface.

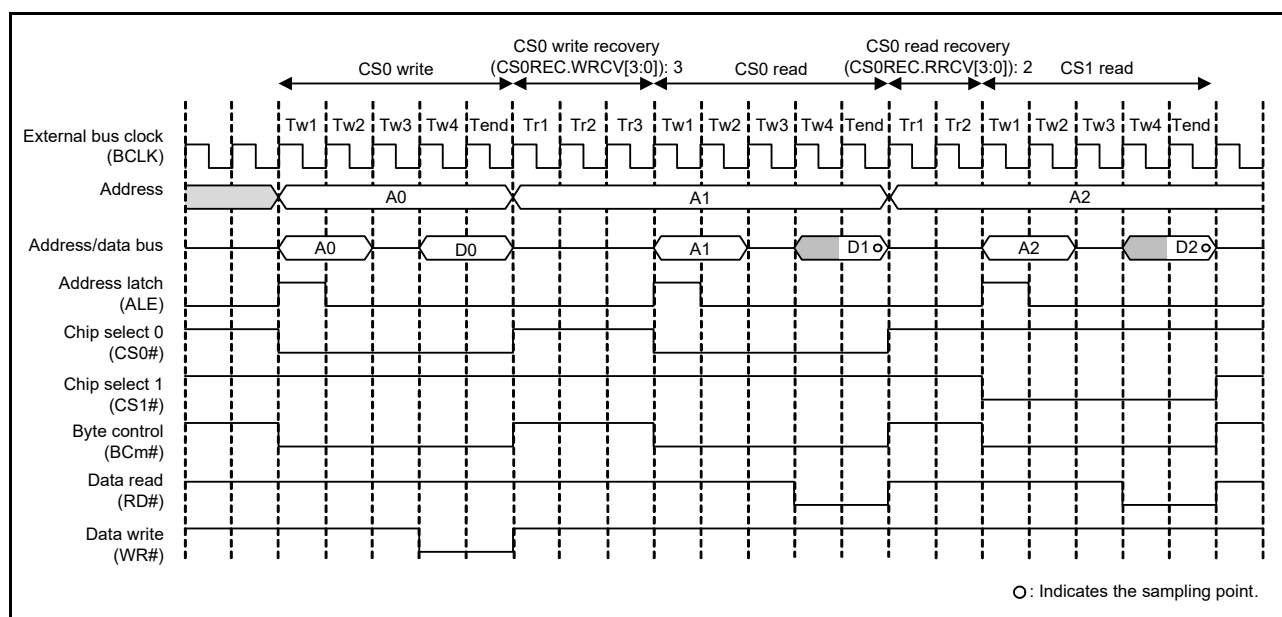


Figure 16.47 Example of Recovery Cycle Insertion with Address/Data Multiplexed I/O Interface (m = 0, 1)

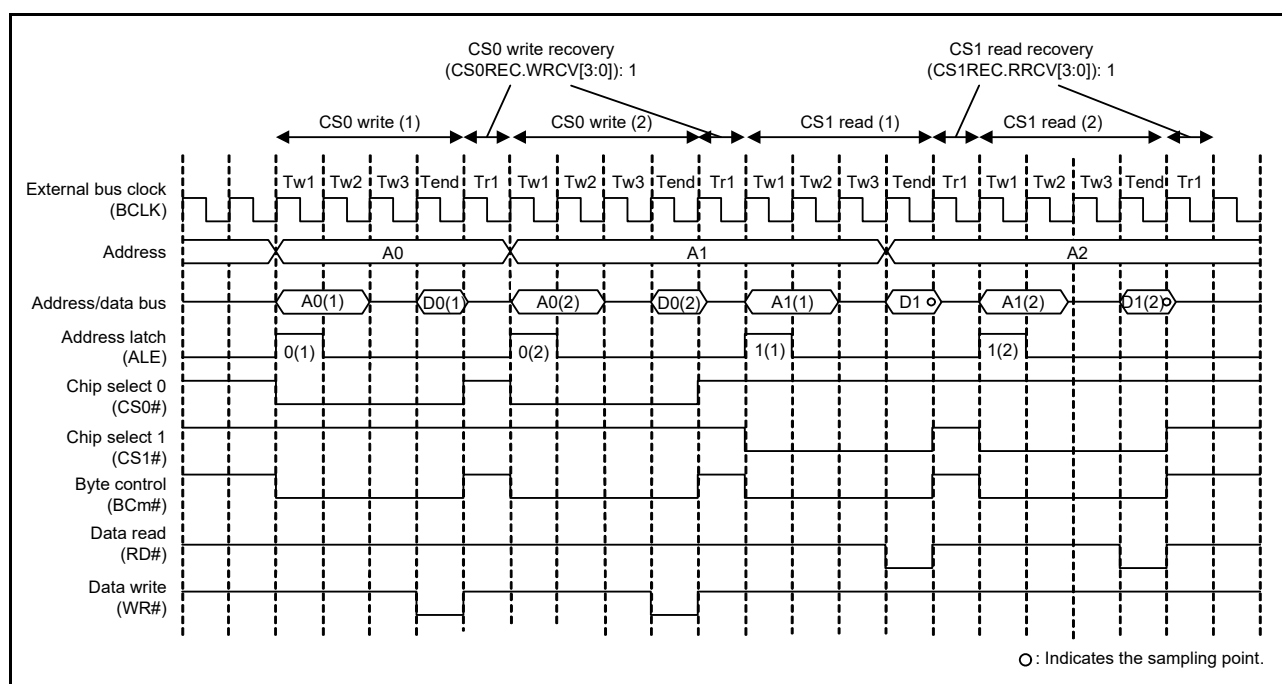


Figure 16.48 Example of Recovery Cycle Insertion When a Bus Access is Split with Address/Data Multiplexed I/O Interface (m = 0, 1)

16.5.5 No Access State

When no external address space is accessed, CSn#, BCn#, WRn#, and RDn# signals are high, ALE signal is low, and D31 to D0 are in the high-impedance state.

16.5.6 Write Buffer Function (External Bus)

The internal main bus is released by writing data to the write buffer before the write access is completed, which allows the next round of bus access to start. However, if the following round of bus access is to an external address space or to a register of the external bus controller, it is suspended until the external bus operations already in progress are completed. Figure 16.49 shows an example of operation when the write-buffer function is in use. When this function is in use, if the next operation after an external write is internal access, the internal access (access to on-chip memory or a peripheral module) is executed in parallel with the external write, i.e. without waiting for completion of the latter operation.

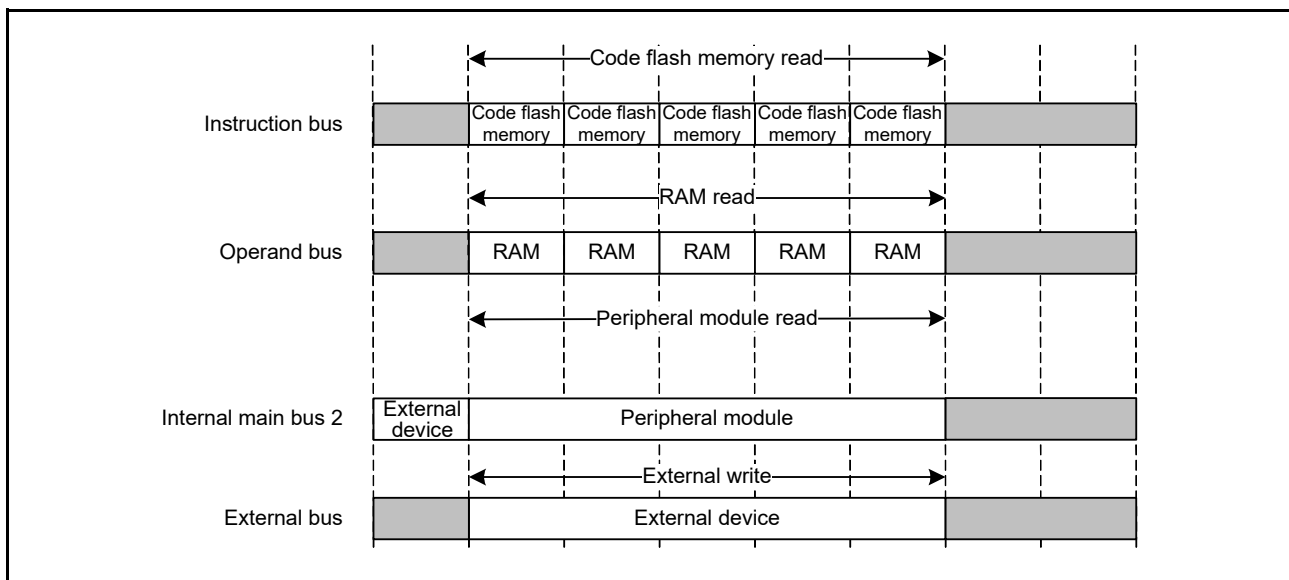


Figure 16.49 Example of Operation when the Write-Buffer Function is in Use

16.5.7 Limitations

(1) Limitations on Using Separate Bus Interface

- Limitations that apply to various bits of CSn wait control register 1 (CSnWCR1) and CSn wait control register 2 (CSnWCR2) at the times of normal and page accesses are listed in Table 16.10.

Even if the setting of the page-read access enable bit in the CSn mode register or the page-write access enable bit in the CSn mode register selects permission (CSnMOD.PRENB = 1 or CSnMOD.PWENB = 1), the first round of access for page access and the access that does not fall within the scope of page access are normal access operation, and thus limitations on normal access must be satisfied.

Table 16.10 Limitations at the Time of Normal and Page Access

Limitations at the Time of Normal Access		Limitations at the Time of Page Access	
Reading	Writing	Reading	Writing
CSON[2:0] ≤ CSRWAIT RDON[2:0] ≤ CSRWAIT CSON[2:0] ≤ RDON	1 ≤ WDON[2:0] CSON[2:0] ≤ CSWWAIT WRON[2:0] ≤ CSWWAIT WDON[2:0] ≤ CSWWAIT WDOFF[2:0] ≤ CSWOFF WDON[2:0] ≤ WRON CSON[2:0] ≤ WRON	CSON[2:0] ≤ CSPRWAIT RDON[2:0] ≤ CSPRWAIT CSON[2:0] ≤ RDON	1 ≤ WDON[2:0] CSON[2:0] ≤ CSPWWAIT WRON[2:0] ≤ CSPWWAIT WDON[2:0] ≤ CSPWWAIT WDOFF[2:0] ≤ CSWOFF WDON[2:0] ≤ WRON CSON[2:0] ≤ WRON

- When two or more external bus access cycles are required for a single transfer request from a bus master and the recovery cycle insertion condition is satisfied with page read access enabled (CSnMOD.PRENB = 1) or page write access enabled (CSnMOD.PWENB = 1), recovery cycles are not inserted between bus access cycles but inserted only after the last bus access cycle of the transfer.

(2) Limitations on Using Address/Data Multiplexed Bus Interface

- In the address/data multiplexed I/O space, page accesses are invalid. If a page access setting is specified, the setting is ignored and the normal read or write operation is performed.
- When the address/data multiplexed I/O interface is set, the BSIZE[1:0] bits in CSnCR should not be set to other than 8- and 16-bit bus spaces. If set, the operation cannot be guaranteed.

Table 16.11 Limitations at the Time of Normal Access

Limitations at the Time of Normal Access	
Reading	Writing
CSO[2:0] ≤ CSRWAIT	CSO[2:0] ≤ CSWWAIT
RDON[2:0] ≤ CSRWAIT	WRON[2:0] ≤ CSWWAIT
CSO[2:0] ≤ RDON	WDON[2:0] ≤ CSWWAIT
AWAIT[1:0] + 2 ≤ RDON	WDOFF[2:0] ≤ CSWOFF
CSO[2:0] ≤ AWAIT	WDON[2:0] ≤ WRON
	CSO[2:0] ≤ WRON
	AWAIT[1:0] + 2 ≤ WRON
	AWAIT[1:0] + 2 ≤ WDON
	CSO[2:0] ≤ AWAIT

(3) Limitation when a Pin is Multiplexed between A0 and BC0# Functions

When the A0 and BC0# pin functions share the same pin, setting the single write strobe mode is prohibited in the 8-bit bus space; otherwise the operation is not guaranteed.

(4) Limitations when 1/2 BCLK is Selected with BCLK Pin Output Select Bit

When 1/2 BCLK is selected through the BCLK pin output select bit, the external bus access cycle starts at the rising edge of the BCLK pin output. However, when two or more external bus access cycles are generated for a single transfer request from a bus master, the second or subsequent external bus access cycle may start at the falling edge of the BCLK pin output depending on the wait cycle settings. Make appropriate register settings according to the specifications of the device to be connected.

(5) Limitations on EXDMAC Single Address Transfer Mode

- During the transfer in EXDMAC single address mode, the EDACK signal can be negated one cycle before the RD# signal is negated for read access or one cycle after the WR# signal is negated for write access through the settings of the EDACKn pin negate wait bit in the EXDMA output set register (EDMOMD.DACKW). Here, the CS# signal assertion and negation timing should be set so that the EDACK signal is enabled while the CS# signal is asserted. Table 16.12 and Table 16.13 list the limitations on the CSnWCR1 and CSnWCR2 register setting during the EXDMAC transfer in single address mode.
- To enable the EDACK signal output during the EXDMAC transfer in single address mode, the external wait function should be disabled (EWENB bit in CSnMOD = 0).
- When the external data read continuous assertion mode is specified (PRMOD bit in CSnMOD = 1) for page read access, the transfer in EXDMAC single address mode is prohibited; if such an attempt is made, correct operation is not guaranteed.
- In the address/data multiplexed I/O space, the transfer in EXDMAC single address mode is prohibited; if such an attempt is made, correct operation is not guaranteed.

Table 16.12 Limitations on EXDMAC Single Address Transfer Mode (EDMOMD.DACKW = 0)

Limitations at the Time of Normal Access		Limitations at the Time of Page Access	
Reading	Writing	Reading	Writing
$CSON[2:0] \leq CSRWAIT$ $RDON[2:0] \leq CSRWAIT$ $CSON[2:0] \leq RDON$ $1 \leq RDON$	$CSON[2:0] \leq CSWWAIT$ $WRON[2:0] \leq CSWWAIT$ $WDON[2:0] \leq CSWWAIT$ $WDOFF[2:0] \leq CSWOFF$ $WDON[2:0] \leq WRON$ $CSON[2:0] \leq WRON$ $1 \leq WRON$	$CSON[2:0] \leq CSPRWAIT$ $RDON[2:0] \leq CSPRWAIT$ $CSON[2:0] \leq RDON$ $1 \leq RDON$	$CSON[2:0] \leq CSPWWAIT$ $WRON[2:0] \leq CSPWWAIT$ $WDON[2:0] \leq CSPWWAIT$ $WDOFF[2:0] \leq CSWOFF$ $WDON[2:0] \leq WRON$ $CSON[2:0] \leq WRON$ $1 \leq WRON$

Table 16.13 Limitations on EXDMAC Single Address Transfer Mode (EDMOMD.DACKW = 1)

Limitations at the Time of Normal Access		Limitations at the Time of Page Access	
Reading	Writing	Reading	Writing
$CSON[2:0] \leq CSRWAIT$ $RDON[2:0] < CSRWAIT$ $CSON[2:0] \leq RDON$ $1 \leq RDON$	$CSON[2:0] \leq CSWWAIT$ $WRON[2:0] \leq CSWWAIT$ $WDON[2:0] \leq CSWWAIT$ $WDOFF[2:0] \leq CSWOFF$ $WDON[2:0] \leq WRON$ $CSON[2:0] \leq WRON$ $1 \leq WRON$ $1 \leq WDOFF$	$CSON[2:0] \leq CSPRWAIT$ $RDON[2:0] < CSPRWAIT$ $CSON[2:0] \leq RDON$ $1 \leq RDON$	$CSON[2:0] \leq CSPWWAIT$ $WRON[2:0] \leq CSPWWAIT$ $WDON[2:0] \leq CSPWWAIT$ $WDOFF[2:0] \leq CSWOFF$ $WDON[2:0] \leq WRON$ $CSON[2:0] \leq WRON$ $1 \leq WRON$ $1 \leq WDOFF$

(6) Prohibition of Access that Spans Areas of Address Space

Single access that spans several areas of the address space is prohibited, and operation of such an access is not guaranteed. Setting must be made so that two areas are not accessed at the same time by a single word or longword access.

Single access that spans two areas of the address space is also prohibited in the EXDMAC block transfer in single address mode or cluster transfer, and operation of such an access is not guaranteed. Setting must be made so that two areas are not accessed at the same time by a single access in the EXDMAC block transfer in single address mode or cluster transfer.

(7) Restrictions on RMPA and String-Manipulation Instructions

- Although the external space has a per-area ending-switching facility (only for data), the allocation of data to be handled by RMPA or string-manipulation instructions to an area where the endian differs from that of the chip is prohibited, and operation is not guaranteed if this restriction is not observed. If data to be handled by RMPA or string-manipulation instructions are allocated to the external space, they must be allocated to areas where the endian setting is the same as that for the chip.
- The allocation of data to be handled by RMPA or string-manipulation instructions to I/O registers is prohibited, and operation is not guaranteed if this restriction is not observed.

(8) Restriction on Instruction Code

When the endian setting for each area is different from that for the chip, no instruction code can be allocated in the area. The instruction code should be allocated to the external space whose endian setting is the same as that for the chip.

16.6 SDRAM Area Controller Operation

The following sections describe how the SDRAM area controller (SDRAMC) is enabled and the SDRAM bus width is set, which is followed by the description of SDRAMC operations including read, write, auto-refresh, self-refresh, initialization sequence, and mode register setting.

16.6.1 Enabling/Disabling SDRAM Access and Setting SDRAM Bus Width

SDRAM access can be enabled or disabled using the SDC control register (SDCCR). The SDRAM bus width can also be set using SDCCR.

Even when the operation of the SDRAM address space is disabled, refresh operation is available as long as self-refresh or auto-refresh operation is enabled.

16.6.2 No Access State

When no external address space is accessed, SDCS#, WEn#, RAS#, and CAS# signals are high.

16.6.3 Insertion of Recovery Cycles

When access to the SDRAM area follows access to the CS area, data recovery cycles are inserted for the CS area controller (CSC). If the number of recovery cycles for the CSC is 0, the ACT command for the next SDRAM access is issued immediately after negation of CSn# signal at the earliest. If the number of recovery cycles are not 0, the ACT command is issued two cycles after the specified recovery cycle period elapsed after negation of CSn# signal at the earliest. Since no data conflicts can occur during access to the SDRAM area, there is no need to set data recovery cycles for the SDRAM (fixed to zero cycle).

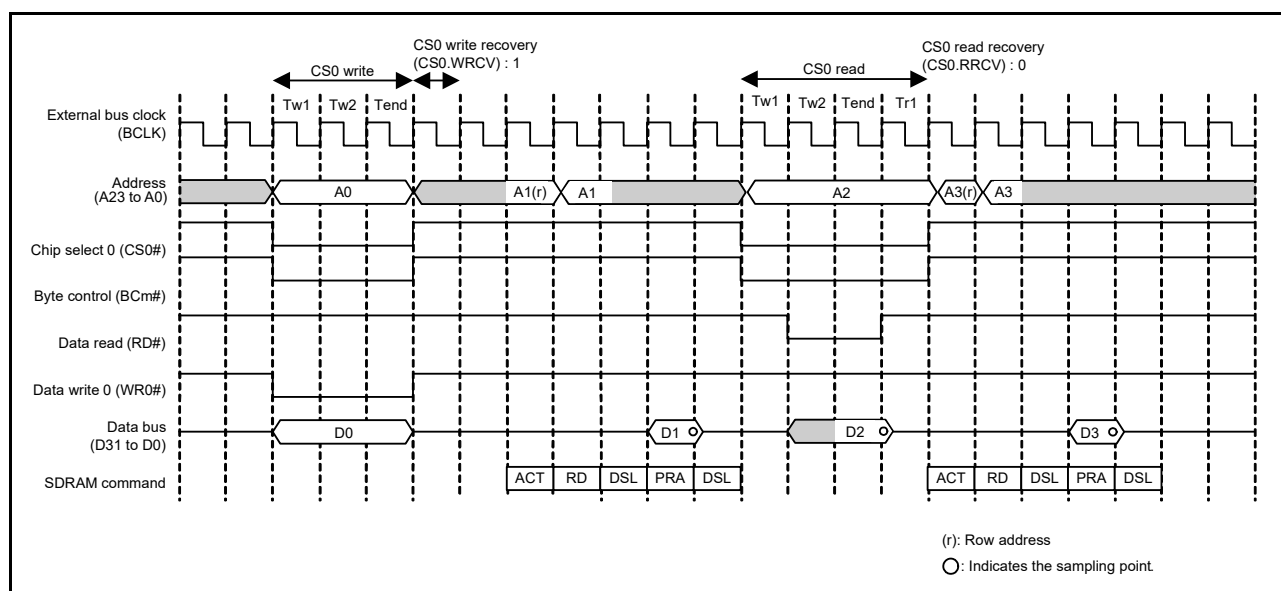


Figure 16.50 Example of Recovery Timing (for SDRAM Access)

16.6.4 Write Buffer Function

In write access, the internal main bus is released by writing data to the write buffer before the write access is completed, which allows the next round of bus access to start. However, if the following round of bus access is to an external address space or to a register of the external bus controller, it is suspended until the external bus operations already in progress are completed.

16.6.5 SDRAM Commands

The SDRAMC issues a command for each bus cycle to control SDRAM. Commands are defined by combination of SDCS#, RAS#, CAS#, WE#, CKE, and other signals.

Table 16.14 lists the commands issued by the SDRAMC.

Table 16.14 List of SDRAMC Commands

Name	Abbreviation	Command	SDCS#	RAS#	CAS#	WE#	CKE	
							n-1	n
DESL	DSL	Device deselect	H	x	x	x	H	x
ACTV	ACT	Bank active	L	L	H	H	H	x
READ	RD	Read	L	H	L	H	H	x
WRIT	WRI	Write	L	H	L	L	H	x
PALL	PRA	All bank precharge	L	L	H	L	H	x
REF	RFA	Auto-refresh	L	L	L	H	H	x
MRS	MRS	Mode register set	L	L	L	L	H	x
SELF	RFS	Self-refresh entry	L	L	L	H	H	L
SELF	RFX	Self-refresh end	H	x	x	x	L	H

Note: H: High level, L: Low level, x: Don't care. (High level or low level)
n: Command issue cycle, n – 1: One cycle before the command is issued.

16.6.6 Conditions for Setting SDRAMC Registers

SDRAMC registers should be modified only when all the corresponding conditions are satisfied as listed in Table 16.15.

Table 16.15 Conditions for Register Modification

Function/Operation	Registers	Conditions
Self-refresh	SDSELF*1	- SDRAM access is disabled. (SDCCR.EXENB = 0*2) - Auto-refresh operation is enabled. (SDRFEN.RFEN = 1)
Auto-refresh	SDRFCR	Self-refresh operation is disabled. (SDSELF.SFEN = 0)
	SDRFEN	Self-refresh operation is disabled. (SDSELF.SFEN = 0)
Initialization sequence	SDIR*1	The SDICR has not been set yet, and the same conditions as SDICR modification should be satisfied.
	SDICR*1	- SDRAM access is disabled. (SDCCR.EXENB = 0*2) - Auto-refresh operation is disabled. (SDRFEN.RFEN = 0) - Self-refresh operation is disabled. (SDSELF.SFEN = 0)
Address register	SDADR	- SDRAM access is disabled. (SDCCR.EXENB = 0*2) - Auto-refresh operation is disabled. (SDRFEN.RFEN = 0) - Self-refresh operation is disabled. (SDSELF.SFEN = 0)
Timing register	SDTR	- During self-refresh operation (SDSELF.SFEN = 1) - or - SDRAM access is disabled. (SDCCR.EXENB = 0*2) - Auto-refresh operation is disabled. (SDRFEN.RFEN = 0) - Self-refresh operation is disabled. (SDSELF.SFEN = 0)
Mode register	SDMOD*1	- SDRAM access is disabled. (SDCCR.EXENB = 0*2) - Self-refresh operation is disabled. (SDSELF.SFEN = 0)
Access mode register	SDAMOD	- SDRAM access is disabled. (SDCCR.EXENB = 0*2) - Auto-refresh operation is disabled. (SDRFEN.RFEN = 0) - Self-refresh operation is disabled. (SDSELF.SFEN = 0)

Note 1. Before modification, confirm that all the status bits in SDSR are 0.

Note 2. After writing 0 to the EXENB bit, confirm that the EXENB bit is set to 0.

16.6.7 Self-Refresh

Transition to or recovery from self-refresh mode can be controlled using the SDRAM self-refresh control register (SDSELF).

Immediately before transition to self-refresh mode, auto-refresh operation is performed. In self-refresh mode, the CKE signal is low. Immediately after recovery from self-refresh mode, the auto-refresh cycle is started.

Figure 16.51 and Figure 16.52 show timing examples of transition to self-refresh mode and recovery from self-refresh mode, respectively.

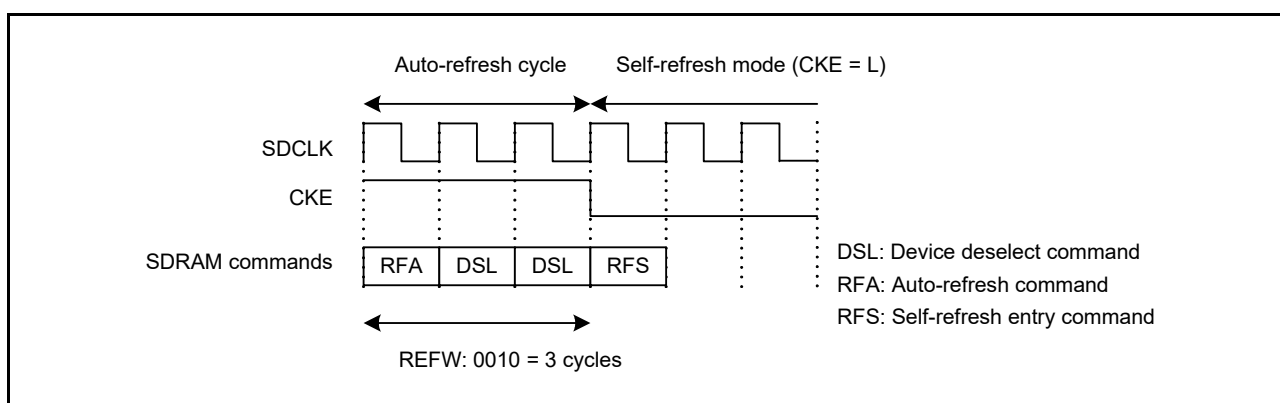


Figure 16.51 Timing Example of Transition to Self-Refresh Mode
(when SDRFCR.REFW[3:0] = 0010b: 3 Cycles)

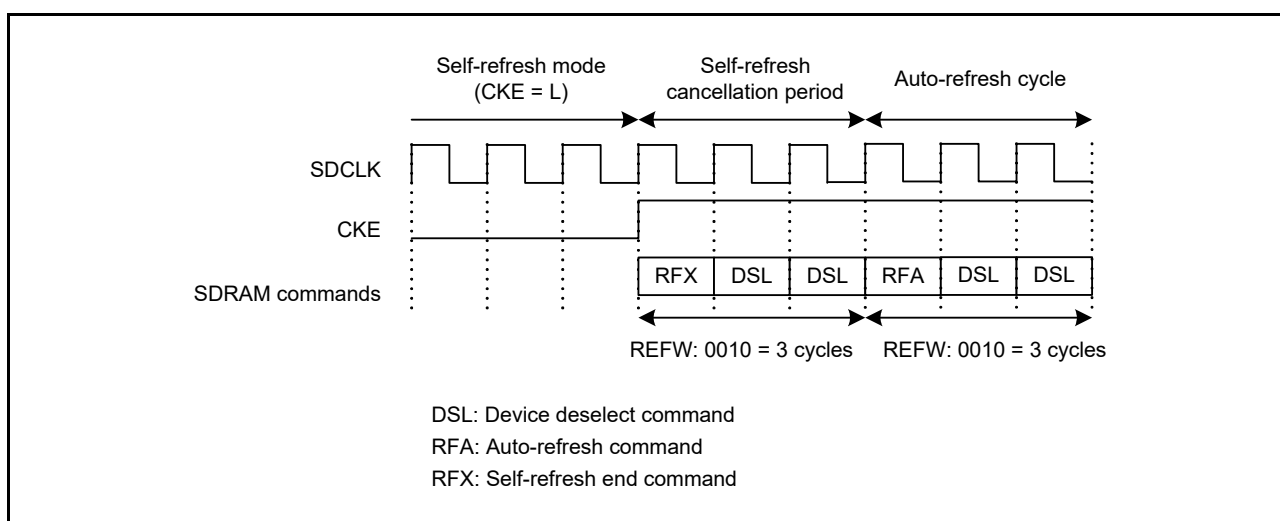


Figure 16.52 Timing Example of Recovery from Self-Refresh Mode

(1) Self-Refresh in All-Module-Clock Stop Mode

When causing transition to self-refresh mode in all-module-clock stop mode, first cause transition to self-refresh mode according to the procedure shown in section 16.6.12.2, Procedure for Transition to and Recovery from Self-Refresh Mode, and then make settings to cause transition to all-module-clock stop mode.

After canceling all-module-clock stop mode, follow the procedure shown in section 16.6.12.2, Procedure for Transition to and Recovery from Self-Refresh Mode.

For details of transition and cancellation of all-module-clock stop mode, refer to section 11, Low Power Consumption.

(2) Self-Refresh in Software Standby Mode

When causing transition to self-refresh mode in software standby mode, first cause transition to self-refresh mode

according to the procedure shown in section 16.6.12.2, Procedure for Transition to and Recovery from Self-Refresh Mode, and then make settings to cause transition to software standby mode. In software standby mode, set the output port enable bit (OPE) in the standby control register (SBYCR) to 1 to hold the output state of the address bus and bus control signals.

After canceling software standby mode, follow the procedure shown in section 16.6.12.2, Procedure for Transition to and Recovery from Self-Refresh Mode.

For details of transition and cancellation of software standby mode, refer to section 11, Low Power Consumption.

(3) Self-Refresh in Deep Software Standby Mode

Transition to deep software standby mode is performed via software standby mode. On transition to deep software standby mode from software standby mode, the state of pins remains unchanged. Therefore, transition to self-refresh mode in deep software standby mode can be made according to the same procedure as that in software standby mode. In deep software standby mode, however, additional setting is necessary to cause transition to self-refresh mode; it is necessary to set the I/O port keep bit (IOKEEP) in the deep software standby control register (DPSBYCR) to 1.

Since the SDRAMC is internally reset by an internal reset signal when deep software standby mode is canceled, the SDRAM control registers need to be set again. After canceling software standby mode, follow the procedure shown below to cancel self-refresh mode. Figure 16.53 shows self-refresh timing in deep software standby mode.

For details of transition and cancellation of deep software standby mode, refer to section 11, Low Power Consumption.

1. In deep software standby mode, the CKE signal output remains low according to the IOKEEP setting in DPSBYCR.
2. Start clock supply to SDRAMC.
3. Set the SDRAM control registers (SDCMOD, SDAMOD, SDADR, and SDTR) again, which have been initialized by an internal reset upon transition to deep software standby mode, and then enable auto-refresh operation (RFEN bit in SDRFEN = 1).
4. Check that all the status bits in SDSR are set to 0 and set the SFEN bit in SDSELF to 1 to set self-refresh mode again.
5. Modify port settings for the SDRAM interface according to the procedure below.
 - (1) Set the enable bits for the SDRAM pins (PFBCR1.MDSDE and PFBCR1.DQM1E in PFBCR1) to 1 to set the ports for SDRAM again.
 - (2) Set the enable bit for the SDCLK pin (PFBCR1.SDCLKE in PFBCR1) to 1 to enable SDCLK pin output again.
 - (3) Set the IOKEEP bit in DPSBYCR to 0 to release the I/O ports from the held state.
6. Set the PSTOP0 bit in SCKCR to 0 to start clock supply to the SDRAM via the SDCLK pin.
7. Check that all the status bits in SDSR are set to 0 and set the SFEN bit in SDSELF to 0 to cancel self-refresh mode.

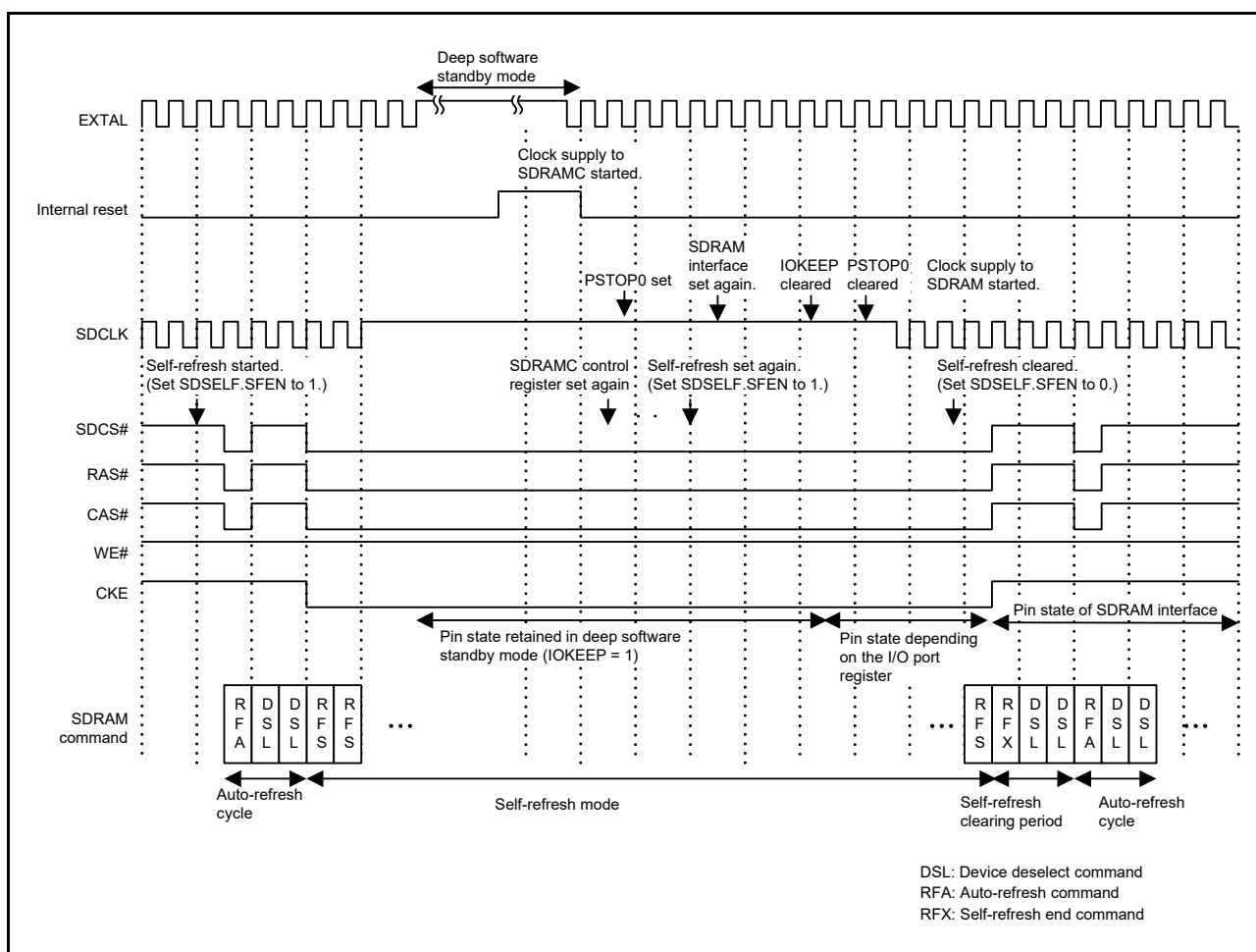


Figure 16.53 Timing Example of Self-Refresh Cycle (in Deep Software Standby Mode)

16.6.8 Auto-Refresh

The auto-refresh cycle can be started by setting the auto-refresh operation enable bit (RFEN) in the SDRAM auto-refresh control register (SDRFEN) to 1. Once the cycle is started, refresh requests are generated at fixed intervals determined by the refresh counter to start the auto-refresh cycle. However, since refresh requests are not accepted during read/write access, the auto-refresh cycle may be suspended. If an auto-refresh request is issued during consecutive access to the SDRAM, the auto-refresh cycle starts after bus access in response to a single transfer request from the bus master is completed.

If an SDRAM access and a refresh request are generated at the same time, the refresh request takes precedence. A CS area access and a refresh request can be made at the same if the SDCS#, RAS#, CAS#, WE#, and CKE signals, which are necessary for issuing the refresh command, are exclusively provided for SDRAM access.

When the RFEN bit in SDRFEN is set to 1 again after the auto-refresh cycle is started, a refresh request is generated.

However, if a request is made during read/write access, a request is actually generated when access is completed.

The refresh counter is halted during self-refresh operation. After recovery from self-refresh mode, the auto-refresh cycle is started and the counter value is reset thus resuming the counter operation.

Figure 16.54 shows an example of the timing of an auto-refresh cycle.

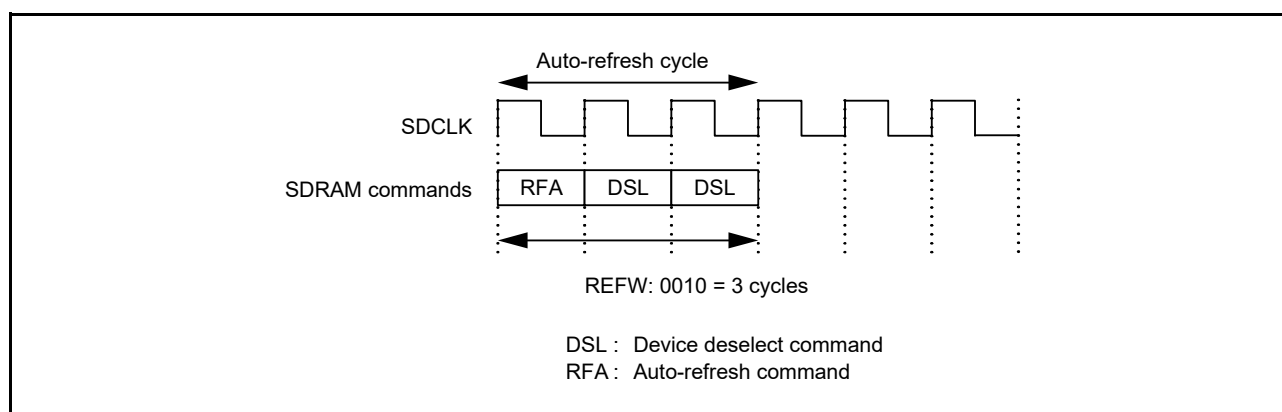


Figure 16.54 Timing Example of Auto-Refresh Cycle (1)

Figure 16.55 and Figure 16.56 show examples of operation when an auto-refresh request is generated during single access and continuous access, respectively.

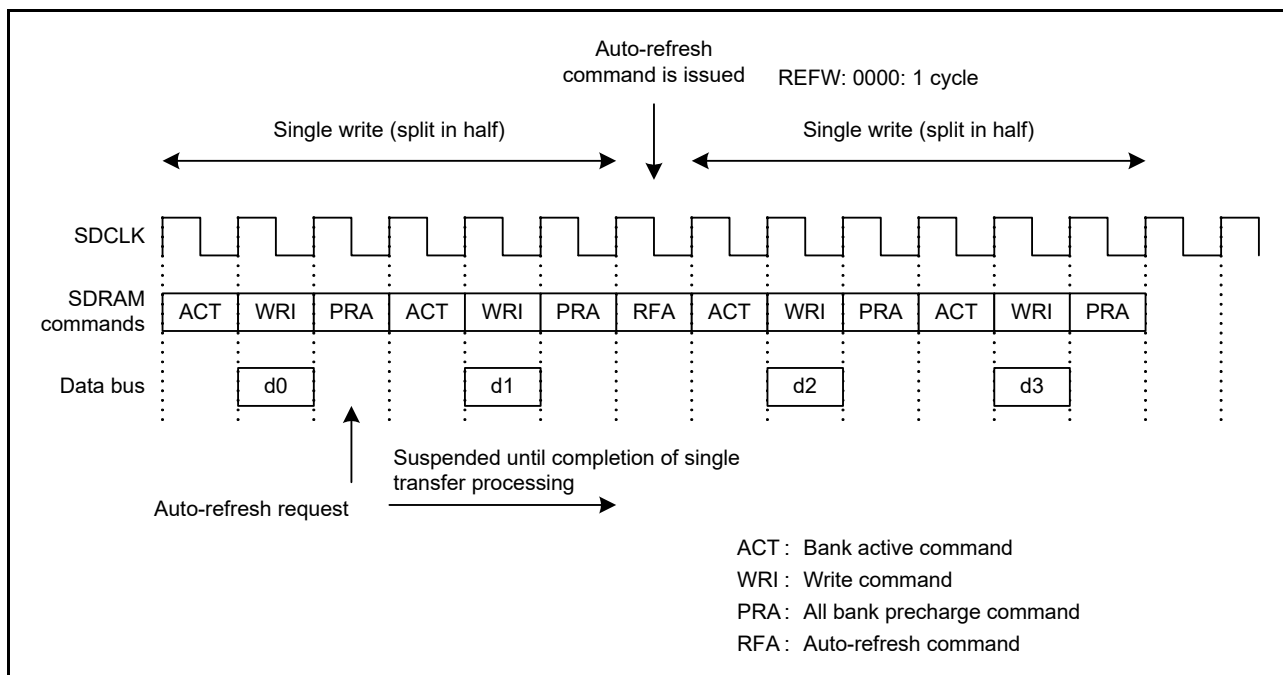


Figure 16.55 Timing Example of Auto-Refresh Cycle (2) (Auto-Refresh Request is Made during Single Access)

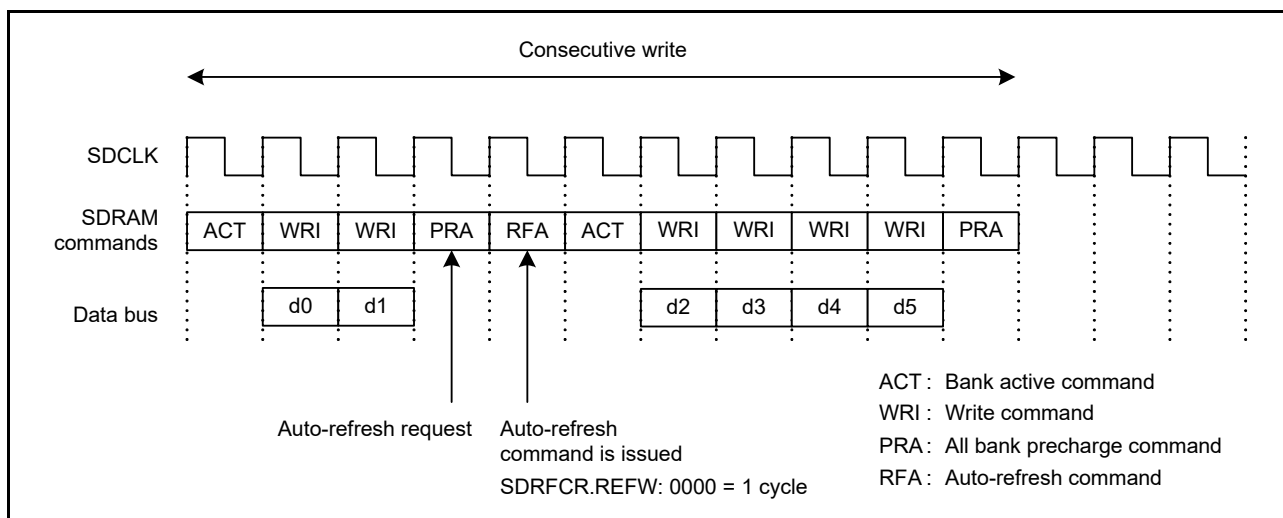


Figure 16.56 Timing Example of Auto-Refresh Cycle (3) (Auto-Refresh Request is Made during Continuous Access)

16.6.9 Initialization Sequencer

The SDRAMC has a sequencer to issue SDRAM initialization commands. After a reset, the initialization sequence must be activated without fail; the operation is not guaranteed if the SDRAM is not initialized.

The SDRAM initialization sequencer issues an all-bank-precharge command followed by auto-refresh commands n times ($n = 1$ to 15). The SDRAM initialization sequence timing can be set using the SDRAM initialization register (SDIR). The SDRAM initialization sequence can be activated using the SDRAM initialization sequence control register (SDICR). These registers should be set only when the conditions listed in Table 16.15, Conditions for Register Modification.

Figure 16.57 shows a timing example of the SDRAM initialization sequence. When the ARFC[3:0] bits in SDIR are set so that auto-refresh operation is performed two or more times, auto-refresh cycles are repeated in the initialization sequence accordingly.

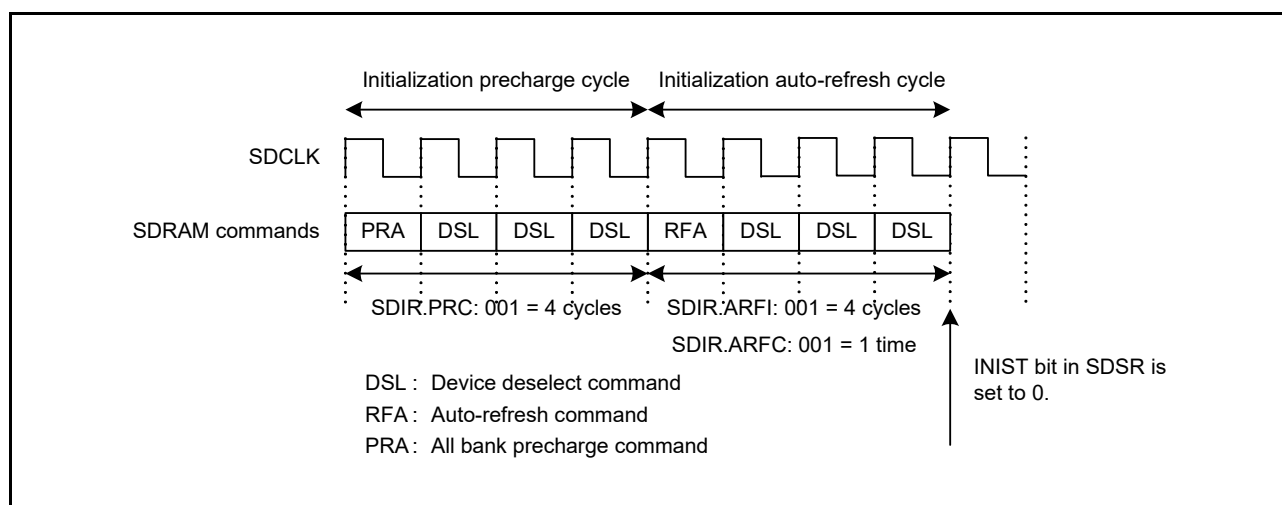


Figure 16.57 Timing Example of SDRAM Initialization Sequence

16.6.10 Read/Write Access

The SDRAMC controls read/write access in the following two modes.

- Single access mode: the row address is output each time data is accessed
- Consecutive access mode: when the same row address is accessed consecutively, only the column address is changed after the row address is output, enabling quick data access.

Consecutive SDRAM access is enabled by setting the continuous access enable bit (BE) in SDRAM access mode register (SDAMOD) to 1 in EXDMAC cluster transfer or block transfer in single address mode.

If the data size for a single transfer by the EXDMAC is less than the width of the external bus, and when bus access for a single transfer request ends once, in the same way as for non-aligned access, operation with consecutive access becomes possible.

When the above condition is not satisfied, the setting for consecutive-access mode is prohibited, and operation is not guaranteed if the setting is made.

Furthermore, setting the SDRAMC column-latency setting bits (CL[2:0]) in SDTR to 1 (CL = 1) in consecutive-access mode is prohibited, and operation is not guaranteed if this setting is made.

When the BE bit in SDAMOD is 0, single access is used in both cluster transfer for the EXDMAC and block transfer in single address mode.

(1) Single Access

Figure 16.58 and Figure 16.59 show examples of the timing of single reading and Figure 16.60 shows an example of the timing of single writing. The specific access timing depends on the SDRAM timing register (SDTR) settings. For details, refer to section 16.6.12.3, Timing Register Settings and Access Timing.

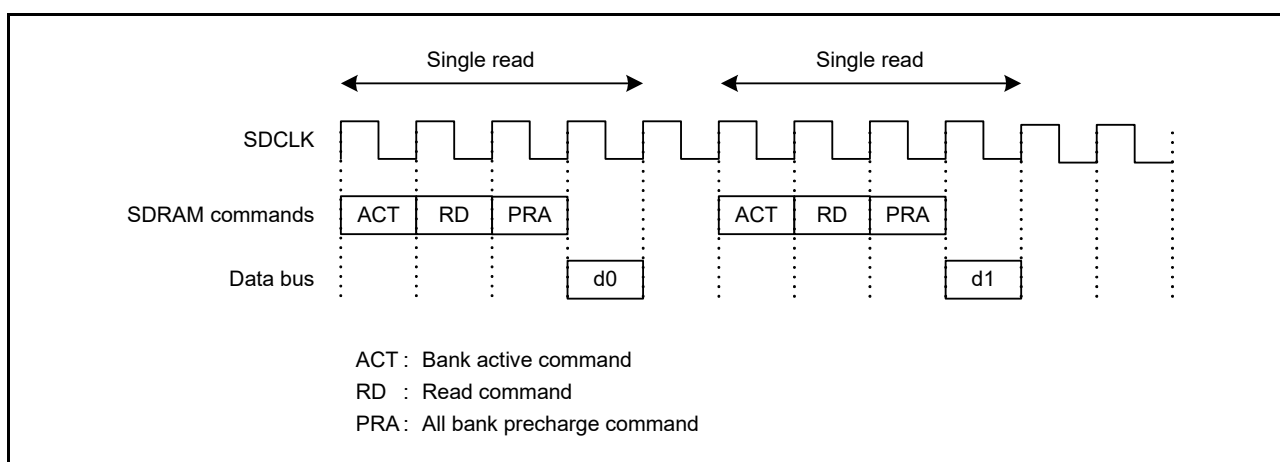


Figure 16.58 Timing Example of Single Read (SDTR.CL[2:0] = 010b: 2 Cycles)

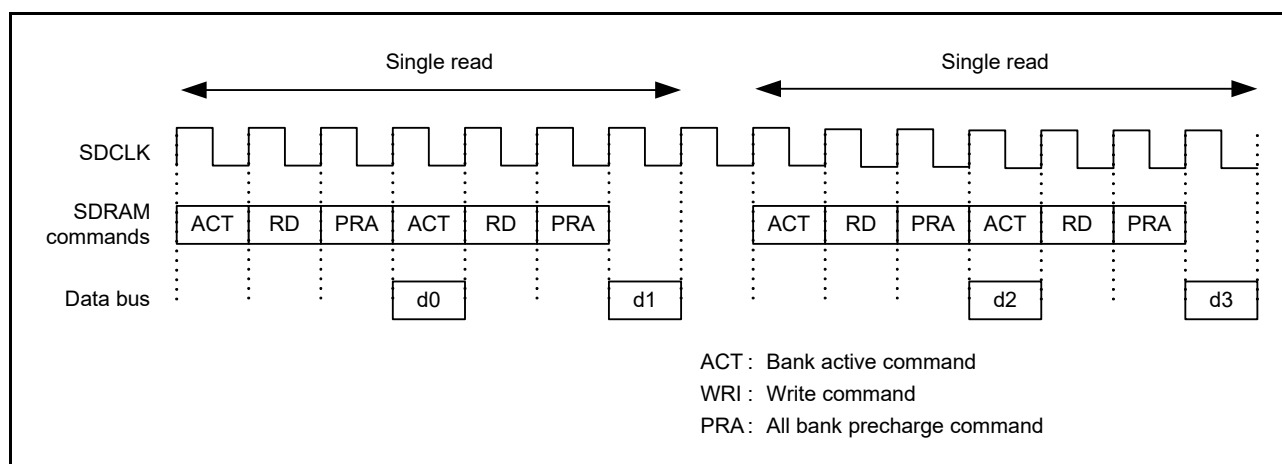


Figure 16.59 Timing Example of Single Read (Cluster Transfer by EXDMAC or Block Transfer in Single Address Mode with SDAMOD.BE = 0 and SDTR.CL[2:0] = 010b: 2 Cycles)

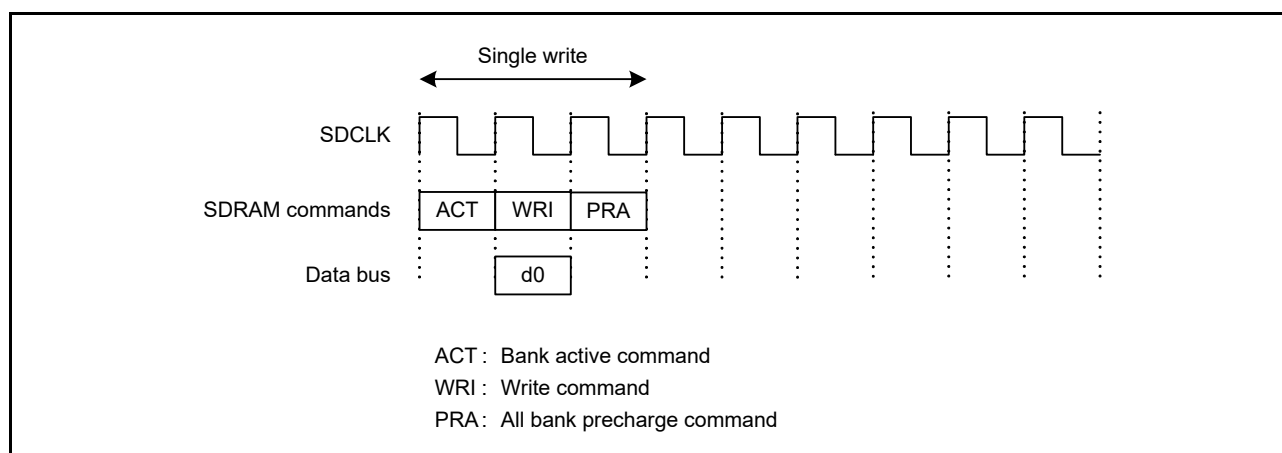


Figure 16.60 Timing Example of Single Write (when the Shortest Timing is Set)

(2) Consecutive Access

Figure 16.61 and Figure 16.62 show timing examples of consecutive read and consecutive write for four data, respectively.

When the SDRAM row address changes during transfer, the pertinent row is automatically deactivated or activated appropriately.

Figure 16.63 shows a timing example of consecutive write in which the row address changes.

The specific access timing depends on the SDRAM timing register (SDTR) settings. For details, refer to section 16.6.12.3, Timing Register Settings and Access Timing.

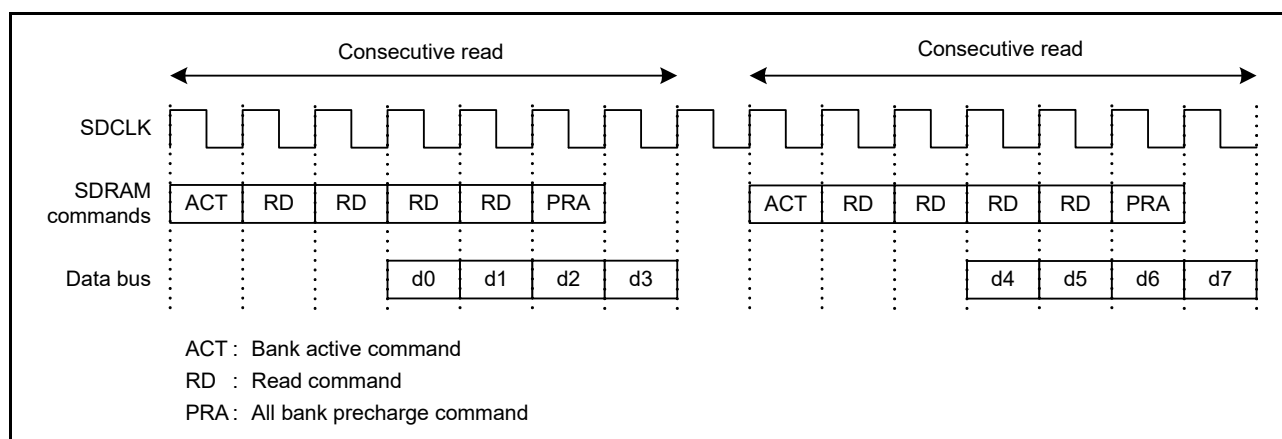


Figure 16.61 Timing Example of Consecutive Read (SDAMOD.BE = 1 and SDTR.CL[2:0] = 010b: 2 Cycles)

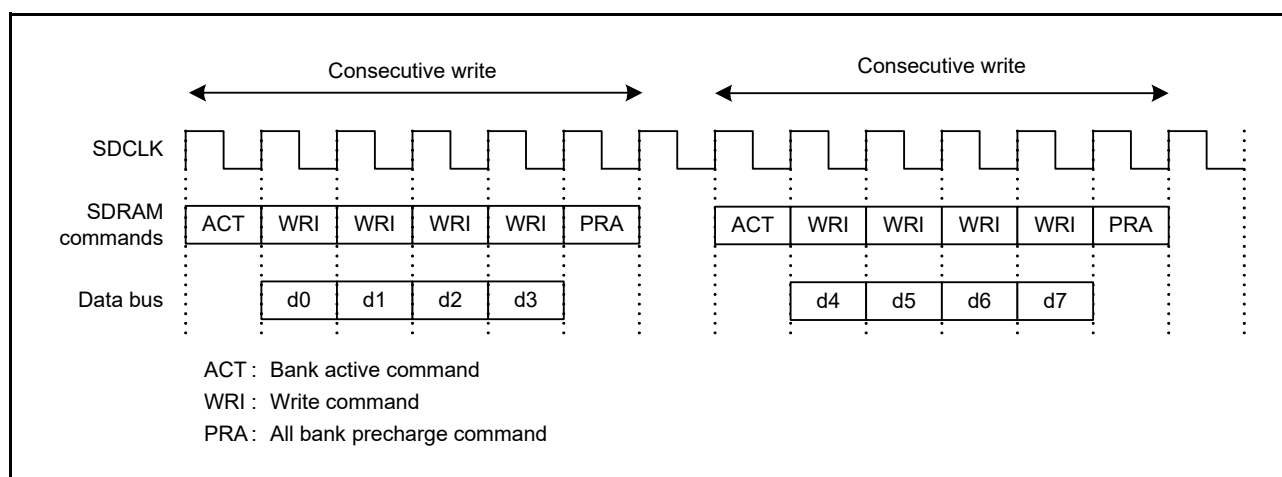


Figure 16.62 Timing Example of Consecutive Write (SDAMOD.BE = 1, when the Earliest Timing is Set)

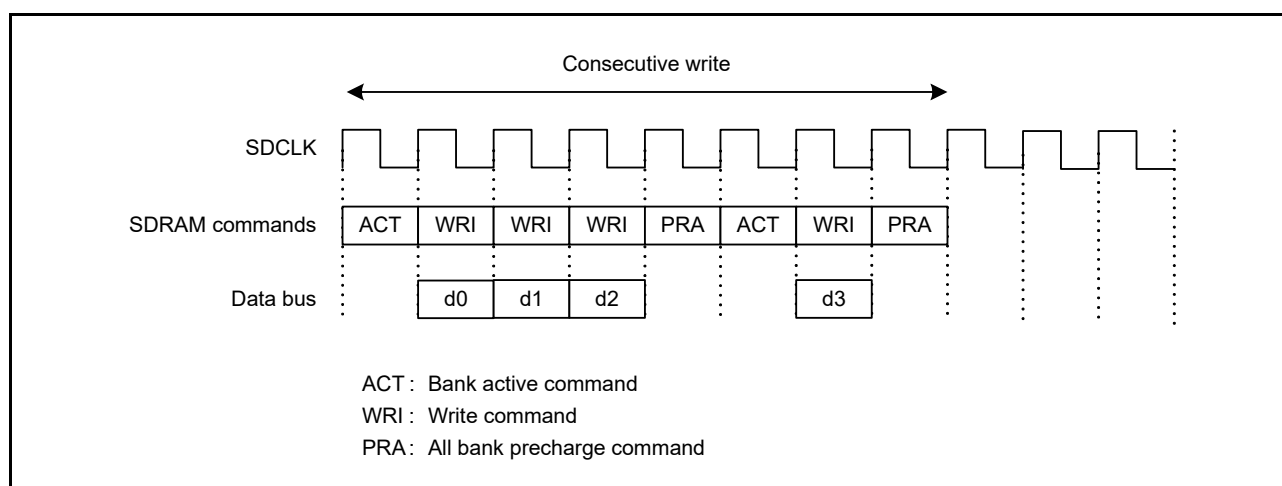


Figure 16.63 Timing Example of Consecutive Write (SDAMOD.BE = 1, when the Earliest Timing is Set) in which the Row Address Changes

16.6.11 Setting Mode Register

Setting the SDRAM mode register (SDMOD) allows the mode register set command to be issued to SDRAM and the value set in the MR[14:0] bits in SDMOD to be output to the lower bits of the address; specifically, to the A14 to A0 for 8-bit bus width, A15 to A1 for 16-bit bus width, and A16 to A2 for 32-bit bus width. Therefore, set the SDCCR.BSIZE[1:0] bits before setting the mode register, to determine the data bus width of the SDRAM.

Figure 16.64 shows the mode register setting timing.

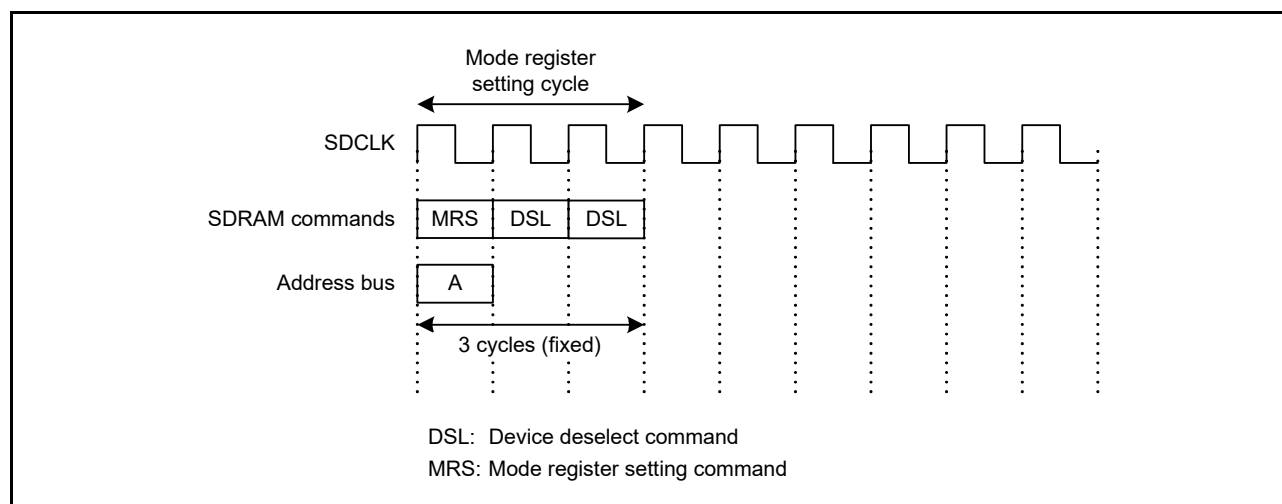


Figure 16.64 Mode Register Setting Timing

16.6.12 SDRAMC Setting Examples

This section describes the SDRAMC setting procedure, timing register setting examples, and procedure for transition to and recovery from self-refresh mode.

16.6.12.1 SDRAMC Access Procedure

Figure 16.65 shows the SDRAMC setting procedure.

The shown specifications including a power-up sequence may be different from that from the specifications of the SDRAM actually used; the system should be designed after reviewing the specifications of the SDRAM.

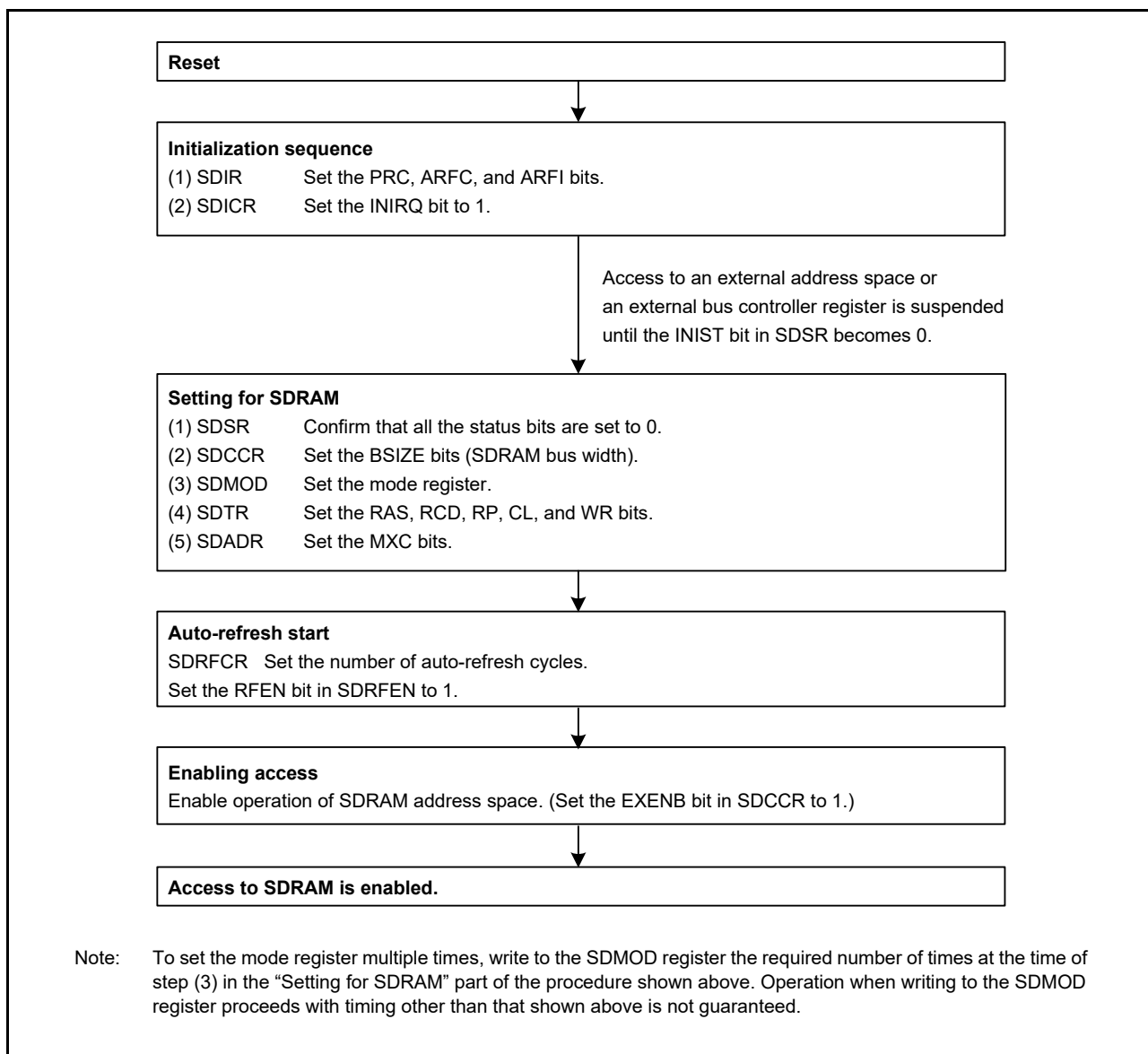


Figure 16.65 SDRAMC Setting Procedure

16.6.12.2 Procedure for Transition to and Recovery from Self-Refresh Mode

Figure 16.66 shows the procedure for transition to and recovery from self-refresh mode.

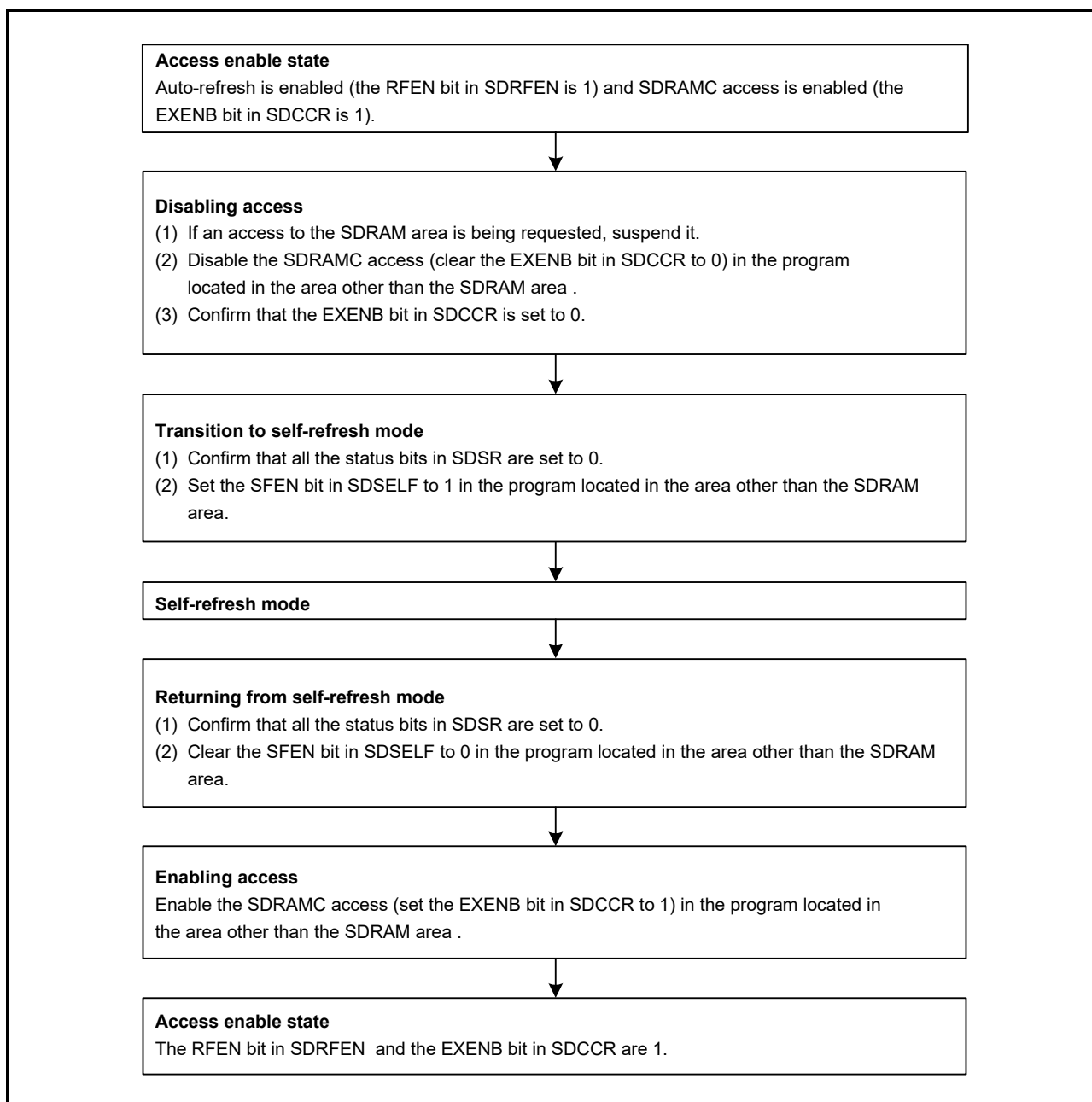


Figure 16.66 Procedure for Transition to and Recovery from Self-Refresh Mode

Note: Transition to and recovery from self-refresh mode requires SDRAM access to be disabled. Accordingly, transition to and recovery from self-refresh mode cannot be made during SDRAM access. The instructions below should be followed in programming.

- Before making transition to self-refresh mode, disable the access to the SDRAM area.
- During transition to self-refresh mode, self-refresh operation, and recovery from self-refresh mode, do not allow any operand access or instruction fetch (including prefetch) to the SDRAM area to be generated.

Figure 16.67 shows the procedure for transition to and recovery from self-refresh mode in deep software standby mode.

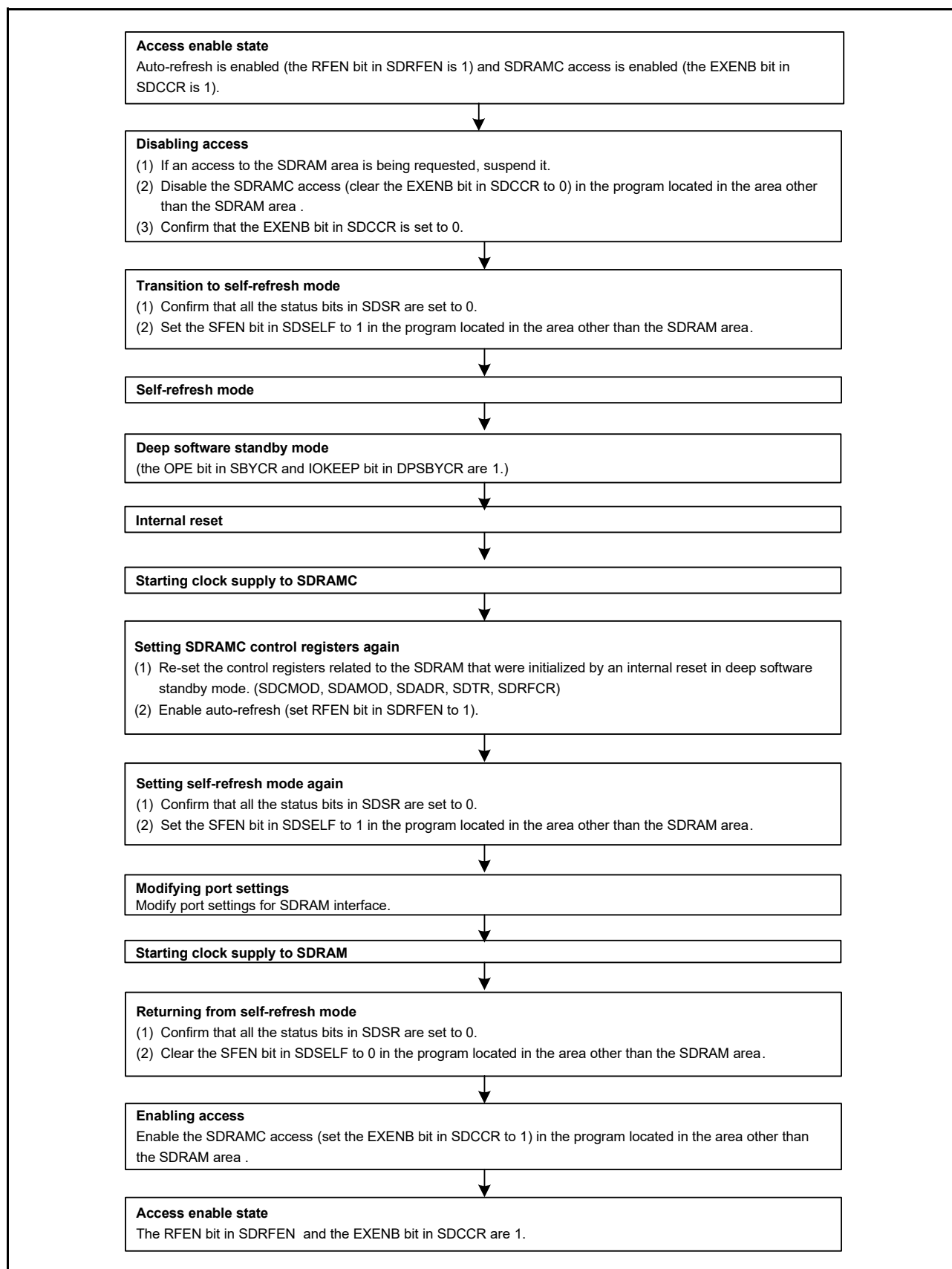


Figure 16.67 Procedure for Transition to and Recovery from Self-Refresh Mode in Deep Software Standby Mode

16.6.12.3 Timing Register Settings and Access Timing

This section describes the relationship between the read/write timing and the settings of the SDRAM timing register (SDTR).

(1) Single Read Timing Examples

Figure 16.68 to Figure 16.72 show the relationship between the single read timing and the SDTR register settings. Table 16.16 lists the correspondence between the figures and the SDTR register settings.

During read access, the next bus access is enabled two cycles after the read data becomes valid at the earliest. However, if two or more accesses occur for one transfer request, the next bus access is enabled one cycle after the read data becomes valid at the earliest, as shown in Figure 16.72.

Table 16.16 Correspondence between Target Figures and SDTR Register Settings (Single Read Timing)

Figure No.	RAS[2:0]	Number of Cycles	RCD[1:0]	Number of Cycles	RP[2:0]	Number of Cycles	CL[2:0]	Number of Cycles
	Settings		Settings		Settings		Settings	
Figure 16.68	010	3	00	1	001	2	010	2
Figure 16.69	000	1	01	2	001	2	010	2
Figure 16.70	000	1	01	2	001	2	011	3
Figure 16.71, Figure 16.72	010	3	00	1	000	1	010	2

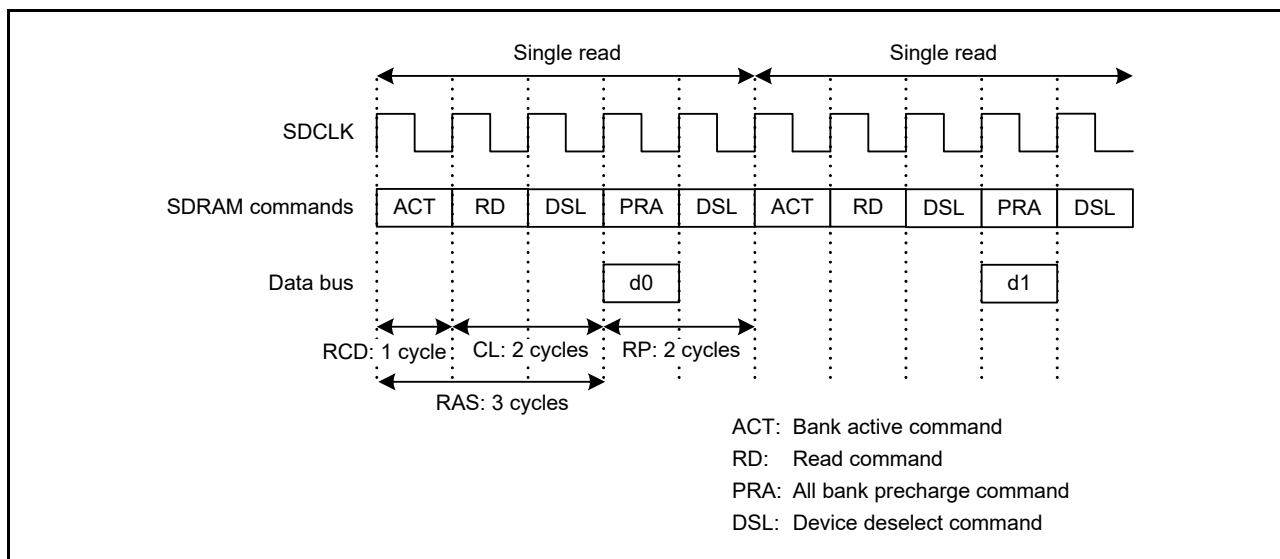


Figure 16.68 Timing Example of Single Read (1)

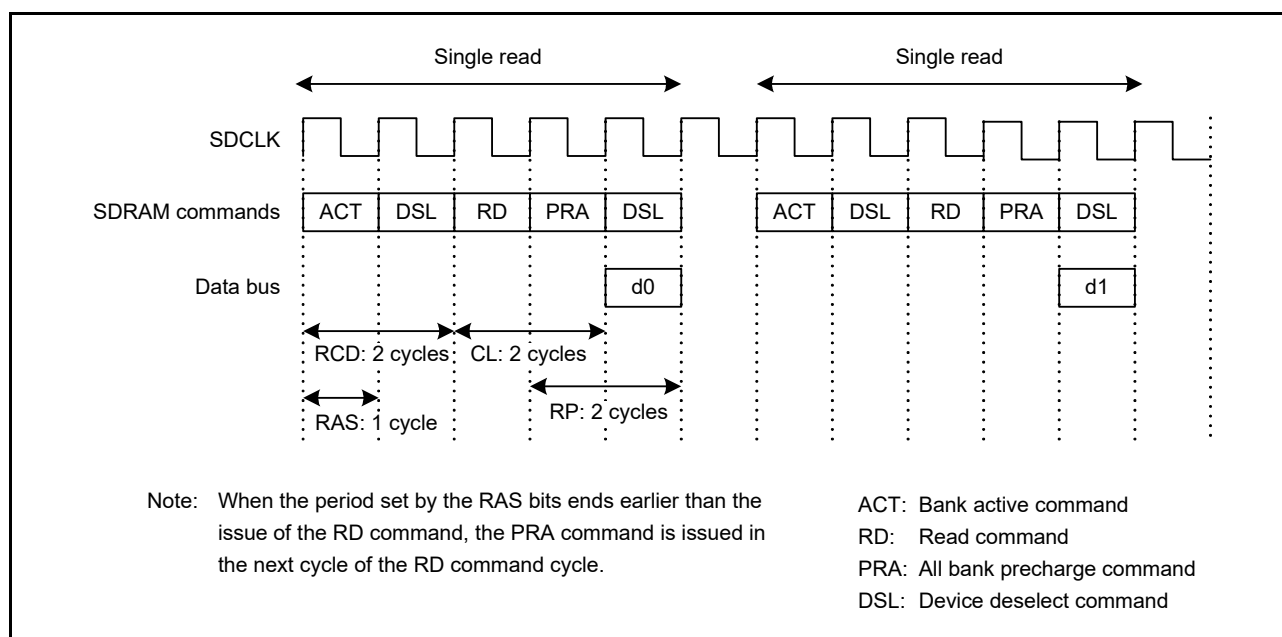


Figure 16.69 Timing Example of Single Read (2)

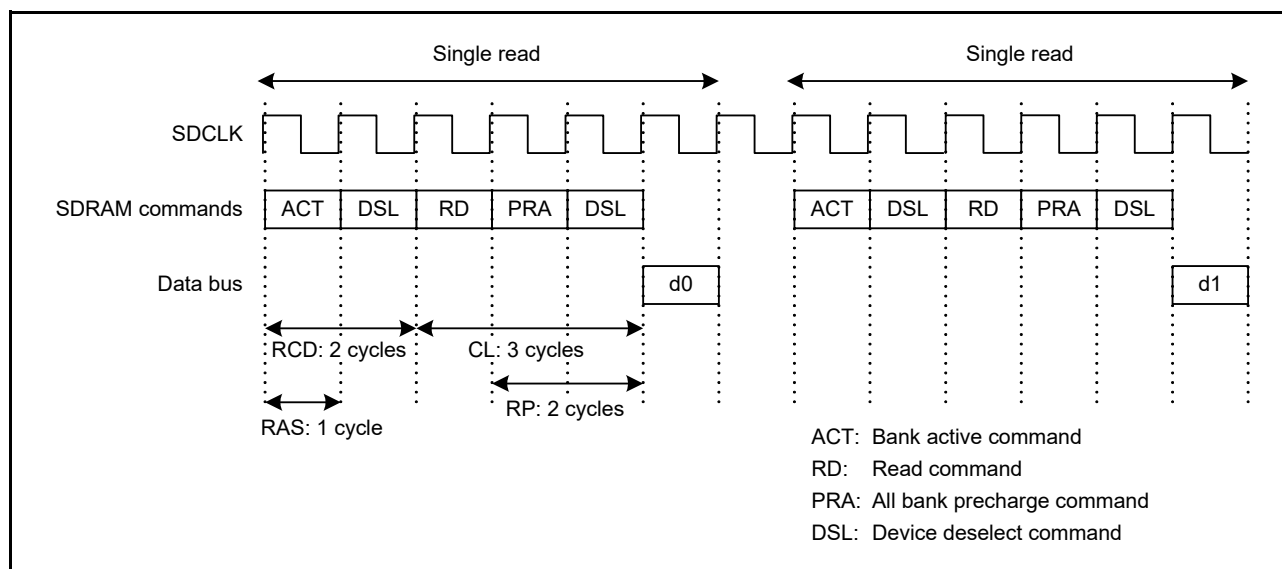


Figure 16.70 Timing Example of Single Read (3)

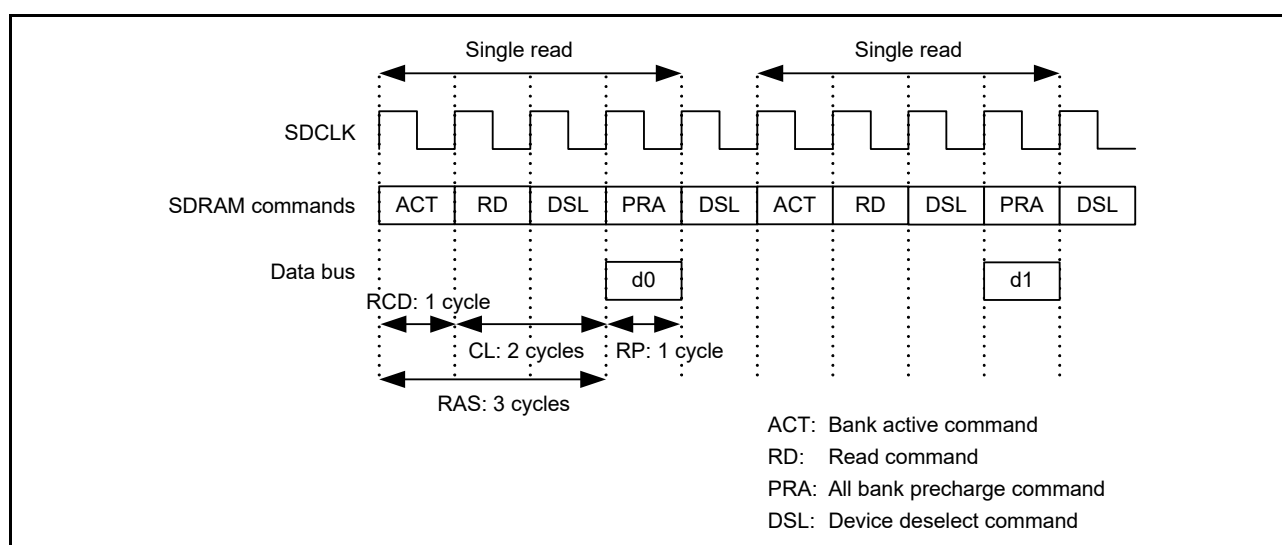


Figure 16.71 Timing Example of Single Read (4)

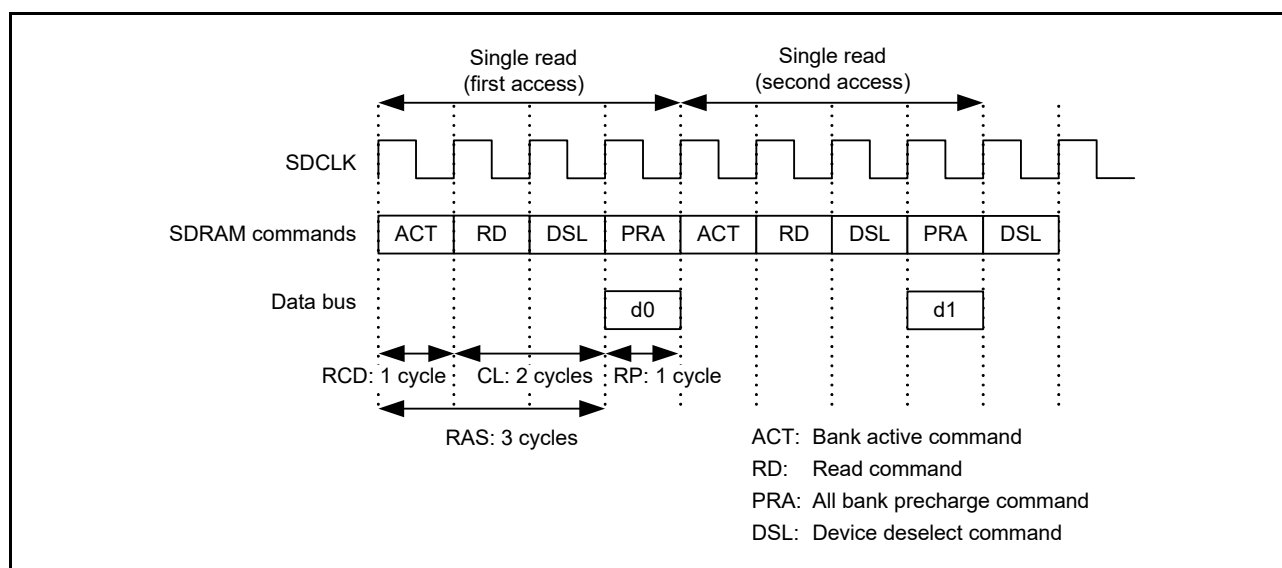


Figure 16.72 Timing Example of Single Read (5) (Two Bus Accesses Occur for One Transfer Request)

(2) Single Write Timing Examples

Figure 16.73 to Figure 16.77 show the relationship between the single write timing and the SDTR register settings.

Table 16.17 lists the correspondence between the figures and the SDTR register settings.

During write access, the next bus access is enabled two cycles after an all-bank-precharge command (PRA) is issued at the earliest. However, if two or more accesses occur for one transfer request, the next bus access is enabled one cycle after the PRA is issued at the earliest, as shown in Figure 16.77.

Table 16.17 Correspondence between Target Figures and SDTR Register Settings (Single Write Timing)

Figure No.	RAS[2:0]	Number of Cycles	RCD[1:0]	Number of Cycles	RP[2:0]	Number of Cycles	WR	Number of Cycles
	Settings		Settings		Settings		Settings	
Figure 16.73	010	3	00	1	001	2	0	1
Figure 16.74	000	1	01	2	001	2	0	1
Figure 16.75	000	1	01	2	001	2	1	2
Figure 16.76, Figure 16.77	010	3	00	1	000	1	0	1

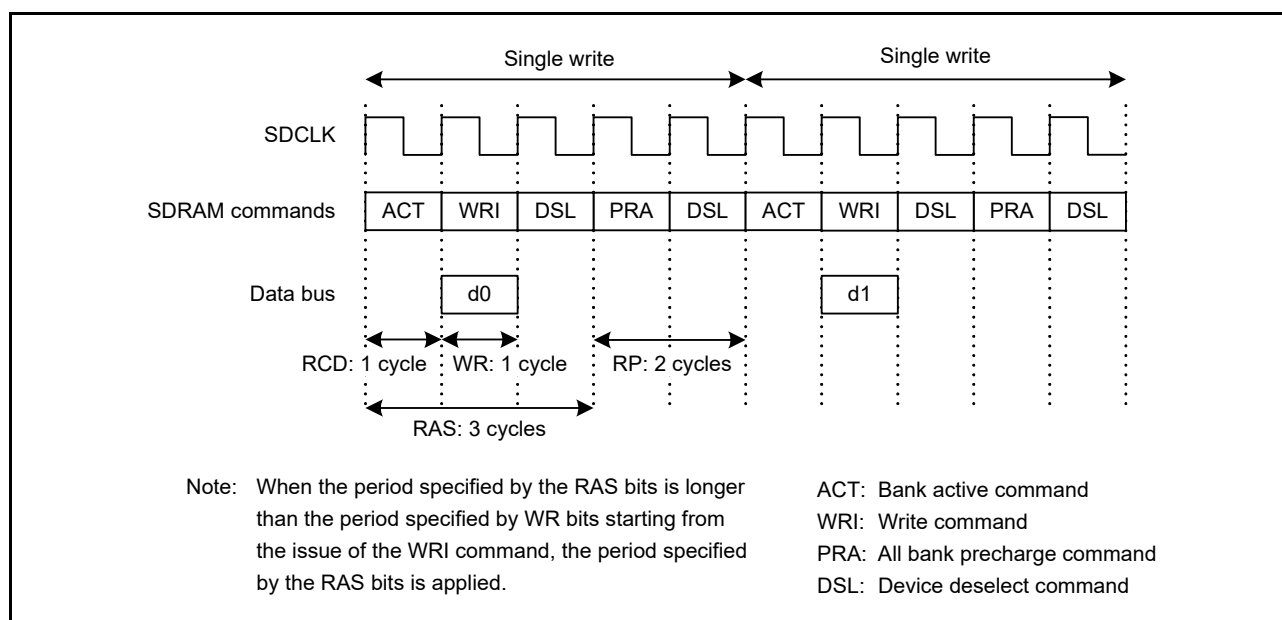


Figure 16.73 Timing Example of Single Write (1)

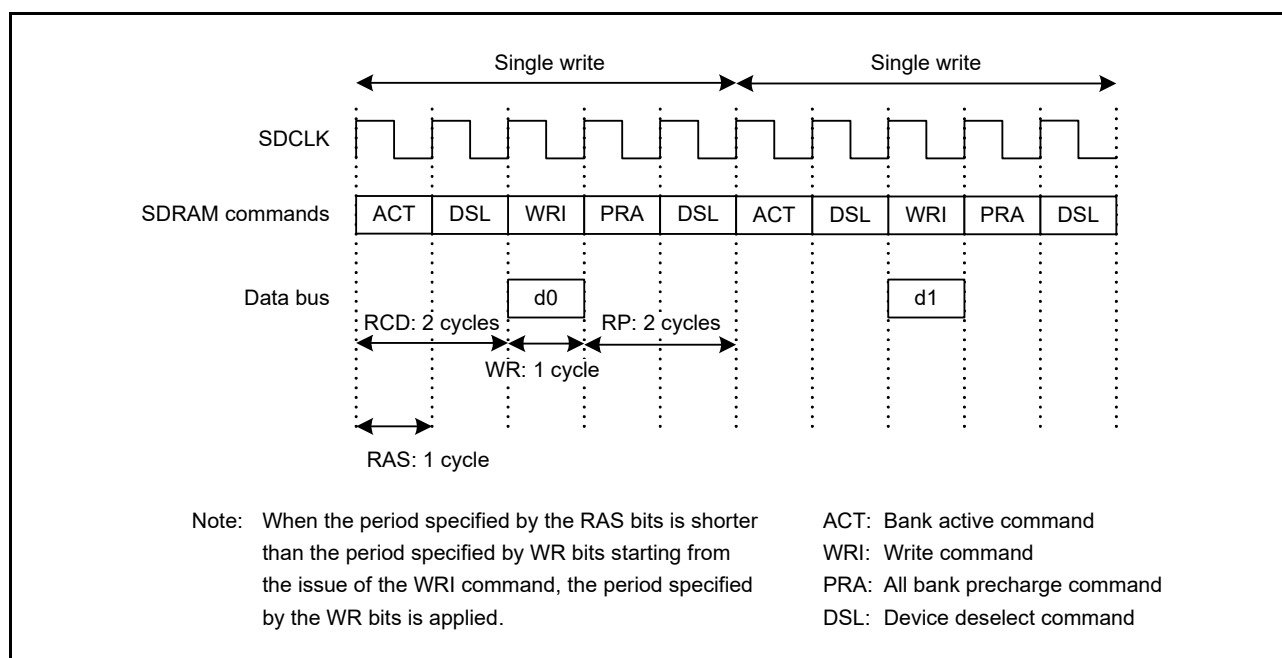


Figure 16.74 Timing Example of Single Write (2)

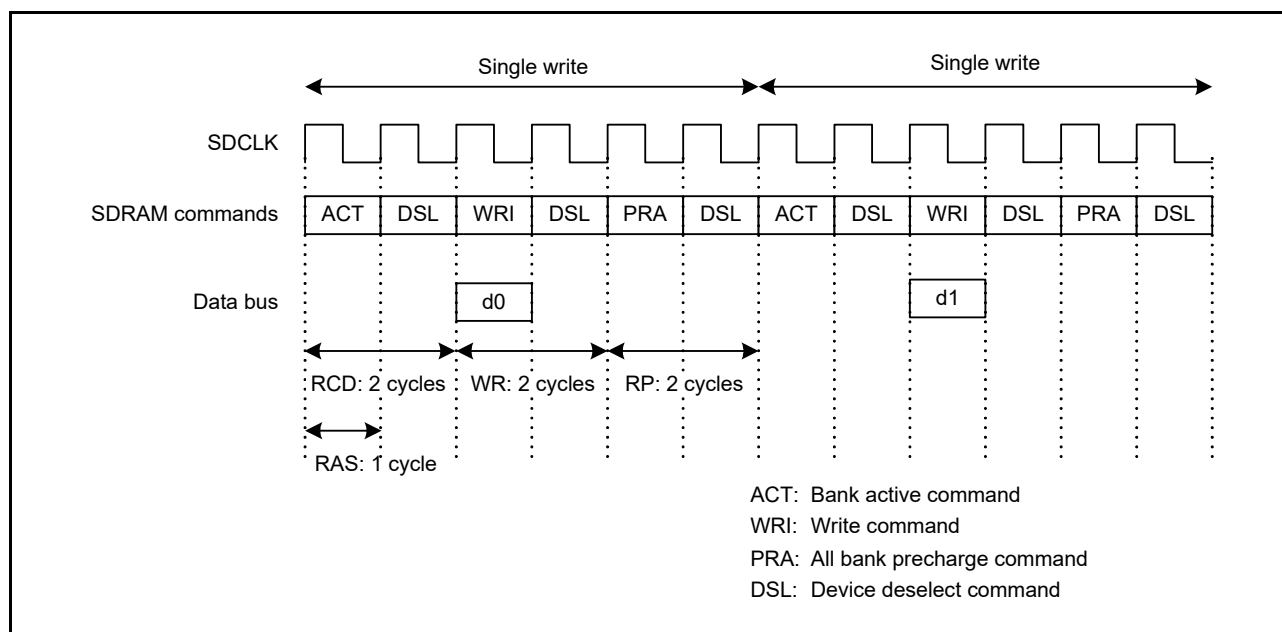


Figure 16.75 Timing Example of Single Write (3)

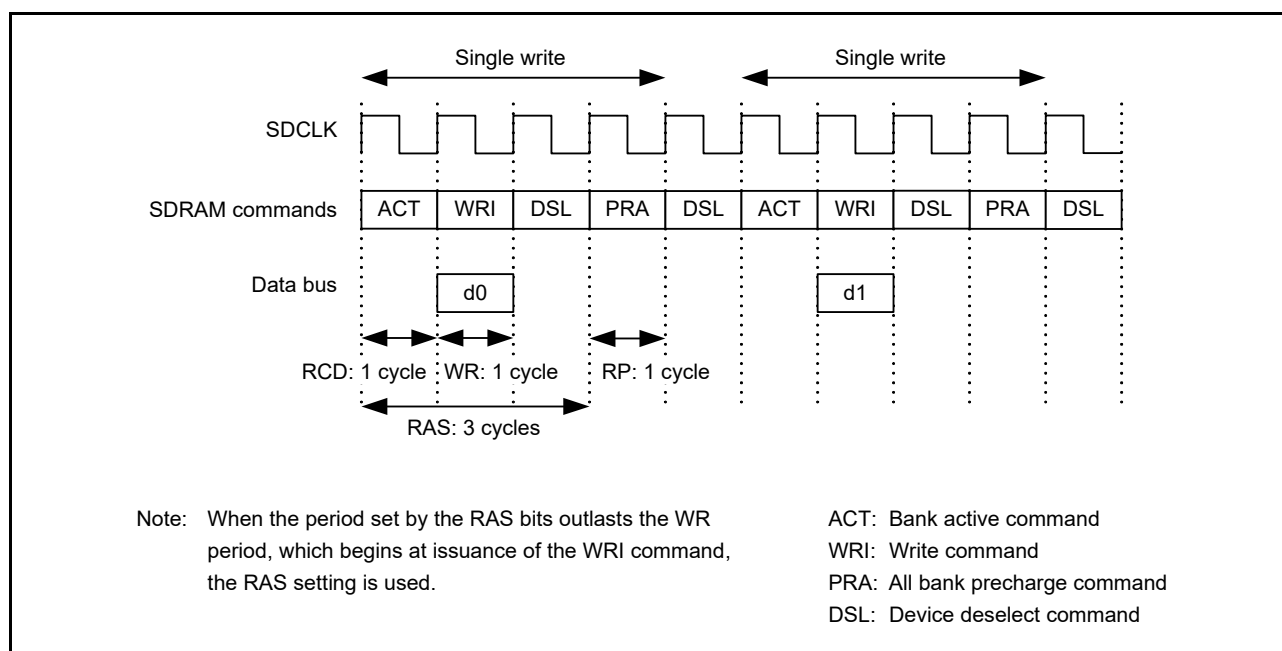


Figure 16.76 Timing Example of Single Write (4)

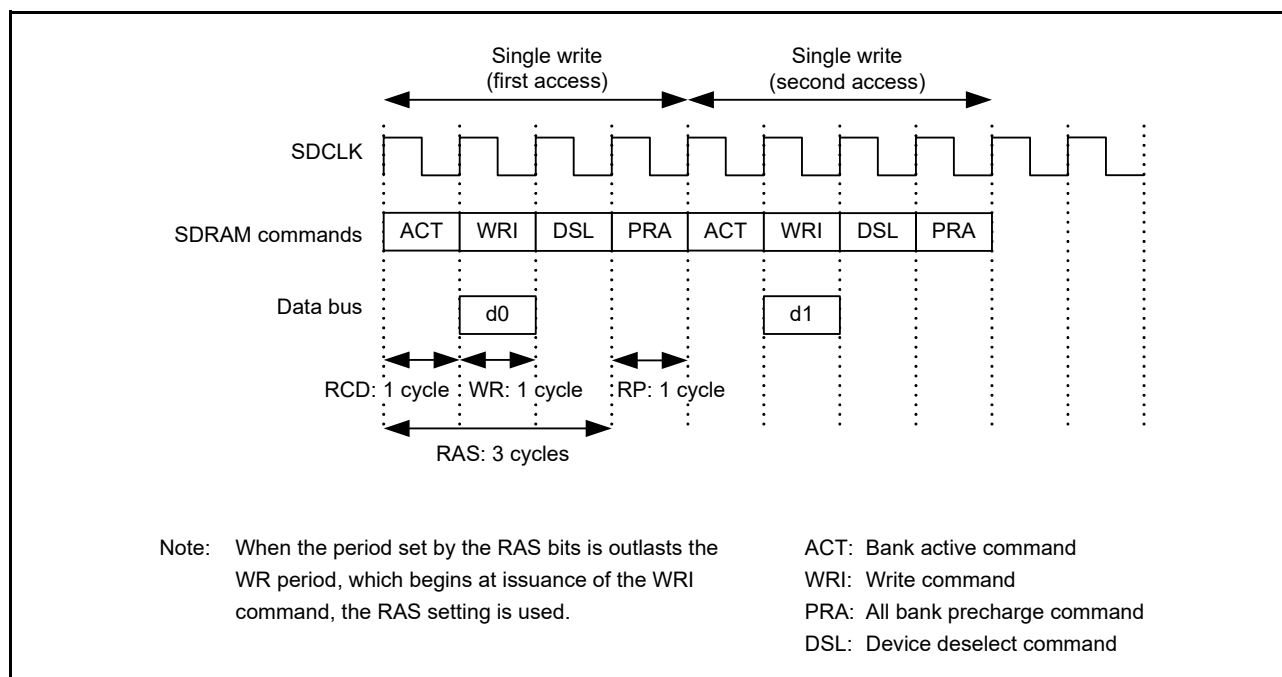


Figure 16.77 Timing Example of Single Write (5) (Two Bus Accesses Occur for One Transfer Request)

(3) Consecutive Read Timing Examples

Figure 16.78 to Figure 16.80 show the relationship between the consecutive read timing for four data and the SDTR register settings. Table 16.18 lists the correspondence between the figures and the SDTR register settings.

Table 16.18 Correspondence between Target Figures and SDTR Register Settings (Consecutive Read Timing)

Figure No.	RAS[2:0]	Number of Cycle	RCD[1:0]	Number of Cycle	RP[2:0]	Number of Cycle	CL[2:0]	Number of Cycle
	Settings		Settings		Settings		Settings	
Figure 16.78	010	3	00	1	001	2	010	2
Figure 16.79	000	1	01	2	001	2	010	2
Figure 16.80	000	1	01	2	001	2	011	3

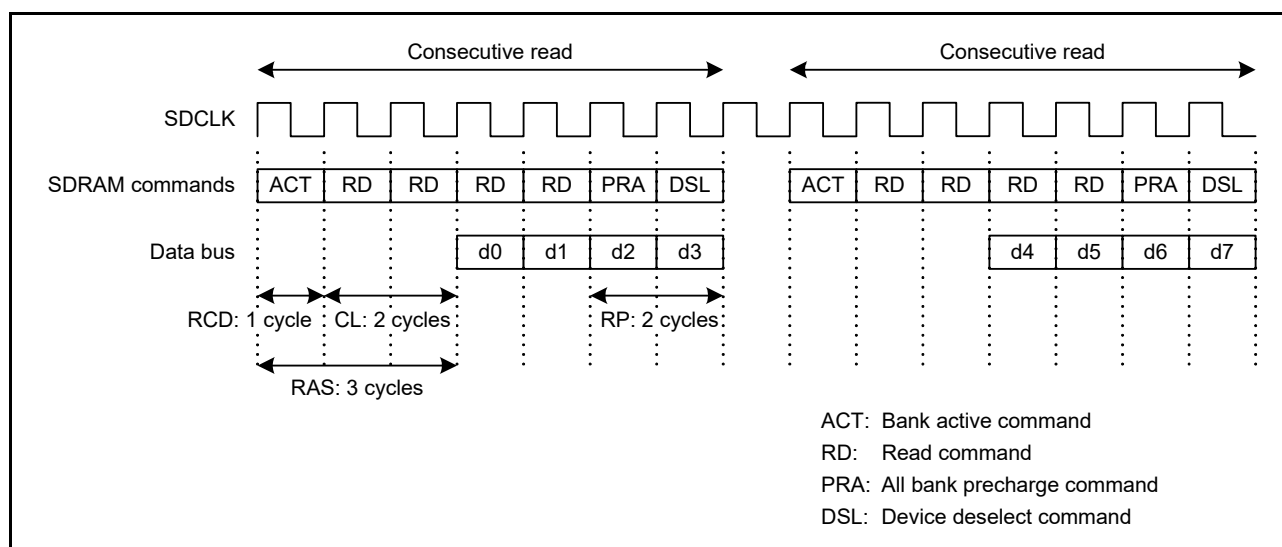


Figure 16.78 Timing Example of Consecutive Read (1)

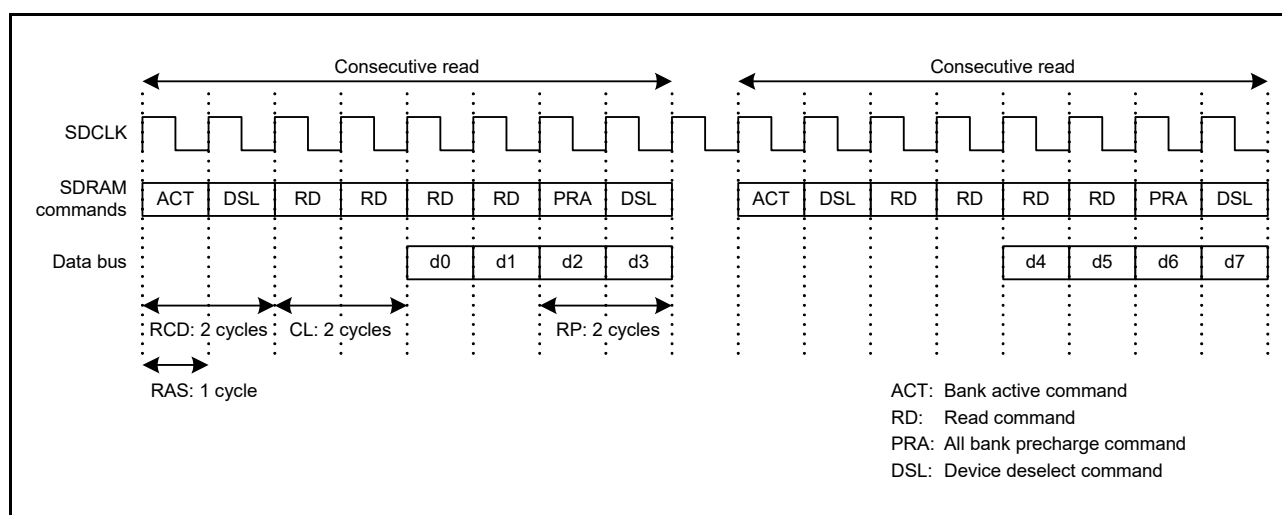


Figure 16.79 Timing Example of Consecutive Read (2)

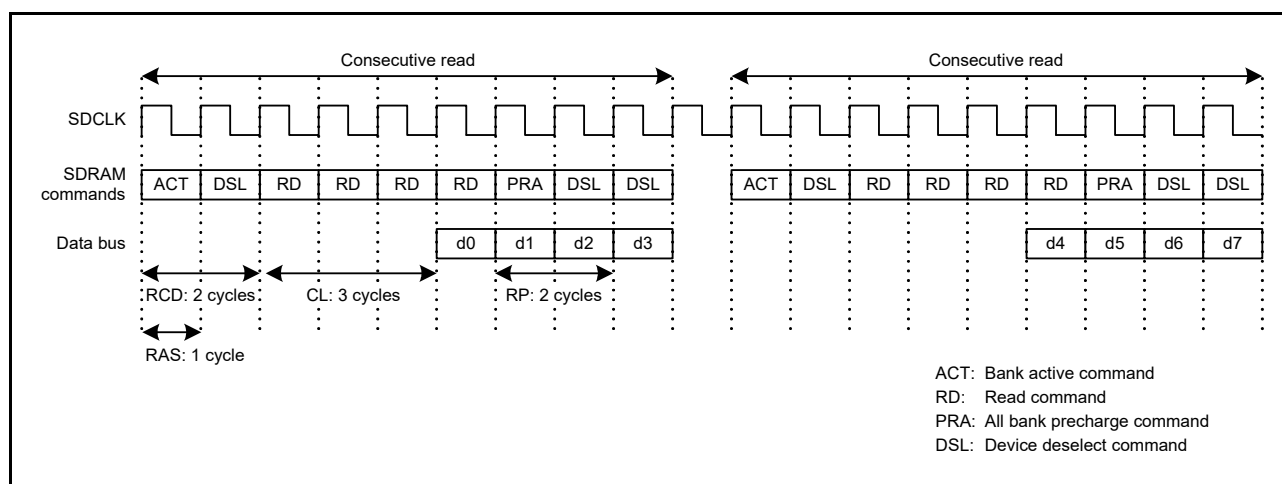


Figure 16.80 Timing Example of Consecutive Read (3)

(4) Consecutive Write Timing Examples

Figure 16.81 to Figure 16.83 show the relationship between the consecutive write timing for four data and the SDTR register settings. Table 16.19 lists the correspondence between the figures and the SDTR register settings.

Table 16.19 Correspondence between Target Figures and SDTR Register Settings (Consecutive Write Timing)

Figure No.	RAS[2:0]	Number of Cycles	RCD[1:0]	Number of Cycles	RP[2:0]	Number of Cycles	WR	Number of Cycles
	Settings		Settings				Settings	
Figure 16.81	010	3	00	1	001	2	0	1
Figure 16.82	000	1	01	2	001	2	0	1
Figure 16.83	000	1	01	2	001	2	1	2

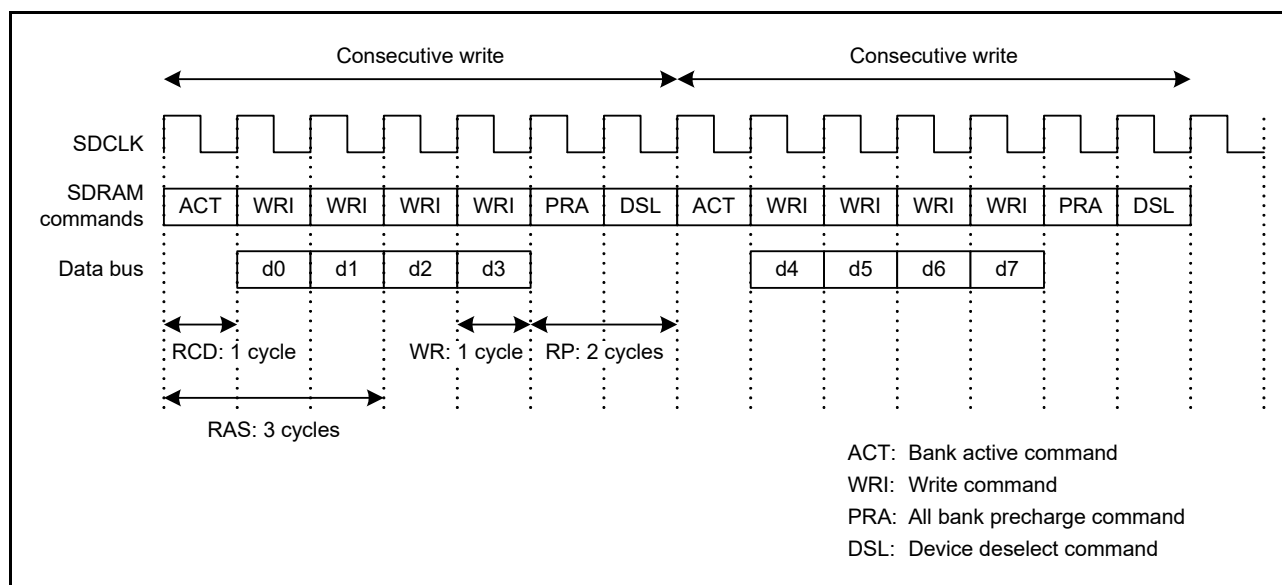


Figure 16.81 Timing Example of Consecutive Write (1)

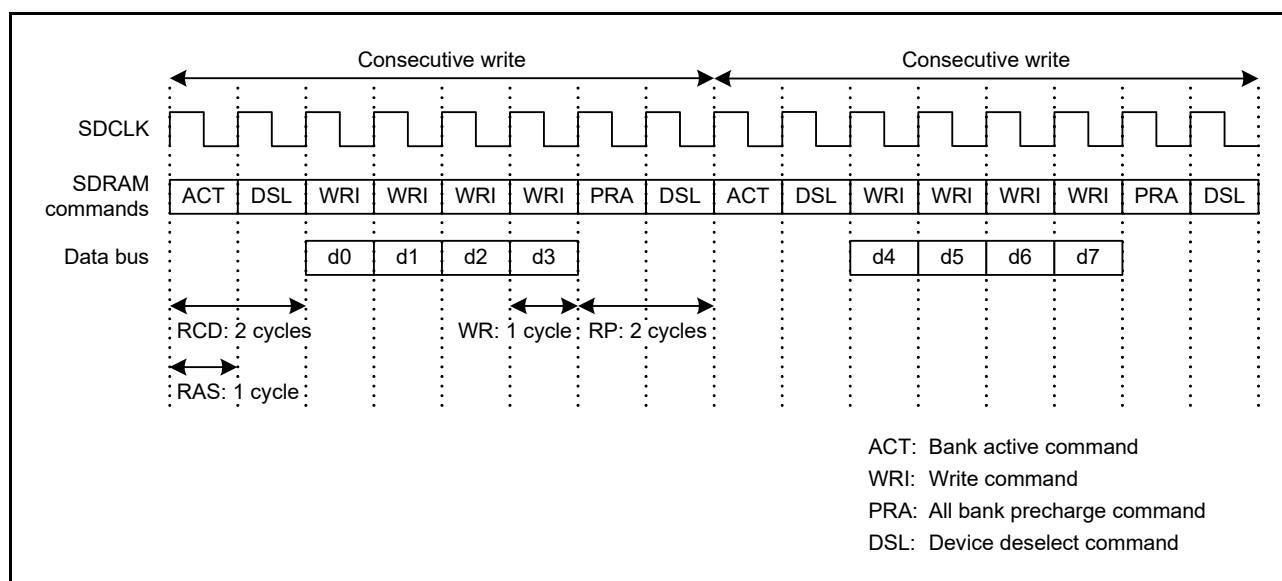


Figure 16.82 Timing Example of Consecutive Write (2)

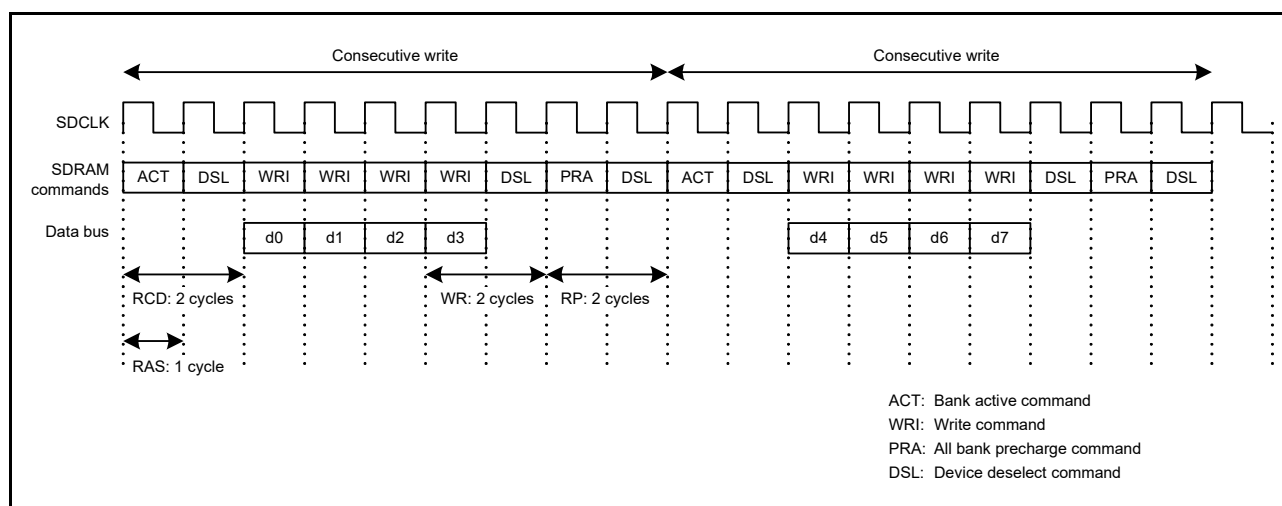


Figure 16.83 Timing Example of Consecutive Write (3)

16.6.13 Address Multiplexing

In the SDRAM space, row and column addresses are multiplexed. The size of the shift in a row address should be specified for address multiplexing by the address multiplex select bits (SDADR.MXC[1:0]) in the SDRAM address register (SDADR). Moreover, in the SDRAM space, the address precharge select command (precharge-sel) is output to the upper bits of column addresses. Table 16.20 lists the relationship between the SDADR.MXC[1:0] settings and the shift amount.

Table 16.20 Address Multiplexing

MXC [1:0]	Shift Amount	Data Bus Width	Address	Address Pins External to the Microcomputer																		
				A18	A17	A16	A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
00	8 bits	8 bits	Row	A26	A25	A24	A23	A22	A21	A20	A19	A18*	A17	A16	A15	A14	A13	A12	A11	A10	A9	A8
			Column	A26	A25	A24	A23	A22	A21	A20	A19	P	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
		16 bits	Row	A26	A25	A24	A23	A22	A21	A20	A19*	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9	A8
			Column	A26	A25	A24	A23	A22	A21	A20	P	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
		32 bits	Row	A26	A25	A24	A23	A22	A21	A20*	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9	A8
			Column	A26	A25	A24	A23	A22	A21	P	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
01	9 bits	8 bits	Row	—	A26	A25	A24	A23	A22	A21	A20	A20*	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9
			Column	—	A26	A25	A24	A23	A22	A21	A20	P	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
		16 bits	Row	—	A26	A25	A24	A23	A22	A21	A20*	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9
			Column	—	A26	A25	A24	A23	A22	A21	P	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
		32 bits	Row	—	A26	A25	A24	A23	A22	A21*	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9
			Column	—	A26	A25	A24	A23	A22	P	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
10	10 bits	8 bits	Row	—	—	A26	A25	A24	A23	A22	A21	A20*	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10
			Column	—	—	A26	A25	A24	A23	A22	A21	P	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
		16 bits	Row	—	—	A26	A25	A24	A23	A22	A21*	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10
			Column	—	—	A26	A25	A24	A23	A22	P	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
		32 bits	Row	—	—	A26	A25	A24	A23	A22*	A21	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10
			Column	—	—	A26	A25	A24	A23	P	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
11	11 bits	8 bits	Row	—	—	—	A26	A25	A24	A23	A22	A21*	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11
			Column	—	—	—	A26	A25	A24	A23	A10	P	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
		16 bits	Row	—	—	—	A26	A25	A24	A23	A22*	A21	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11
			Column	—	—	—	A26	A25	A24	A11	P	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
		32 bits	Row	—	—	—	A26	A25	A24	A23*	A22	A21	A20	A19	A18	A17	A16	A15	A14	A13	A12	A11
			Column	—	—	—	A26	A25	A12	P	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0

P: Precharge setting command (Precharge-sel) is output.

*: When the PALL command is issued, Precharge-sel = 1 (High) is output. When the Active command is issued, the corresponding address is output.

—: Don't care

16.6.14 Examples for Connecting with SDRAMs

16.6.14.1 32-Bit Bus Space

Figure 16.84 shows an example for connecting to two 512-Mbit SDRAMs with 13-bit row address, 10-bit column address and 16-bit bus.

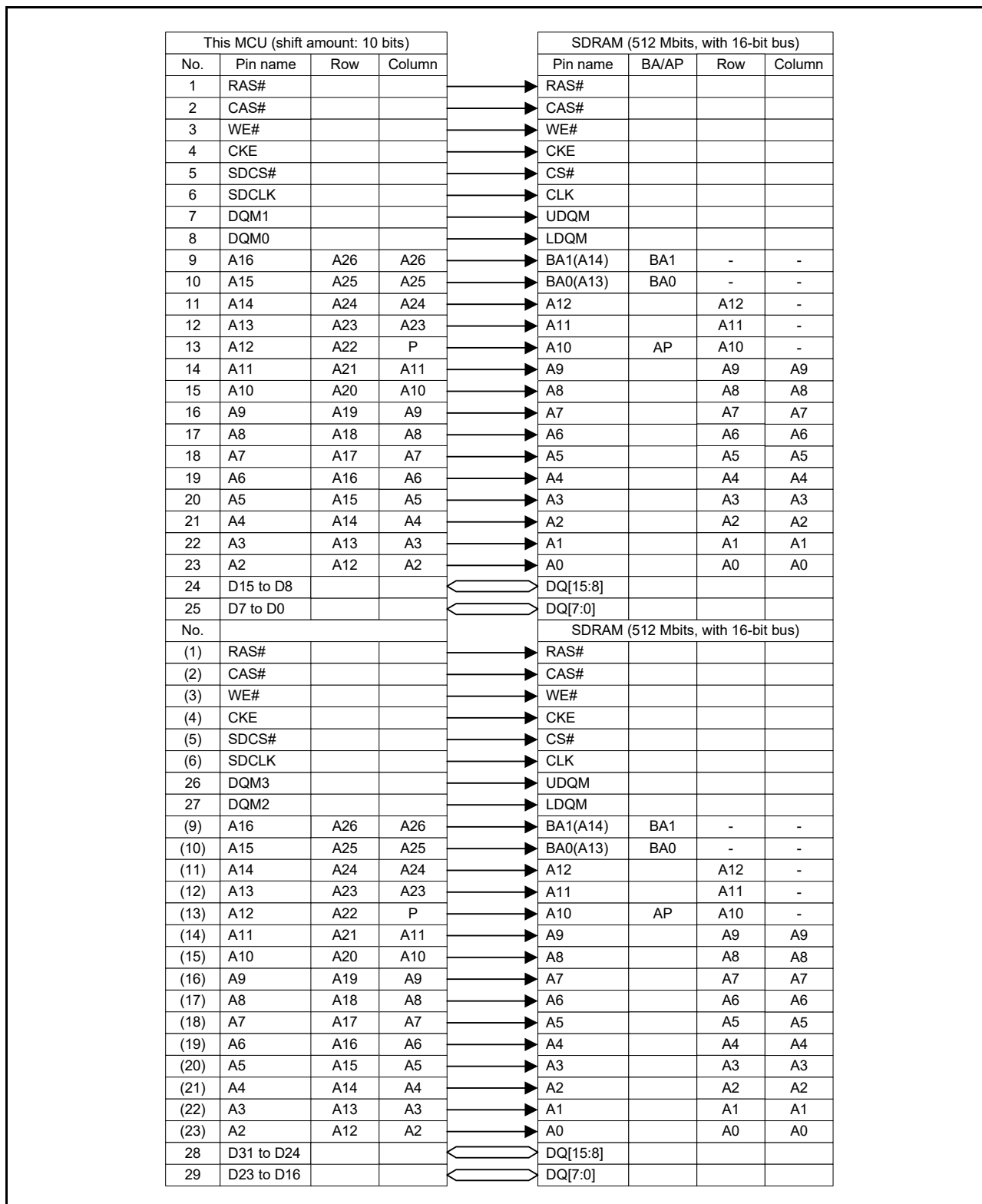


Figure 16.84 SDRAM Connection Example (512-Mbit x 2, with 16-Bit Bus)

Figure 16.85 shows an example for connecting to a 256-Mbit SDRAM with 12-bit row address, 9-bit column address and 32-bit bus.

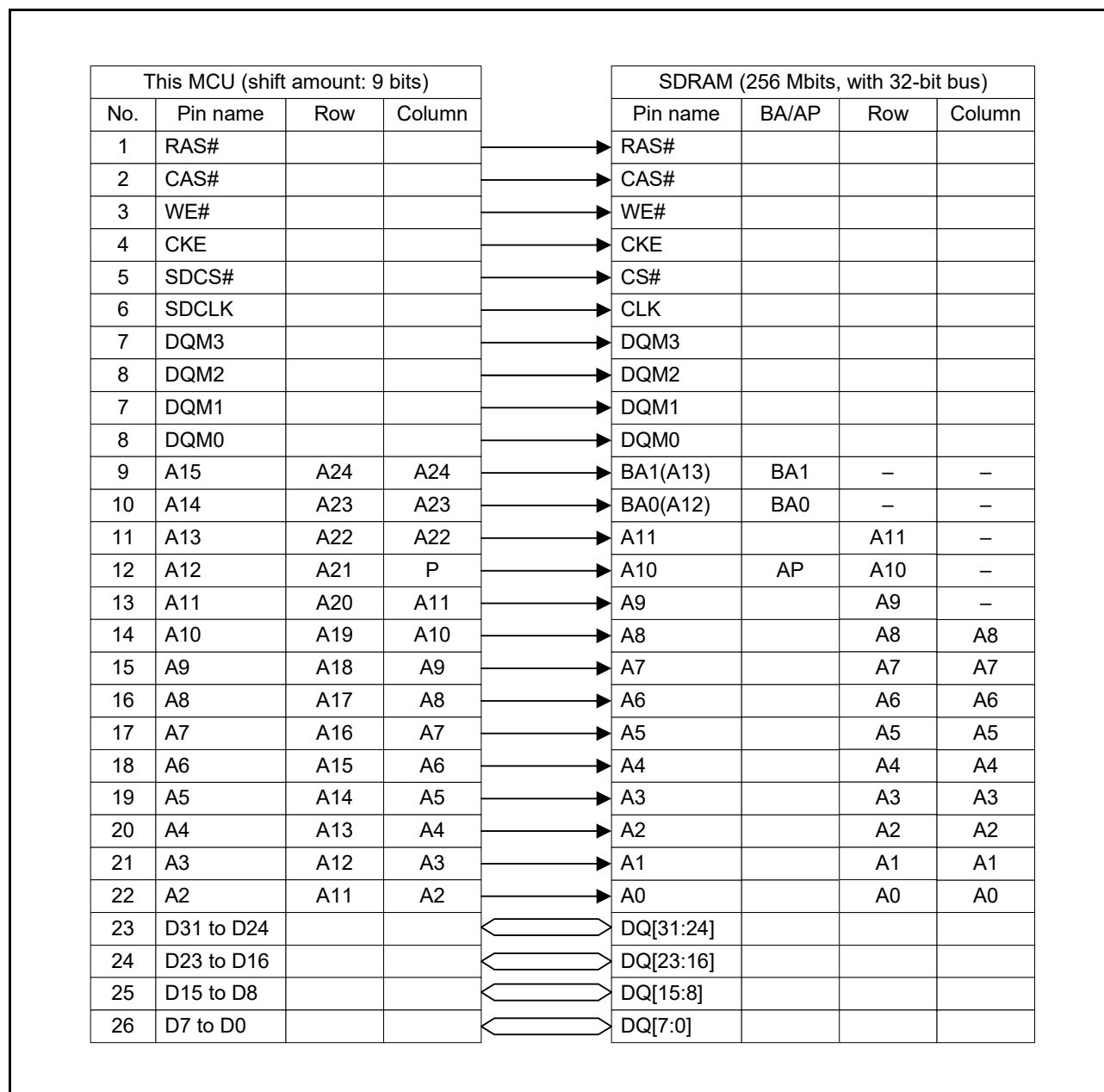


Figure 16.85 SDRAM Connection Example (256-Mbit × 1, with 32-Bit Bus)

Figure 16.86 shows an example for connecting to two 128-Mbit SDRAMs with 12-bit row address, 9-bit column address and 16-bit bus.

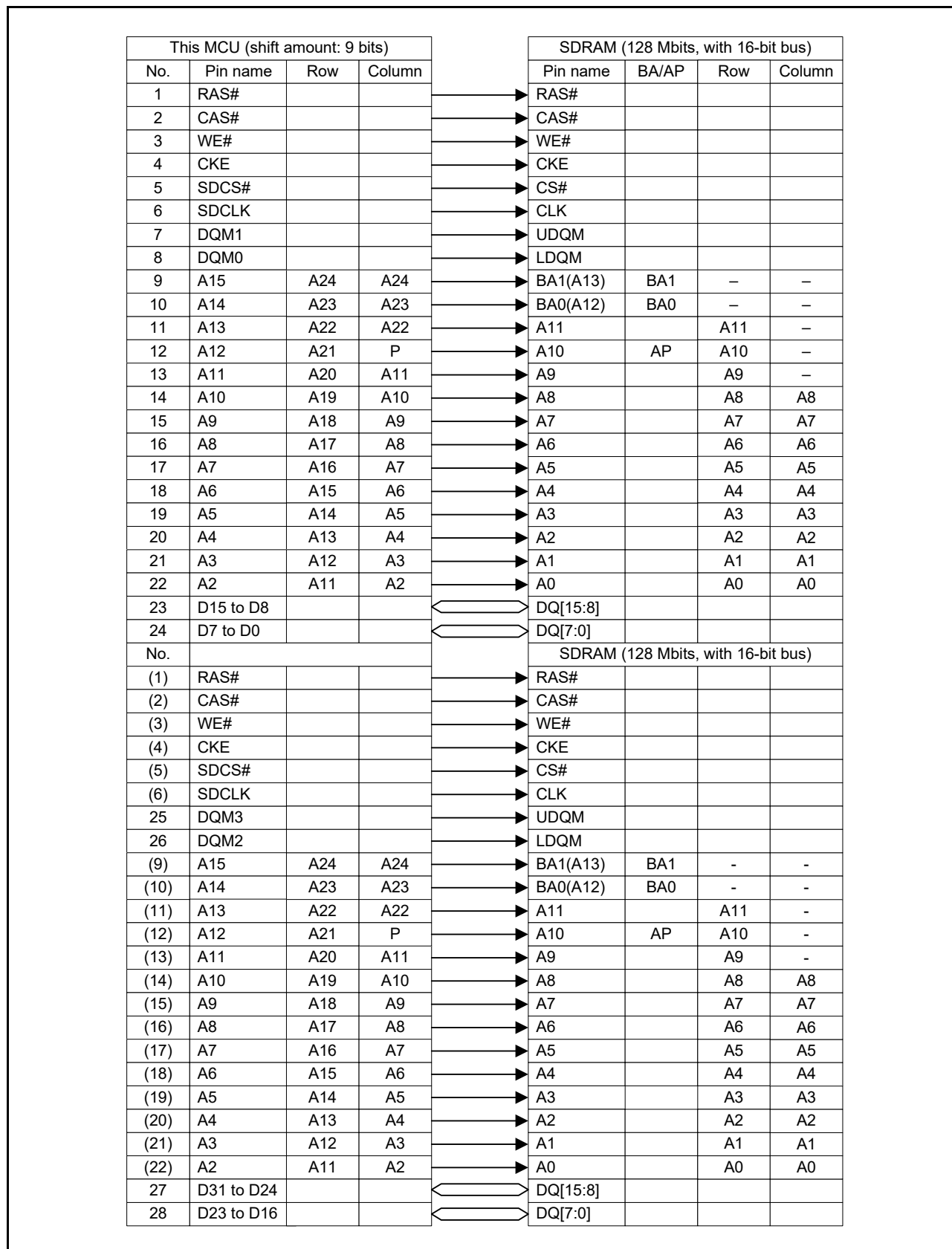


Figure 16.86 SDRAM Connection Example (128-Mbit × 2, with 16-Bit Bus)

16.6.14.2 16-Bit Bus Space

Figure 16.87 shows an example for connecting to two 512-Mbit SDRAMs with 13-bit row address, 11-bit column address and 8-bit bus.

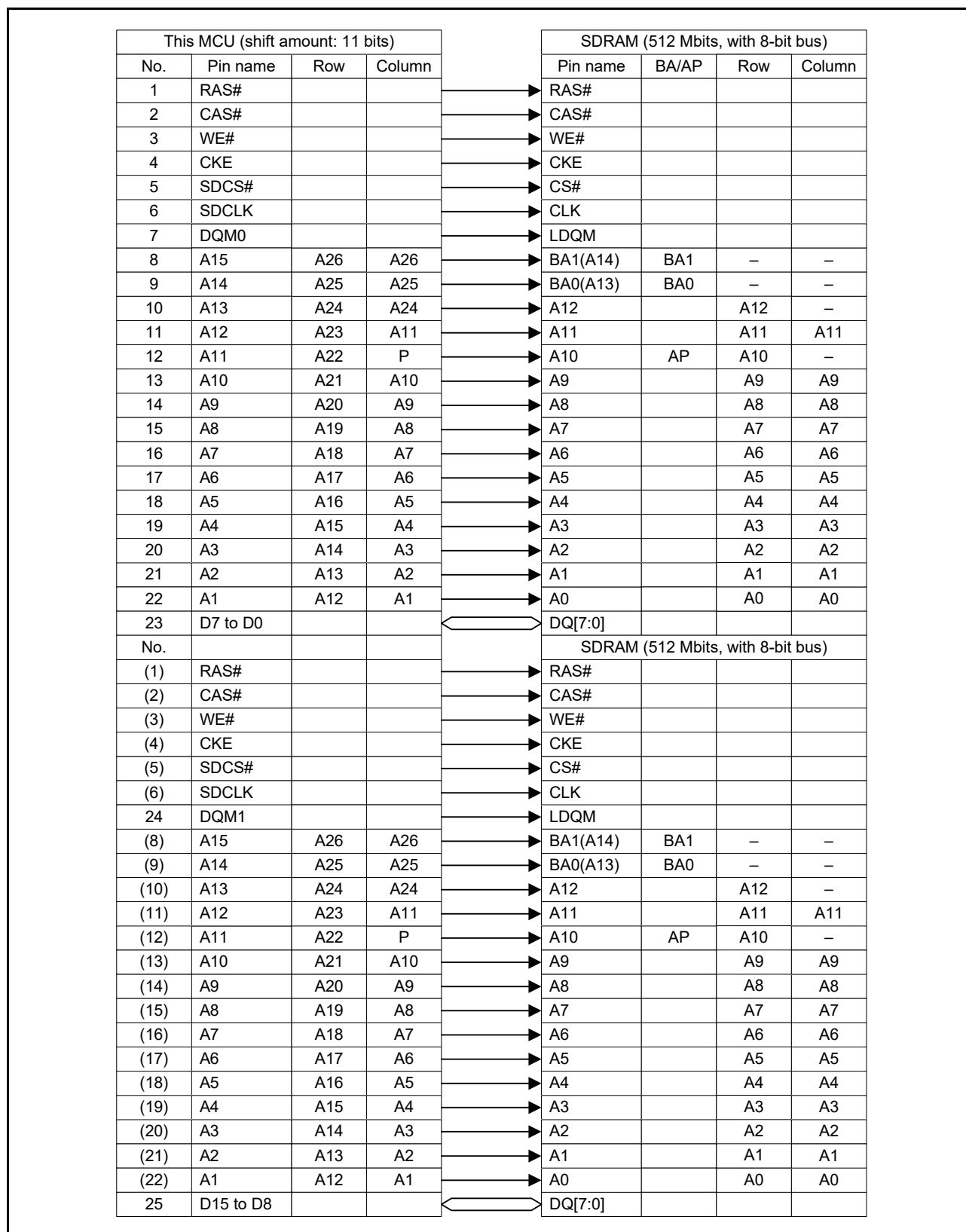


Figure 16.87 SDRAM Connection Example (512-Mbit x 2, with 8-Bit Bus)

Figure 16.88 shows an example for connecting to a 512-Mbit SDRAM with 13-bit row address, 10-bit column address and 16-bit bus.

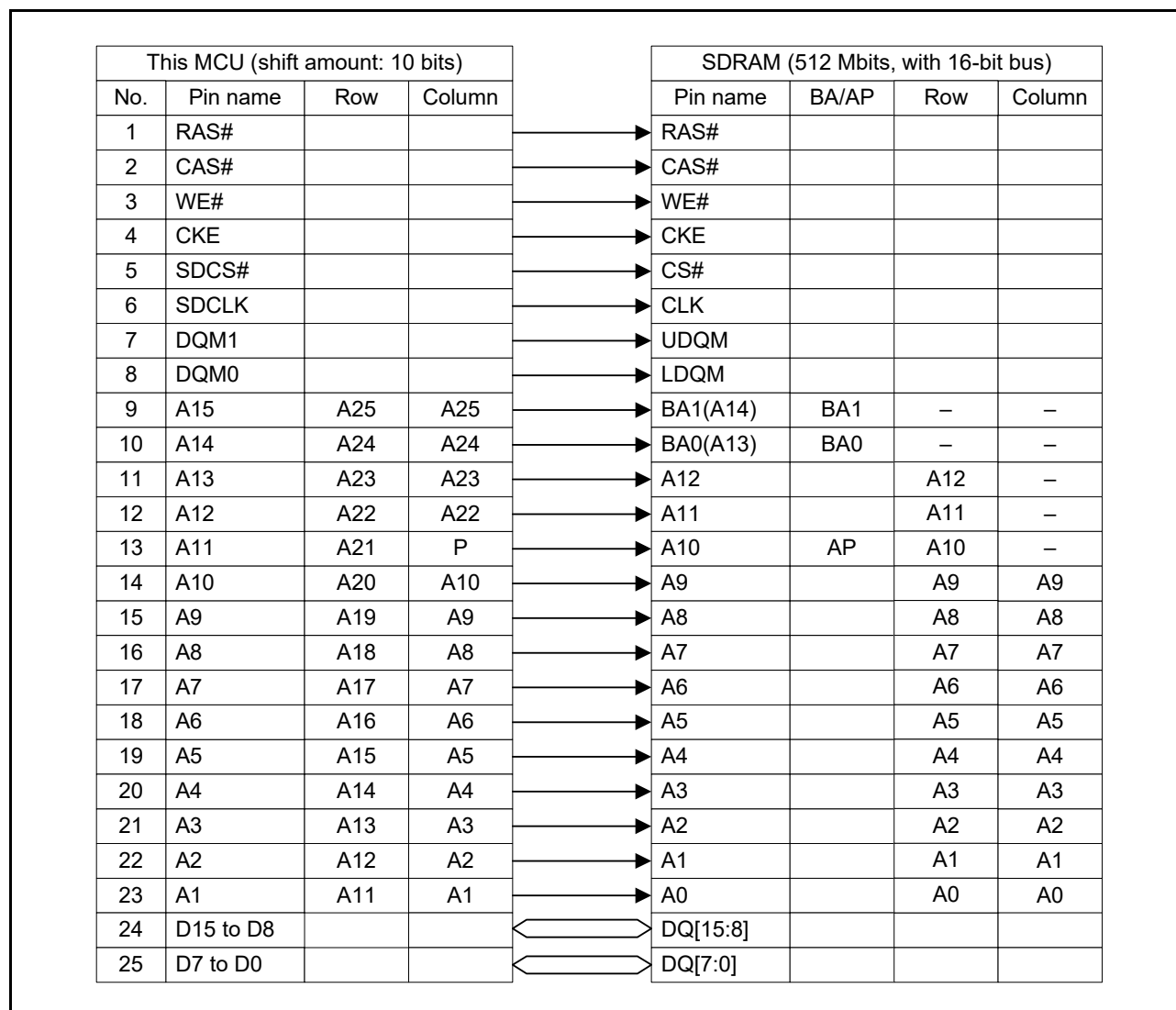


Figure 16.88 SDRAM Connection Example (512-Mbit × 1, with 16-Bit Bus)

Figure 16.89 shows an example for connecting to a 256-Mbit SDRAM with 13-bit row address, 9-bit column address and 16-bit bus.

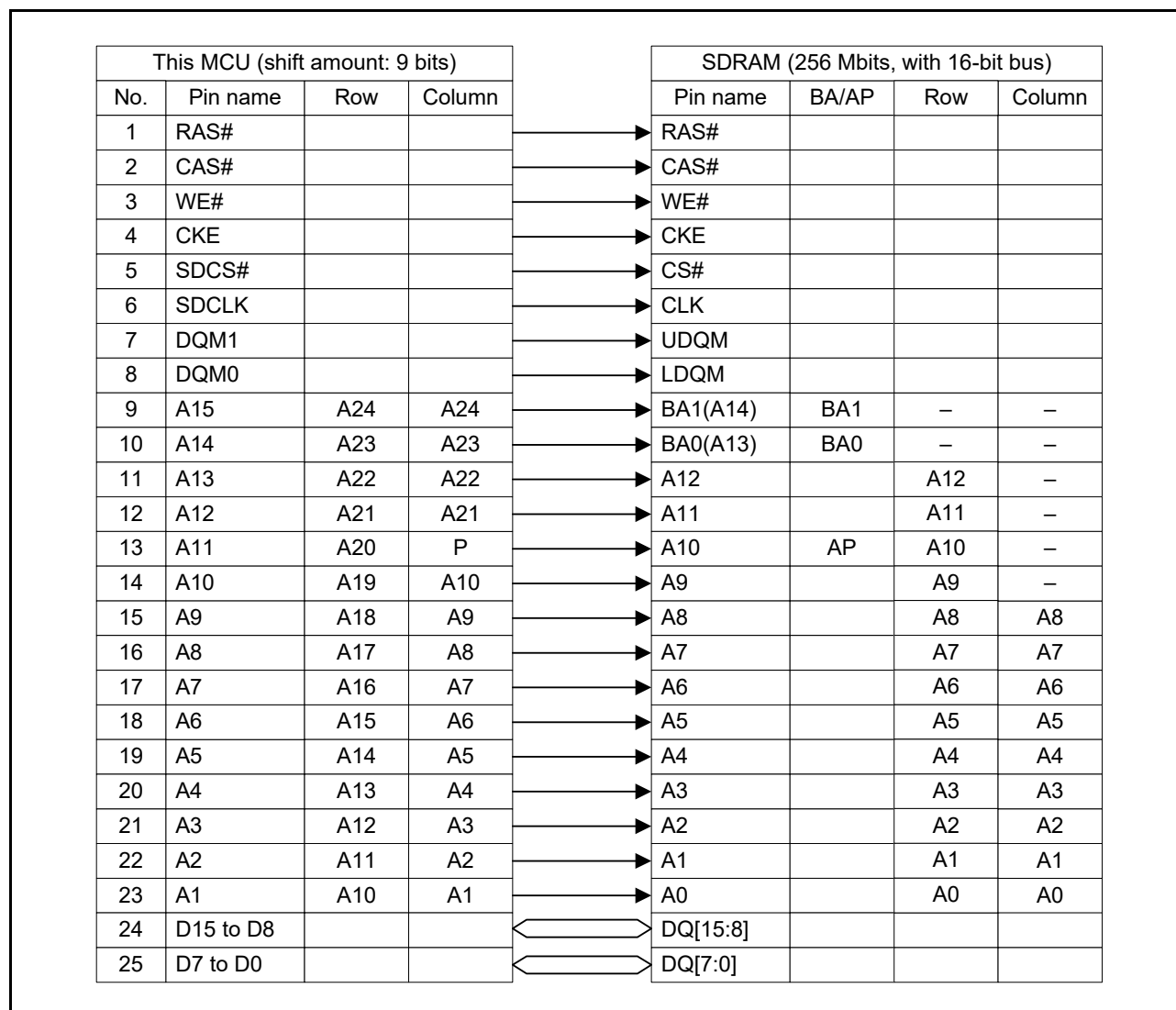


Figure 16.89 SDRAM Connection Example (256-Mbit × 1, with 16-Bit Bus)

16.6.15 Restrictions

(1) Prohibition of Access that Spans Areas of External Address Space

Single access that spans two areas of the external address space is prohibited, and operation of such an access is not guaranteed. Setting must be made so that two areas are not accessed at the same time by a single word or longword access.

Single access that spans two areas of the address space is also prohibited in the EXDMAC block transfer in single address mode or cluster transfer, and operation of such an access is not guaranteed. Setting must be made so that two areas are not accessed at the same time by a single access in the EXDMAC block transfer in single address mode or cluster transfer.

(2) Restrictions on RMPA and String-Manipulation Instructions

- Although the external space has a per-area ending-switching facility (only for data), the allocation of data to be handled by RMPA or string-manipulation instructions to an area where the endian differs from that of the chip is prohibited, and operation is not guaranteed if this restriction is not observed. If data to be handled by RMPA or string-manipulation instructions are allocated to the external space, they must be allocated to areas where the endian setting is the same as that for the chip.
- The allocation of data to be handled by RMPA or string-manipulation instructions to I/O registers is prohibited, and operation is not guaranteed if this restriction is not observed.

(3) Low Power Consumption State

In all-module clock stop mode, software standby mode, and deep software standby mode, auto-refresh operation is not available since the clock supply to SDRAMC is stopped. To retain the data in the SDRAM when the SDRAM is externally connected, use the self-refresh function. For the procedure for transition to and recovery from self-refresh mode, refer to section 16.6.7, Self-Refresh.

(4) Consecutive-Access Mode

For block transfer or cluster transfer by the EXDMAC in single-address mode, the setting CL = 1 is prohibited, and operation is not guaranteed if this setting is made.

(5) Setting the SDRAM Timing Register

Set the RAS[2:0] bits in the SDRAM timing register (SDTR) to a value less than or equal to the sum of the row column latency (SDTR.RCD[1:0]) and column latency (SDTR.CL[2:0]) settings. Operation is not guaranteed if this condition is not satisfied.

(6) Restriction on Instruction Code

When the endian setting for each area is different from that for the chip, no instruction code can be allocated in the area. The instruction code should be allocated to the external space whose endian setting is the same as that for the chip.

16.7 Bus Error Monitoring Section

The bus error monitoring section monitors the individual areas for bus errors, and when a bus error occurs, the error is indicated to the bus master.

16.7.1 Types of Bus Error

There are two types of bus error: illegal address access and timeout.

Illegal address access is the detection of illegal access to an area, and time-out is the detection of a bus-access operation not being completed within 768 cycles.

16.7.1.1 Illegal Address Access

When the illegal address access detection enable bit (IGAEN) in the bus error monitoring enable register (BEREN) is set to 1, access of the following types leads to illegal address access errors.

- Access to areas of external space for which operation has been disabled (CSnCR.EXENB = 0, SDCCR.EXENB = 0)
- With respect to areas other than those described above, access to illegal address ranges

The address ranges where access will lead to illegal address access errors are indicated in Table 16.21.

16.7.1.2 Timeout

When the timeout detection enable bit (TOEN) in the bus error monitoring enable register (BEREN) is set to 1, bus access that is not completed within 768 cycles leads to a timeout error.

- CS areas (CS0 to CS7): Bus access is not completed (the WAIT# signal is not negated) within 768 external bus clock (BCLK) cycles from the start of the access.
Once a timeout error occurs, accesses from the bus master are rejected for 256 BCLK cycles. If multiple external bus accesses are generated with a single request from the bus master during the transfer, the bus accesses cannot be stopped by a timeout. At this time, timeout errors may occur repeatedly.
- Internal peripheral buses (2 and 3): Bus access is not completed within 768 peripheral module clock (PCLKB) cycles from the start of the access. In this MCU, a timeout error does not occur.
Once a timeout error occurs, accesses from the bus master are rejected for 256 PCLKB cycles.
- Internal peripheral buses (4 and 5): Bus access is not completed within 768 peripheral module clock (PCLKA) cycles from the start of the access.
Once a timeout error occurs, accesses from the bus master are rejected for 256 PCLKA cycles. If either of the MSTPB13, MSTPB14, and MSTPB15 bits in MSTPCRB are set to release the corresponding modules from the module stop state, set the BEREN.TOEN bit to 1.
- Internal peripheral bus (6): Bus access is not completed within 768 FlashIF clock (FCLK) cycles from the start of the access.
Once a timeout error occurs, accesses from the bus master are rejected for 256 FCLK cycles. In this MCU, a timeout error does not occur.

16.7.2 Operations When a Bus Error Occurs

When a bus error occurs, the error is indicated to the CPU. Operation is not guaranteed when a bus error occurs.

- Bus error indication to the CPU
An interrupt is generated. The IERn register in the ICU can specify whether to generate an interrupt in the case of a bus error.

16.7.3 Conditions Leading to Bus Errors

Table 16.21 lists the types of bus errors for each area in the respective address space.

If an illegal address access error or timeout is detected when no bus error has occurred (bus error status register n (BERSRn; n = 1 or 2) is cleared), the detected error is reflected on the BERSRn. Once a bus error occurs, no subsequent bus errors are reflected on the register unless the register is cleared.

If bus errors are simultaneously caused by two or more bus masters, error information of only one bus master is reflected. Once a bus error occurs, the status is retained until BERSRn is cleared.

Table 16.21 Types of Bus Errors

Address	Type of Area		Type of Error				
			Illegal Address Access		Timeout		
	On-Chip ROM Enabled	On-Chip ROM Disabled	On-Chip ROM Enabled	On-Chip ROM Disabled	On-Chip ROM Enabled	On-Chip ROM Disabled	
0000 0000h to 0007 FFFFh	Memory bus 1		—		—		
0008 0000h to 0008 7FFFh	Internal peripheral bus 1		—		—		
0008 8000h to 0009 FFFFh	Internal peripheral bus 2		Δ		—		
000A 0000h to 000B FFFFh	Internal peripheral bus 3		Δ		—		
000C 0000h to 000D FFFFh	Internal peripheral bus 4		Δ		✓		
000E 0000h to 000F FFFFh	Internal peripheral bus 5		Δ		—		
0010 0000h to 0011 FFFFh	Internal peripheral bus 6	Reserved area	—	✓	—	—	
0012 0000h to 007F FFFFh			Δ	✓	—	—	
0080 0000h to 00FF FFFFh	Memory bus 3		—	—	—	—	
0100 0000h to 07FF FFFFh	External bus (CS1 to CS7)		[IA]		[TO]		
0800 0000h to 0FFF FFFFh	External bus (SDRAM area)		[IA]		—		
1000 0000h to 7FFF FFFFh	Reserved area		✓		—	—	
8000 0000h to FFFF FFFFh	Memory bus 2	Reserved area	—	✓	—	—	
FF00 0000h to FF7F FFFFh		External bus (CS0)	—	[IA]	—	[TO]	
FF80 0000h to FFFF FFFFh			—		—		

—: A bus error does not result.

Δ: A bus error may or may not result.

✓: A bus error results.

[IA]: Access to this area leads to detection of a bus error if operation for this area is disabled (CSnCR.EXENB = 0; n = 0 to 7, SDCCR.EXENB = 0).

[TO]: Bus access not being completed within 768 cycles leads to detection of a bus error.

Note: The capacity of the code flash memory differ depending on the product. For details, refer to section 62, Flash Memory (FLASH).

16.8 Interrupt

16.8.1 Interrupt Source

An illegal address access error or detection of a timeout leads to a bus error signal for the interrupt controller.

Table 16.22 Interrupt Source

Name	Interrupt Source	DTC Trigger	DMAC Trigger
BUSERR	Illegal address access error or timeout	Not possible	Not possible